Selected Topics in Probability Theory

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Vorwort

Der nachfolgende Text ist meine Mitschrift zur Vorlesung „Selected Topics in Probability Theory". Die Notizen sind auf Englisch abgefasst, also in der Sprache der Vorlesung. Die Vorlesung basiert auf folgender Literatur:

- G. Grimmett: Percolation, Springer 2000
- G. Grimmett: Probability on Graphs 2010
- H. Duminil-Copin: Introduction to percolation theory
- B. Bollobás and O. Riordan: Percolation, CUP 2007
- M. Barlow: Random Walks and Heat Kernels on Graphs, CUP 2017
- A.-S. Sznitman: Topics in occupation times and the Gaussian free field, EMS 2012

Diese Notizen sind weder fertiggestellt noch korrekturgelesen. Insbesondere im mittleren Teil habe ich einfach die handgeschriebenen Notizen des Professors eingefügt. Auch am Ende fehlen noch Vorlesungen.
Die letzte inhaltliche Änderung wurde am 6. Juni 2026 vorgenommen. Der LaTeX-Quellcode kann hier heruntergeladen werden (vorsicht: Code-Qualität ist schlecht und das Projekt ist sehr chaotisch aufgebaut).

Contents

0 Introduction: What this lecture is about

[todo: Muss noch abgetippt werden]

1 Bernoulli Percolation

1.1 Definition and Basics

We start with a few reminders from graph theory.

Definition 1.1. A (unoriented) graph is a tuple $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where $\mathcal{V}$ is the set of *vertices/sites* and $\mathcal{E}$ is the set of *edges/bonds*, i.e.,

$$\mathcal{E} \subseteq \{\{x, y\} \mid x, y \in \mathcal{V}\} .$$

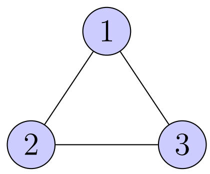
If $e = \{x, y\} \in \mathcal{E}$ is an edge we call x, y *endpoints*. In this case we say that x, y are *neighbours* in the graph $\mathcal{G}$ and write $x \sim y$.

We will mostly denote edges by e, f and vertices by x, y, z.

Remark. Graphs are almost always represented in pictorial way. For example the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with

$$\mathcal{V} = \{1, 2, 3\}, \quad \mathcal{E} = \Big\{\{1, 2\}, \{2, 3\}, \{3, 1\}\Big\}$$

may be represented by



Throughout the following chapter (and the whole lecture) we will mostly use a certain type of graph called *hypercubic lattice*, that is $\mathcal{G} = (\mathbb{Z}^d, \mathcal{E}_d)$, where

$$\mathcal{E}_d = \{\{x, y\} \mid \|x - y\|_1 = 1\}$$

(see Figure 1.1).

Definition 1.2. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph. A path from $x \in \mathcal{V}$ to $y \in \mathcal{V}$ of length ℓ is a sequence $\gamma = (\gamma_0, \dots, \gamma_\ell)$ such that $\gamma_0 = x$, $\gamma_\ell = y$ with $\{\gamma_i, \gamma_{i+1}\} \in \mathcal{E}$. If γ_i are distinct

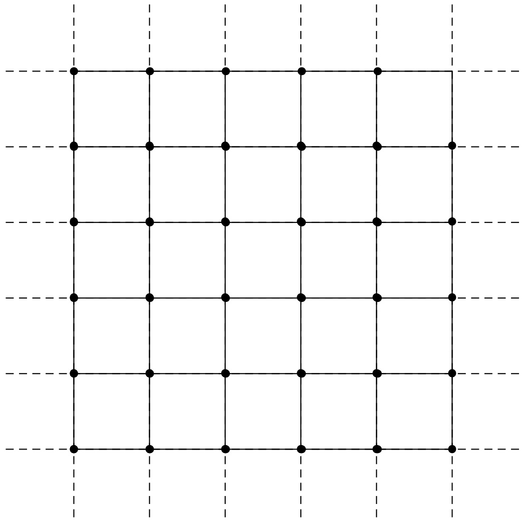


Figure 1.1: Graphical representation of $(\mathbb{Z}^2, \mathcal{E}_d)$.

we call the path *self-avoiding*. The distance between two points $x, y \in \mathcal{V}$ is given by

$$d(x,y) = \inf \{\text{length}(\gamma) \mid \gamma \text{ is a (self-avoiding) path from } x \text{ to } y\}.$$

On $\mathcal{G} = (\mathbb{Z}^d, \mathcal{E}_d)$ the graph distance coincides with the distance induced by the 1-norm $\|\cdot\|_1$.

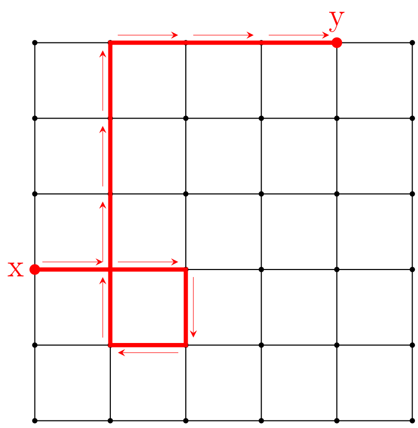


Figure 1.2: This is *not* a self-avoiding path, because we have a loop

Definition 1.3. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph and let $A \subseteq \mathcal{V}$. We define

$$\begin{aligned}
\partial A &:= \left\{ x \in A \,\middle|\, \exists y \in A^{\complement} \text{ s.t. } x \sim y \right\}, && \text{the } \textit{boundary of } A, \\
\partial A &:= \left\{ x \in A \,\middle|\, \exists y \in A^{\complement} \text{ s.t. } x \sim y \right\}, && \text{the } \textit{external boundary of } A, \\
\Delta A &:= \left\{ e \in \mathcal{E} \,\middle|\, e = \{x, y\}, x \in A, y \in A^{\complement} \right\} && \text{the set of edges crossing the boundary}
\end{aligned}$$

and if $\mathcal{G} = (\mathbb{Z}^d, \mathcal{E}_d)$,

$$\Lambda_n = [-n, n]^d \cap \mathbb{Z}^d.$$

We now introduce the probability space associated with percolation: Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph, $\Omega = \{0, 1\}^{\mathcal{E}}$ which we call the *percolation configuration*. Given a realization $\omega \in \Omega$, if $\omega(e) = 1$ we say that the edge e is *open* (in ω) otherwise, if $\omega(e) = 0$ we say that e is *closed* (in ω). Let $\mathcal{A}$ be the σ-algebra on Ω, generated by events depending on the state of only finitely many edges (cylinder σ-algebra). Moreover, given $p \in [0, 1]$ we define the *percolation measure* by

$$\mathbb{P}_p = \mu^{\otimes E}, \quad \text{where} \quad \mu(\{1\}) = 1 - \mu(\{0\}) = p.$$

This measure is characterized as follows: let $e_1, \dots, e_n \in \mathcal{E}$ and $\omega_1, \dots, \omega_n \in \{0, 1\}$, then

$$\mathbb{P}_p\left(\bigcap_{i=1}^{n} \{\omega(e_i) = \omega_i\} \right) = p^{\sum_{i=1}^{n} \omega_i} (1 - p)^{\sum_{i=1}^{n} (1 - \omega_i)}.$$

This construction corresponds to opening and closing edges of the graph $\mathcal{G}$ independently according to a Bernoulli-distribution with parameter p.[1] We denote

$$\{x \leftrightarrow y\} := \{\omega \in \Omega : \text{there is an open path from } x \text{ to } y \text{ in } \omega\}.$$

An open path is a path that only contains open edges. For $A, B \subseteq \mathcal{V}$ we define

$$\{A \leftrightarrow B\} := \bigcup_{x \in A} \bigcup_{y \in B} \{x \leftrightarrow y\}$$

and, if the graph $\mathcal{G}$ is infinite,

$$\{x \leftrightarrow \infty\} = \bigcap_{n \in \mathbb{N}} \{\exists y \in \mathcal{V} : d(x, y) = n, x \leftrightarrow y\}.$$

Exercise 1.4. Show that these sets are events.

[1] That is, the family of random variables $(X_e)_{e \in \mathcal{E}}$ with $X_e : \Omega \to \mathbb{R}$ given by $X_e(\omega) = e(\omega)$ for $\omega \in \Omega$ is i.i.d..

Definition 1.5. Given a realization $\omega \in \Omega$, a connected component of

$$(\mathcal{V}, \{e \in \mathcal{E} \mid \omega(e) = 1\})$$

is called *cluster*. For $x \in \mathcal{V}$ we write $\mathcal{C}_x = \mathcal{C}_x(\omega)$ for the cluster containing x.

A big part of percolation theory is to study the beharviour of the map

$$[0,1] \ni p \mapsto \mathbb{P}_p(x \leftrightarrow y).$$

Is this map continuous? Increasing? etc. In order to do this we have to introduce an ordering on Ω: Given $\eta, \omega \in \Omega$ we define

$$\omega < \eta \quad \text{if } \omega(e) \leq \eta(e) \text{ for all } e \in \mathcal{E}.$$

This is clearly not a total order! Using this ordering we can talk about increasing or decrasing functions: A function $f : \Omega \to \mathbb{R}$ is *increasing* if $f(\omega) \leq f(\eta)$ for all $\omega < \eta$. Similarly, an event $A \in \mathcal{A}$ is increasing if $\omega \mapsto \mathbf{1}_A(\omega)$ is an incrasing function. This is equivalent to (check this as an exercise)

$$\omega \in A, \eta > \omega \implies \eta \in A.$$

Example 1.6. Show the following: The event $\{x \leftrightarrow y\}$ is increasing, as well as $\{|\mathcal{C}_x| \geq 17\}$. The event $|\mathcal{C}_x| = 10$ is neither increasing nor decreasing.

Proposition 1.7. Let $(\Omega, \mathcal{A}, \mathbb{P}_p)$ be the probability space we just introduced. The following holds:

(a) Let $A \in \mathcal{A}$ be an increasing event, then the map $[0,1] \ni p \mapsto \mathbb{P}_p(A)$ is non-decreasing.

(b) If $f : \Omega \to \mathbb{R}$ is increasing, then the map $[0,1] \ni p \mapsto \mathbb{E}_p[f]$ is non-decreasing.

Proof. We only prove (b) as it implies (a). To this end, let $(U_e)_{e \in \mathcal{E}}$ be a family of i.i.d. random variables on a probability space $(\hat{\Omega}, \Sigma, \mu)$ that are uniformly distributed on $[0,1]$. Now, define $X_e^p(\hat{\omega}) = \mathbf{1}_{U_e(\hat{\omega}) \leq p}(\hat{\omega}) \in \{0,1\}$ for each $e \in \mathcal{E}$, $\hat{\omega} \in \hat{\Omega}$. Then for $p \in [0,1]$ fixed, $(X_e^p)_{e \in \mathcal{E}}$ is a collection of i.i.d. random variables with

$$\mu\Big(X_e^p = 1\Big) = \mu\Big(U_e \leq p\Big) = \int_{-\infty}^{p} \mathbf{1}_{[0,1]}(t)\, \mathrm{d}t = p.$$

That is, the distribution of $X^p = (X_e^p)_{e \in \mathcal{E}}$ is $\mathbb{P}_p$.

If $p < p'$ then $X_e^p \leq X_e^{p'}$ pointwise and thus for any increasing function $f : \Omega \to \mathbb{R}$ that is also integrable (or non-negative)

$$\mathbb{E}_p[f] = \int_{\hat{\Omega}} f\Big(X^p(\hat{\omega})\Big)\mu(\mathrm{d}\hat{\omega}) \leq \int_{\hat{\Omega}} f\Big(X^{p'}(\hat{\omega})\Big)\mu(\mathrm{d}\hat{\omega}) = \mathbb{E}_{p'}[f],$$

where $\Omega = [0,1]^{\mathcal{E}}$ and we regard X^p as a function $\hat{\Omega} \to \Omega$. This concludes the proof. $\square$

1.2 Critical probability

We define our next object of interest, the *percolation probability* $\theta(p)$, by

$$\theta(p) := \mathbb{P}_p(0 \leftrightarrow \infty) = \mathbb{P}_p(|\mathcal{C}_x| = \infty) = \lim_{n\to+\infty} \mathbb{P}_p(0 \leftrightarrow \partial\Lambda_n)$$

Note that $0 \leftrightarrow \partial\Lambda_n$ is an event as it depends only on finitely many edges meaning that all of the involved sets are measurable, i.e., all probabilities are well defined.

In the following the goal is to study the critical value

$$p_c := \inf\{p \in [0,1] : \theta(p) > 0\}.$$

Question 1.8. Does $0 < p_c < 1$ occur?

Exercise 1.9. Prove that in $d = 1$,

$$\theta_1(p) = \begin{cases} 0, & \text{if } p < 1 \text{ and} \\ 1, & \text{if } p = 1. \end{cases}$$

Consider the event

$$I = \{\omega \in \Omega \mid \omega \text{ contains an infinite cluster}\} = \bigcup_{x\in\mathbb{Z}^d} \{|\mathcal{C}_x| = +\infty\}. \tag{1.1}$$

What is the probability of I?

Proposition 1.10. Consider percolation on $\mathcal{G} = (\mathbb{Z}^d, \mathcal{E}_d)$ and let I be given by (1.1). The following statements hold:

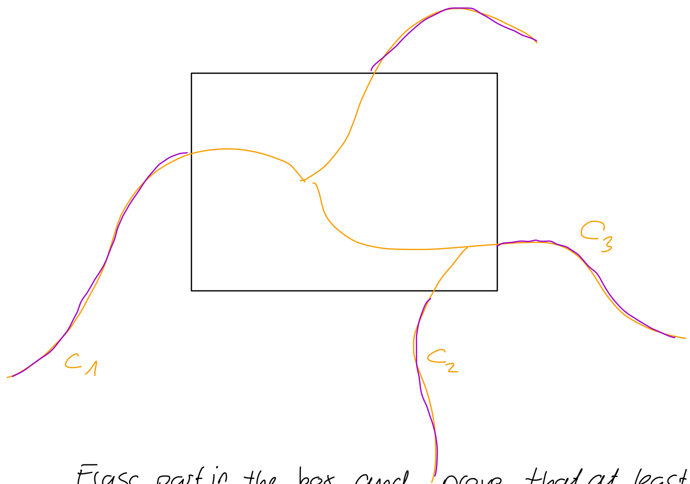
(a) It holds that $\mathbb{P}_p(I) \in \{0,1\}$.

(b) $\mathbb{P}_p(I) = 0$ if and only if $\theta(p) = 0$.

Proof. *(a):* Enumerate edges of $\mathcal{E}_d$ by increasing distance from the origin and write $\mathcal{E}_d = \{\omega_1, \omega_2, \ldots\}$. Define

$$I_n := \left\{\omega \in \Omega \;\middle|\; \text{there is infinite cluster in } \mathbb{Z}^d \setminus \Lambda_n\right\}$$

We now claim that $I = I_n$. Indeed, if we show this, the claim follows by Kolmogorov's 0-1-law. The implication $I \supseteq I_n$ is obvious. To prove the opposite inclusion assume that $\omega \in I$ is fixed. Close all edges in Λ_n, which divides our initial infinite cluster into sub-clusters $C_1, \ldots, C_n$. Then at least one of the resulting clusters has to be infinite (note that the cluster divides into finitely many sub-clusters, because all, because else



Erase part in the box and prove that at least one of the remaining clusters is infinite and that the number of clusters that are generated by erasing is finite.

Figure 1.3:

we would have infinitely many clusters connected to Λ_n which is not possible as $\Delta\Lambda_n$ is finite). This shows that $\omega \in I_n$ (also see Figure 1.2)

(b): We estimate

$$\begin{aligned}
\theta(p) &= \mathbb{P}_p(|\mathcal{C}_0| = \infty) \\
&\le \mathbb{P}_p(I) \\
&= \mathbb{P}_p\Bigg(\bigcup_{x \in \mathbb{Z}^d} \{|\mathcal{C}_x| = \infty\} \Bigg) \\
&\le \sum_{x \in \mathbb{Z}^d} \mathbb{P}_p(|\mathcal{C}_x| = \infty) \\
&\overset{(!)}{=} \sum_{x \in \mathbb{Z}^d} \theta(p),
\end{aligned}$$

where (!) follows from the translation invariance of $\mathbb{P}_p(|\mathcal{C}_x| = \infty)$. From this the claimed equivalence follows. □

To prove the next theorem we need to introduc the concept of dual graphs.

Dual graphs. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a *planar* graph (that is a graph that can be drawn in the plane without intersecting edges). A region in the plane that is enclosed by vertices of the graph we call *face of the graph* $\mathcal{G}$. The dual graph of $\mathcal{G}$, $\mathcal{G}^\star = (\mathcal{V}^\star, \mathcal{E}^\star)$ has an edge for each pair of faces in $\mathcal{G}$ that are separated from each other by an edge, and a self-loop when the same face appears on both sides of an edge (see Figure 1.2).

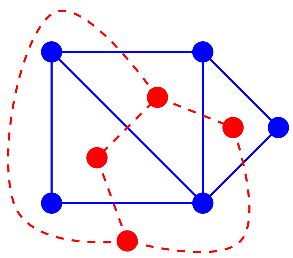


Figure 1.4: Graph in blue and in red its dual graph.

Claim. There is a bijection between $\mathcal{E}$ and $\mathcal{E}^\star$ via $e \mapsto e^\star$. This is achieved by mapping each edge to the dual edges that crosses it.

Theorem 1.11. Consider percolation on $\mathcal{G} = (\mathbb{Z}^d, \mathcal{E}_d)$. For $d \geq 2$, $p_c(d) \in (0, 1)$.

Proof. *Step 1* . We show that $\theta_d(p) = 0$ for any $p \leq \frac{1}{2d}$ which implies that $p_c(d) \geq \frac{1}{2d}$. Let

$$\mathcal{P}_n := \{\gamma : \gamma \text{ is a self-avoiding path starting in 0 of length } n\}$$

and observe that for all $n \in \mathbb{N}$,

$$\{|\mathcal{C}_0| = \infty\} \subseteq \{\exists \gamma \in \mathcal{P}_n \text{ where all edges of } \gamma \text{ are open}\}$$

which implies by monotonicity of the measure

$$\theta(p) = \mathbb{P}_p(|\mathcal{C}_0| = \infty) \leq \mathbb{P}_p\left(\exists \gamma \in \mathcal{P}_n,\ \gamma \text{ is open }\right) \leq \sum_{\gamma \in \mathcal{P}_n} \underbrace{\mathbb{P}_p\left(\gamma \text{ is open}\right)}_{=p^n} = p^n \left|\mathcal{P}_n\right|$$

for all $n \in \mathbb{N}$. Noting that 0 has $2d$ neighbours and every γ_i has $2d - 1$ neighbours that aren't used to build the path γ we obtain the upper bound

$$|\mathcal{P}_n| \leq (2d)(2d-1)^{n-1}.$$

Plugging this into the above line and letting $n \to +\infty$ with $p \leq \frac{1}{2d}$, we obtain that $\theta(p) = 0$.

Step 2 . Let $d = 2$. We claim that for any $p > \frac{3}{4}$, $\theta_2(p) > 0$ (which again implies that $p_c(2) \leq \frac{3}{4}$). We will use something called *Peierls argument*.

Note that $\mathbb{Z}^2$ is self-dual (that is $\mathbb{Z}^2$ is its own dual). We now define a *dual configuration* living on the dual graph. Let $\omega \in \Omega$. Define

$$\omega^\star(e^\star) = 1 - \omega(e) = \begin{cases} 1, & \text{if } \omega(e) = 0 \\ 0, & \text{if } \omega(e) = 1 \end{cases}$$

If $\omega \sim \mathbb{P}_p$, then $\omega^\star \sim \mathbb{P}_{1-p}$ and if $\mathcal{C}_0$ is finite, then there is a cycle in $\omega^\star$ surrounding 0 (we are not going to show this, the reader is advised to carefully check this as an exercise). Now,

$$\begin{aligned} 1 - \theta(p) &= \mathbb{P}_p(|\mathcal{C}_0| < \infty) \\ &= \mathbb{P}_p\left(\bigcup_{n\in\mathbb{N}_0} \{\omega^\star \text{ contains an open cycle of length } n \text{ surrounding } 0\}\right) \\ &\le \sum_{n\in\mathbb{N}_0} \underbrace{\mathbb{P}_p(\dots)}_{(1-p)^n} \\ &\le \sum_{n\in\mathbb{N}} (1-p)^n \cdot \#\left\{\text{cycles around 0 of length } n\right\}. \end{aligned}$$

If n is odd or $n = 2$ there are no such cycles. A cycle of length n intersects the positive x-axis at most at distance $n/2$ from the origin, that is we have at most $n/2$ possibilities to cross. Moreover, in every step we have at most 3 possibilities to continue our path, thus

$$\#\left\{\text{cycles around 0 of length } n\right\} \le \frac{n}{2} \cdot 3^{n-1}.$$

Using all these observations one obtains (in the sum we have $2k = n$)

$$1 - \theta(p) \le \sum_{k\ge 2} (1-p)^{2k} k \cdot 3^{2k-1} < 1$$

if $p > \frac{3}{4}$ from which the claim follows.

Step 3 . We claim

$$p_c(d) \le p_c(2) \quad \text{for all } d \ge 2. \tag{1.2}$$

Using that $\mathbb{Z}^2$ can be embedded in $\mathbb{Z}^d$ for $d \ge 2$ we have that

$$\mathbb{P}_p(|\mathcal{C}_0| = \infty) \ge \mathbb{P}_p(|\mathcal{C}_0 \cap \mathbb{Z}^2| = \infty). \qquad \square$$

1.3 Harris-FKG-inequality

To motivate our next result consider two events A, B. What can we say about the probability of the intersection $\mathbb{P}(A \cap B)$?

As an example take $x, y, u, v \in \mathbb{Z}^d$ and let $e \in \mathcal{E}$ be an edge. One would expect

$$\mathbb{P}_p(x \leftrightarrow y) \le \mathbb{P}_p\left(x \leftrightarrow y \mid e \text{ open }\right) \quad \text{or} \quad \mathbb{P}_p(x \leftrightarrow y) \le \mathbb{P}_p\left(x \leftrightarrow y \mid u \leftrightarrow v\right).$$

Theorem 1.12 (Harris-FKG-inequality). Let $p \in [0,1]$. The following holds:

(a) If A, B are increasing events then $\mathbb{P}_p(A \cap B) \geq \mathbb{P}_p(A)\mathbb{P}_p(B)$.

(b) If $f, g \in L^2(\mathbb{P}_p)$ are increasing then $\mathbb{E}_p[fg] \geq \mathbb{E}_p[f]\mathbb{E}_p[g]$.

Remark.
- If we replace by f by $-f$ and g by $-g$ we can also put decreasing and the equality still holds.
- The proof does not use the geometry of $\mathbb{Z}^d$, it only uses the properties of the product measure. However, FKG even holds for some non-product measure.

Proof of Theorem 1.12. We only show (b), as it implies (a).

Step 1 . Assume that f, g only depend on finitely many edges. We enumerate the edges $\mathcal{E}_d = \{e_1, e_2, \ldots\}$ and write $\omega_i := \omega(e_i)$. If $f, g : \{0,1\}^n \to \mathbb{R}$ are increasing we claim that

$$\mathbb{E}_p[f(\omega_1, \ldots, \omega_n)g(\omega_1, \ldots, \omega_n)] \geq \mathbb{E}_p[f(\omega_1, \ldots, \omega_n)]\mathbb{E}_p[g(\omega_1, \ldots, \omega_n)]. \tag{1.3}$$

We prove this claim by induction: For $n = 1$, $a, b \in \{0,1\}$ we have

$$\Big(f(a) - f(b)\Big)\Big(g(a) - g(b)\Big) \geq 0$$

so multiplying by $\mathbb{P}_p(\omega_1 = a)\mathbb{P}_p(\omega_1 = b) \geq 0$ and summing over all a, b gives us

$$2(\mathbb{E}_p[fg] - \mathbb{E}_p[f]\mathbb{E}_p[g]) \geq 0.$$

Assume that (1.3) holds for $n \leq k$. Then, using that ω_i are independent, and writing $\boldsymbol{\omega} = (\omega_1, \ldots, \omega_{k+1}) = (\boldsymbol{\omega}', \omega_{k+1})$, as well as $\tilde{f}(k) = \mathbb{E}_p[f(\boldsymbol{\omega}', k)]$ and $\tilde{g}(k) = \mathbb{E}_p[g(\boldsymbol{\omega}', k)]$ for $k \in \{0,1\}$ we find

$$\begin{aligned}
\mathbb{E}_p\Big[(f \cdot g)(\boldsymbol{\omega})\Big] &= p \cdot \mathbb{E}_p\Big[(f \cdot g)(\boldsymbol{\omega}', 1)\Big] + (1-p) \cdot \mathbb{E}_p\Big[(f \cdot g)(\boldsymbol{\omega}', 0)\Big] \\
&\geq p \cdot \mathbb{E}_p\Big[f(\boldsymbol{\omega}', 1)\Big]\mathbb{E}_p\Big[g(\boldsymbol{\omega}', 1)\Big] + (1-p) \cdot \mathbb{E}_p\Big[f(\boldsymbol{\omega}', 0)\Big]\mathbb{E}_p\Big[g(\boldsymbol{\omega}', 0)\Big] \\
&= \mathbb{E}_p[\tilde{f}\tilde{g}] \\
&\geq \mathbb{E}_p[\tilde{f}]\mathbb{E}_p[\tilde{g}] \\
&= \mathbb{E}_p[f]\mathbb{E}_p[g].
\end{aligned}$$

Step 2 . Take $f, g \in L^2(\mathbb{P}_p)$ possibly not depending on finitely many edges and set

$$f_n := \mathbb{E}_p\left[f \mid \sigma(\omega_1, \ldots, \omega_n)\right] \quad \text{and} \quad g_n = \mathbb{E}_p\left[g \mid \sigma(\omega_1, \ldots, \omega_k)\right].$$

Then f_n, g_n only depend on $\omega_1, \ldots, \omega_n$ and are increasing. Also f_n, g_n, $n \in \mathbb{N}$ are martingales bounded in $L^2(\mathbb{P}_p)$ so evoking the martingale convergence theorem we find

$$f_n \xrightarrow{L^2} f, \quad g_n \xrightarrow{L^2} g.$$

Since convergence in L^2 implies convergence in L^1 we obtain, using the previous step,

$$\mathbb{E}_p[fg] = \lim_{n \to +\infty} \mathbb{E}_p[f_n g_n] \geq \lim_{n \to +\infty} \mathbb{E}_p[f_n]\mathbb{E}[g_n] = \mathbb{E}_p[f]\mathbb{E}_p[g]. \qquad \square$$

The next corollary is also very useful:

Corollary 1.13 (square-root-trick)**.** For n increasing events $A_1, \dots, A_n \in \mathcal{A}$ it holds

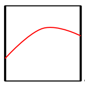
$$\max_{k=1,\dots,n} \mathbb{P}_p(A_k) \geq 1 - \Big(1 - \mathbb{P}_p\left(A_1 \cup \dots \cup A_n\right)\Big)^{1/n}.$$

Proof. Using Theorem 1.12 inductively we get

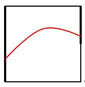
$$\begin{aligned} \mathbb{P}_p(A_1 \cup \dots \cup A_n) &= 1 - \mathbb{P}_p(A_1^\complement \cap \dots \cap A_n^\complement) \\ &\leq 1 - \mathbb{P}_p(A_1^\complement) \dots \mathbb{P}_p(A_n^\complement) \\ &\leq 1 - \Big(1 - \max_{k=1,\dots,n} \mathbb{P}_p(A_k)\Big)^n, \end{aligned}$$

and the claim follows by rearranging. □

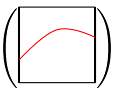
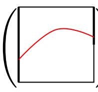
Application: percolation on $\mathbb{Z}^2$. Consider percolation on $(\mathbb{Z}^2, \mathcal{E}_2)$. Consider a rectangle and the event that there is a path from left to right in that rectangle. We denote this event by

.

We want to compare this event to the event that the path from left to right ends up in the upper half of the square denoted by

.

Note that, by symmetry, this has the same probability that the path ends up in the lower half. Using the square-root-trick, Corollary 1.13 we obtain the following implication:

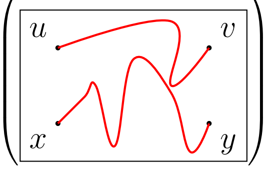
$\mathbb{P}_p\Big($  $\Big) \geq 1 - \varepsilon \implies \mathbb{P}_p\Big($  $\Big) \geq 1 - \sqrt{\varepsilon}.$

1.4 BK inequality

In various situations the FKG inequality is useless since it goes in the wrong direction. We thus need a complementary opposite inequality, which of course cannot hold for the intersection $A \cap B$ of two increasing events A, B. We are, thus, going to define a new event, so called disjoint occurrence of A, B, denoted by $A \circ B$. t is useful to start with an example.

Example 1.14. Consider four vertices $x, y, u, v \in \mathcal{V}$ and consider the event that there exist γ_1, γ_2 two disjoint paths connecting x, y and u, v. t seems plausible that the fact that we cannot use the same edges for connecting x, y and u, v makes the probability of the event smaller than $\mathbb{P}_p(x \leftrightarrow y)\mathbb{P}_p(u \leftrightarrow v)$. Informally we can write this as

$\mathbb{P}_p\Bigg($



$\Bigg) \le \mathbb{P}_p(x \leftrightarrow y)\mathbb{P}_p(u \leftrightarrow v)$

disjoint paths

Definition 1.15. Let $A \in \mathcal{A}$ and $\omega \in \Omega$. A set $I \subseteq \mathcal{E}$ is a *witness of A in ω* if

$$\omega \in A \text{ and } \omega' \in \Omega \text{ s.t. } \omega\Big|_I = \omega'\Big|_I \implies \omega' \in A.$$

Remark. Note that I may depend on ω.

Example 1.16.
- If γ is a path from x to y and ω is such that all edges of γ are open, then the set containing all edges of γ is a witness of $\{x \leftrightarrow y\}$.
- If $\omega \in A$, then $\mathcal{E}$ is always a witness for A in ω.

Exercise 1.17.
(a) Find a witness for $|\mathcal{C}_0| = 3$, $|\mathcal{C}_0| \geq 3$.

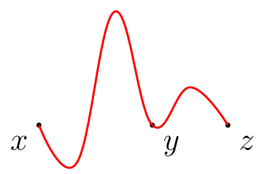
(b) If A is increasing, $\omega \in A$, then there is I witnessing A in ω such that $\omega|_I \equiv 1$.

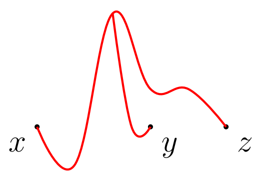
Definition 1.18. Let $A, B \in \mathcal{A}$. Set

$$A \circ B := \left\{ \omega \in \Omega \;\middle|\; \exists I, J \subseteq \mathcal{E} \text{ disjoint s.t.} \begin{matrix} I \text{ witnesses } A \text{ in } \omega \\ J \text{ witnesses } B \text{ in } \omega \end{matrix} \right\}.$$

When $A \circ B$ occurs, we say A, B *occur disjointly.*

Example 1.19.
- $\{e \text{ is open}\} \circ \{f \text{ is open}\} = \begin{cases} \emptyset, & \text{if } e = f \\ \{e,\ f \text{ are open}\}, & \text{else} \end{cases}.$

- The set $\{x \leftrightarrow y\} \circ \{y \leftrightarrow z\}$ contains 

but not 

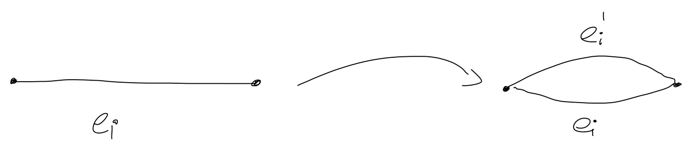
Remark.
- It holds that $A \circ B \subseteq A \cap B$ for any $A, B \in \mathcal{A}$.
- If A and B depend on disjoint sets of edges, then $A \circ B = A \cap B$.
- If A is increasing and B is decreasing, then $A \circ B = A \cap B$ (Exercise!).
- If A and B are increasing, then

$$A \circ B = \left\{ \omega \in \Omega \;\middle|\; \exists I, J \text{ disjoint, } \omega\Big|_{I \cup J} \equiv 1 \text{ and } \begin{matrix} I \text{ witnesses } A \text{ in } \omega, \\ J \text{ witnesses } B \text{ in } \omega \end{matrix} \right\}.$$

Theorem 1.20. Let A, B be events (depending on finitely many edges). Then

$$\mathbb{P}_p(A \circ B) \leq \mathbb{P}_p(A)\mathbb{P}_p(B)$$

Proof. We only show the theorem for increasing events as this is complicated enough. Let $A, B \subseteq \{0,1\}^{\mathcal{E}}$ increasing and $\mathcal{E} = \{e_1, \ldots, e_n\}$ finite. We "duplicate" edges by adding to every e_i a parallel edge e_i' like



Throughout the proof we assume that all edges in witnessing sets are always open. We consider percolation on the "duplicated graph". Configurations are $\bar{\omega} = (\omega, \omega')$ with

$$\begin{aligned} \omega &= (\omega_1, \ldots, \omega_n) = (\omega(e_1), \ldots, \omega(e_n)) \\ \omega' &= (\omega_1', \ldots, \omega_n') = (\omega(e_1'), \ldots, \omega(e_n')). \end{aligned}$$

Denote by $\bar{\mathbb{P}_p}$ the corresponding measure and set $\bar{\mathcal{E}} = \mathcal{E} \cup \mathcal{E}'$.

We set

$$\omega^{(i)} := (\omega_1', \ldots, w_i', \omega_{i+1}, \ldots, \omega_n)$$

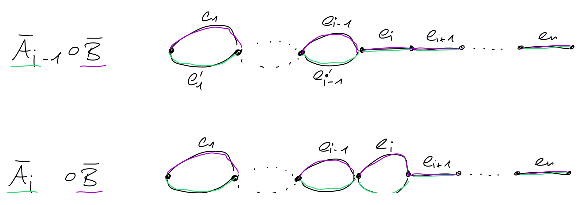


Figure 1.5: Edges "used" by $\bar{A}_{i-1} \circ \bar{B}$ and $\bar{A}_i \circ B$

which interpolates between $\omega^{(0)} = \omega$ and $\omega^{(n)} = \omega'$. Moreover, we define

$$\bar{A}_i := \left\{\bar{\omega} \,\middle|\, \omega^{(i)} \in A\right\}$$
$$\bar{B} := \{\bar{\omega} \mid \omega \in B\}.$$

Now, note that

$$\bar{\mathbb{P}}_p(\bar{A}_n \circ \bar{B}) = \bar{\mathbb{P}}_p(\bar{A}_n)\bar{\mathbb{P}}_p(\bar{B}) = \mathbb{P}_p(A)\mathbb{P}_p(B),$$

where we used in the first equality that $\bar{A}_n$ depends on ω' and $\bar{B}$ on ω, so $\bar{A}_n \circ \bar{B} = \bar{A}_n \cap \bar{B}$. Using that $\bar{A}_0$ and $\bar{B}$ only depend on ω,

$$\bar{\mathbb{P}}_p(\bar{A}_0 \circ \bar{B}) = \mathbb{P}_p(A \circ B).$$

Hence, BK is equivalent to

$$\bar{\mathbb{P}}_p(\bar{A}_n \circ B) \geq \bar{\mathbb{P}}_p(\bar{A}_0 \circ \bar{B}).$$

We thus show

$$\boxed{\text{for all } i = 1, \dots, n, \quad \bar{\mathbb{P}}_p(\bar{A}_{i-1} \circ \bar{B}) \leq \bar{\mathbb{P}}_p(\bar{A}_i \circ \bar{B}).} \tag{1.4}$$

We will decompose $\bar{A}_{i-1} \circ \bar{B}$: Let

$$C_1 := \left\{\exists I, J \subseteq \bar{\mathcal{E}} \setminus \{e_i, e_i'\} \text{ disjoint, open s.t. } \begin{matrix} I \text{ witnesses } \bar{A}_{i-1} \text{ in } \bar{\omega} \text{ and} \\ J \text{ witnesses } \bar{B} \text{ in } \bar{\omega} \end{matrix}\right\},$$

i.e., the event "$A_{i-1} \circ B$" occurs regardless of ω_i.

$$C_2 := C_1^{\complement} \cap \left\{\exists I, J \subseteq \bar{\mathcal{E}} \setminus \{e_i, e_i'\} \text{ disjoint, open s.t. } \begin{matrix} I \text{ witnesses } \bar{A}_{i-1} \text{ in } \bar{\omega} \text{ and} \\ J \cap \{e_i\} \text{ witnesses } \bar{B} \text{ in } \bar{\omega}^{e_i} \end{matrix}\right\},$$

where $\bar{\omega}^{e_i}$ is given by

$$(\bar{\omega}^{e_i})_f = \begin{cases} \bar{\omega}, & \text{if } f \neq e_i \\ 1, & \text{if } f = e_I. \end{cases}$$

The event C_2 is described by the following: If $\omega_i = 1$, then there is a witness for $\bar{B}$ using e_i and a disjoint witness for $\bar{A}_{I-1}$, if $\omega_i = 0$, $\bar{A}_{i-1} \cap \bar{B}$ does not occur. Lastly, define

$$C_3 := C_1^{\complement} \cap C_2^{\complement} \cap \left\{\exists I, J \subseteq \bar{\mathcal{E}} \setminus \{e_i, e_i'\} \text{ disjoint, open s.t. } \begin{matrix} I \cup \{e_i\} \text{ witnesses } \bar{A}_{i-1} \text{ in } \bar{\omega}^{e_i} \text{ and} \\ J \text{ witnesses } \bar{B} \text{ in } \bar{\omega} \end{matrix}\right\},$$

that is, the only way $\bar{A}_{i-1} \circ \bar{B}$ occurs is that $\omega_i = 1$ and the witness for $\bar{A}_{i-1}$ uses e_i. Now, observe that C_1, C_2, C_3 are independent of ω_i and ω_i' and that

$$\bar{A}_{i-1} \circ \bar{B} = C - 1 \cup (C_2 \cap \{\omega_i = 1\}) \cup (C_3 \cap \{\omega_i = 1\}),$$

which is a disjoint union. Hence

$$\bar{\mathbb{P}}_p(\bar{A}_{i-1} \circ \bar{B}) = \bar{\mathbb{P}}_p(C_1) + p\Big(\bar{\mathbb{P}}_p(C_2) + \bar{\mathbb{P}}_p(C_3)\Big). \tag{1.5}$$

Similarly,

$$\bar{A}_i \circ \bar{B} \supseteq C_1 \cup (C_2 \cap \{\omega_i\}) \cup (C_3 \cap \{\omega_i' = 1\}).$$

This is only an inclusion as it could also be that the witnesses for $\bar{A}_i$ and $\bar{B}$ both contain e_i'. Hence,

$$\bar{\mathbb{P}}_p(\bar{A}_i \circ B) \geq \bar{\mathbb{P}}_p(C_1) + p\Big(\bar{\mathbb{P}}_p(C_2) + \bar{\mathbb{P}}_p(C_3)\Big). \tag{1.6}$$

As (1.5) and (1.6) imply (1.4) we conclude the proof. □

Corollary 1.21. Consider percolation on $\mathbb{Z}^d$. Then

$$\mathbb{P}_p(x \leftrightarrow y \circ y \leftrightarrow z) \leq \mathbb{P}_p(x \leftrightarrow y)\mathbb{P}_p(y \leftrightarrow z).$$

Proof. Note that we need to be careful here as $x \leftrightarrow y$ depends on infinitely many edges contrary to the statement of Theorem 1.20. Note that

$$\{x \leftrightarrow y\} = \bigcup_{n \in \mathbb{N}} \{x \overset{\Lambda_n}{\longleftrightarrow} y\},$$

which is a union of increasing sets. Using Theorem 1.20 and passing to the limit $n \to +\infty$ we conclude. □

1.5 Russo's formula

If A is increasing, then by monotonicity

$$[0,1] \ni p \mapsto \mathbb{P}_p(A)$$

is increasing. In the following we will discuss Russo's formula which allows us to take the derivative of this map. To this end, we need another definition:

Definition 1.22. Let $A \in \mathcal{A}$ be an event, $\omega \in \Omega$ and $e \in \mathcal{E}$. We say that e is pivotal for A in ω if

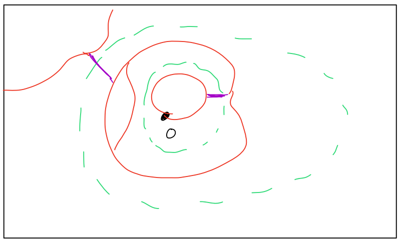
$$\mathbf{1}_A(\omega_e) \neq \mathbf{1}_A(\omega^e),$$

where

$$\omega_e(f) = \begin{cases} \omega(f), & f \neq e \\ 0, & f = e \end{cases} \quad \text{and} \quad \omega^e(f) = \begin{cases} \omega(f), & f \neq e \\ 1, & f = e. \end{cases}$$

The edge e is pivotal for A in ω if changing $\omega(e)$ decides if $\omega \in A$.

Example 1.23. Consider $A = \{0 \leftrightarrow \partial\Lambda_n\}$ and ω as on the figure:



The two edges marked ▬ are pivotal for A in ω

They are the only edges open in ω crossing „blocking surfaces"
marked in green.

Remark. Note that the event "e is pivotal" is independent of $\omega(e)$.

Theorem 1.24 (Russo). Let A be an increasing event depending only on the state of edges in a finite set $\mathcal{E}_0 \subseteq \mathcal{E}$. Then

$$\frac{\mathrm{d}}{\mathrm{d}p}\mathbb{P}_p(A) = \sum_{e \in \mathcal{E}_0} \mathbb{P}_p(e \text{ is pivotal for } A) = \mathbb{E}_p[\#\text{edges pivotal for } A].$$

Remark. When A depends on the state of infinitely many edges, then $p \mapsto \mathbb{P}_p(A)$ does not need to be differentiable (or even continuous). Consider, e.g., $A = \{0 \leftrightarrow \infty\}$ on $(\mathbb{Z}, \mathcal{E}_1)$.

Proof of Theorem 1.24. As A depends only on edges in $\mathcal{E}_0$, we consider $\Omega = \{0,1\}^{\mathcal{E}_0}$, view A as $A \subseteq \Omega$ and for $\omega \in \Omega$ set

$$|\omega| := |\{e \in \mathcal{E}_0 \mid \omega(e) = 1\}|\,.$$

Note that Ω and A are finite. Then

$$p \mapsto \mathbb{P}_p(A) = \sum_{\omega \in A} p^{|\omega|}(1-p)^{|\mathcal{E}_0| - |\omega|}$$

is a polynomial in p and differentiable as such. We write $\mathcal{E}_0 = \{e_1, \ldots, e_n\}$ and $\omega_i = \omega(e_i)$.

For $p_i \in [0,1]$ for $i = 1, \dots, k$ and $\omega \in \Omega$ we set

$$\mathbb{P}_{p_1,\dots,p_k}(\omega) := \prod_{i=1}^{k} p_i^{\omega_i}(1-p_i)^{1-\omega_i},$$

that is $\mathbb{P}_p(\omega) = \mathbb{P}_{p,\dots,p}(\omega)$. Moreover set

$$f(p_1,\dots,p_k) := \mathbb{P}_{p_1,\dots,p_k}(A) = \sum_{\omega\in A}\prod_{i=1}^{k} p_i^{\omega_i}(1-p_i)^{1-\omega_i}, so$$

Then we have that

$$\frac{\mathrm{d}}{\mathrm{d}p}\mathbb{P}_p(A) = \frac{\mathrm{d}}{\mathrm{d}p} f(p,\dots,p) = \sum_{i=1}^{k} \partial_{p_i} f(p,\dots,p). \tag{1.7}$$

But,

$$\begin{aligned} \partial_{p_i} f(p,\dots,p) =& \frac{\partial}{\partial p_i}\left(\sum_{\omega\in A}\mathbf{1}_A(\omega)\left(\prod_{j\neq i} p_j^{\omega_j}(1-p_j)^{1-\omega_j}\right)p_i^{\omega_i}(1-p_i)^{1-\omega_i}\right)\Bigg|_{p_1,\dots,p_n=p} \\ =& \sum_{\omega\in\Omega}\mathbf{1}_A(\omega)\left(\prod_{j\neq i} p^{\omega_j}(1-p)^{1-\omega_j}\right)\cdot\left(\mathbf{1}_{\omega_i=1}-\mathbf{1}_{\omega_i=-1}\right) \end{aligned} \tag{1.8}$$

Consider $\omega' \in \{0,1\}^{\mathcal{E}_0\setminus e_i}$ with $\omega = (\omega', \omega_i)$.

- If ω' is such that $(\omega',0) \in A$, $(\omega',1) \in A$, then contributions of $(\omega',1)$ and $(\omega',0)$ cancel each other out.
- If $(\omega',0) \notin A$, $(\omega',1) \in A$ (that is e_i is pivotal for A in ω) only $\mathbf{1}_{\omega_i\neq 1}$ remains.
- If $(\omega',0) \notin A$ and $(\omega',1) \notin A$ we get 0.
- The case $(\omega',0) \in A$ and $(\omega',1) \notin A$ is impossible since A is increasing.

Hence, we get

$$\begin{aligned} (1.8) =& \sum_{\omega\in\Omega}\mathbf{1}\{e_i \text{ is pivotal for } A \text{ in } \omega,\, \omega_i = 1\}\prod_{i\neq j}\left(p^{\omega_i}(1-p)^{1-\omega_i}\right) \\ =& \,\mathbb{P}_p(e_i \text{ is pivotal for } A). \end{aligned}$$

Inserting this back into (1.7) completes the proof. $\square$

Example 1.25 (Application of the previous theorem)**.** Let A be an increasing event. Then, noting that the events "e is pivotal for A" and "e is open" are independent we have

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}p}\mathbb{P}_p(A) &= \sum_{e\in\mathcal{E}_0}\mathbb{P}_p(e \text{ is pivotal for } A)\\ &= \frac{1}{p}\sum_{e\in\mathcal{E}_0}\mathbb{P}_p(e \text{ is pivotal for } A,\, e \text{ is open})\\ &= \frac{1}{p}\sum_{e\in\mathcal{E}_0}\mathbb{P}_p(A\cap\{e \text{ is pivotal for } A\})\\ &= \frac{1}{p}\sum_{e\in\mathcal{E}_0}\mathbb{P}_p\left(e \text{ is pivotal for } A \mid A\right)\mathbb{P}_p(A)\\ &= \frac{1}{p}\mathbb{E}_p\left[\text{number of pivotal edges for } A \mid A\right]\cdot\mathbb{P}_p(A).\end{aligned}$$

By integrating this over $p\in[p_1,p_2]$, we obtain

$$\mathbb{P}_{p_2}(A) = \mathbb{P}_{p_1}(A)\exp\left\{\int_{p_1}^{p_2}\frac{1}{p}\mathbb{E}_p\left[\#\text{pivotals}\mid A\right]\mathrm{d}p\right\}.$$

Using the trivial bound $\#\text{pivots}\le|\mathcal{E}_0|$ we deduce

$$\mathbb{P}_{p_2}(A)\le\mathbb{P}_{p_1}(A)\left(\frac{p_2}{p_1}\right)^{|\mathcal{E}_0|}.$$

Note that the coupling, FKG, BK and Russo work on *any* graph, not just $\mathbb{Z}^d$.

1.6 Ergodicity and Mixing

We now work on $\mathcal{G}=(\mathbb{Z}^d,\mathcal{E}_d)$ and formalize the "translation invariance". The additive group $(\mathbb{Z}^d,+)$ acts

- on $\mathbb{Z}^d$ by the usual translation,
- on $\mathcal{E}_d$ by $z\cdot\{x,y\}=\{x+z,x+y\}$.
- on $\Omega=\{0,1\}^{\mathcal{E}_d}$ by $(z\cdot\omega)\{x,y\}=\omega(\{x-z,y-z\})$. Note that e is open in ω iff $z\cdot e$ is open in ω.
- on $\mathcal{A}$ via $z\cdot A=\{z\cdot\omega:\omega\in A\}$.

Lemma 1.26 ($\mathbb{P}_p$ is translation invariant)**.** For any $A\in\mathcal{A}$ and for any $z\in\mathbb{Z}^d$

$$\mathbb{P}_p(z\cdot A)=\mathbb{P}_p(A).$$

Proof. Define the translated measure $(z \cdot \mathbb{P}_p)(A) := \mathbb{P}_p(z \cdot A)$. Consider A that depends on the state of finitely many elements (cylinder events) i.e., we take

$$A = \bigcap_{i=1}^{n} \{\omega \in \Omega : \omega(e_i) = \omega_i\}, \quad e_{1,}, \dots, e_n \in E, \quad \omega_1, \dots, \omega_n \in \{0,1\}, \tag{1.9}$$

so we have that

$$(z \cdot \mathbb{P}_p)(A) = p^{\sum_{i=1}^{n} \omega_i}(1-p)^{\sum_{i=1}^{n}(1-\omega_i)} = \mathbb{P}_p(A).$$

Since the sets of the form A from (1.9) make up a π-system we conclude by Dynkin's lemma. □

Proposition 1.27 (Mixing)**.** Consider percolation on $\mathbb{Z}^d$ and let $A, B \in \mathcal{A}$. Then

$$\lim_{\|z\|_1 \to +\infty} \mathbb{P}_p(A \cap (z \cdot B)) = \mathbb{P}_p(A)\mathbb{P}_p(B).$$

We first need a lemma, approximating the above by cylinder events.

Lemma 1.28. For any $A \in \mathcal{A}$ and $\varepsilon > 0$ there exists A_ε of the form (1.9) (we denote the collection of all such sets as $\mathcal{C}$) such that $\mathbb{P}_p(A \triangle A_\varepsilon) \le \varepsilon$.

Proof. Define

$$\mathcal{D} := \{A \in \mathcal{A} : \forall \varepsilon > 0 \exists A_\varepsilon \in \mathcal{C} \text{ s.t. } \mathbb{P}_p(A \triangle A_\varepsilon) \le \varepsilon\}.$$

Then one can prove that $\Omega \in \mathcal{D}$, $\mathcal{C} \subseteq \mathcal{D}$ and $\mathcal{D}$ is a λ-system so we conclude by Dynkin's lemma. □

Proof of Proposition 1.27. Fix $\varepsilon > 0$ and $A_\varepsilon, B_\varepsilon \in \mathcal{A}$ such that

$$\mathbb{P}_p(A \triangle A_\varepsilon), \mathbb{P}_p(B \triangle B_\varepsilon) \le \varepsilon.$$

As $A_\varepsilon, B_\varepsilon \in \mathcal{C}$, the formula holds for them. So we have for z large,

$$\begin{aligned} \mathbb{P}_p(A \cap (z \cdot B)) &\le \mathbb{P}_p(A_\varepsilon \cap (z \cdot B)) + 2\varepsilon \\ &= \mathbb{P}_p(A_\varepsilon)\mathbb{P}_p(B_\varepsilon) + 2\varepsilon \\ &\le \mathbb{P}_p(A)\mathbb{P}_p(B) + 4\varepsilon. \end{aligned}$$

The lower bound follows analogously. □

Theorem 1.29. Let $A \in \mathcal{A}$ be invariant, that is $z \cdot A = A$ for any $z \in \mathbb{Z}^d$. Then $\mathbb{P}_p(A) \in \{0,1\}$.

Proof. Since $A = z \cdot A$ we have by Proposition 1.27,

$$\mathbb{P}_p(A) = \lim_{\|z\|_1 \to +\infty} \mathbb{P}(A \cap (z \cdot A)) = \mathbb{P}_p(A)\mathbb{P}_p(A).$$ □

Example 1.30.
- Obivously $\{\omega \in \Omega : \omega \text{ contains infinite cluster}\} = I$ is translation invariant so by our previous theorem we have $\mathbb{P}_p(I) \in \{0, 1\}$.
- Define $I_k := \{\omega \in \Omega : \omega \text{ contains exactly } k \text{ infinite clusters}\}$, $k \in \mathbb{N}_0 \cup \{\infty\}$ is again translation invariant. So also $\mathbb{P}_p(I_k) \in \{0, 1\}$. This means the random variable

$$N(\omega) = \#\text{of infinite clusters in } \omega.$$

is $\mathbb{P}_p$-a.s. constant. We will cover this in more depth in the next chapter.

Question 1.31. Find k such that $\mathbb{P}_p(N = k) = 1$.

2 Uniqueness of infinite clusters

We have already proved that an infinite cluster exists with positive probability for $p > p_c$ (c.f. Proposition 1.10). It is thus natural to ask how many of these clusters exists. Suprisingly, $\mathbb{P}_p$-a.s. there exists at most one such cluster.

Theorem 2.1 (Aizuman, Kesten, Newman, 1987)**.** For any $d \in \mathbb{N}$, $p \in [0,1]$ consider Bernoulli-percolation on $(\mathbb{Z}^d, E_d)$. Let N denote the number of infinite clusters in $\mathbb{Z}^d$. Then either

- $\mathbb{P}_p(N = 0) = 1$ or
- $\mathbb{P}_p(N = 1) = 1$.

That is, if $p < p_c$ we have a.s. no infinite cluster and if $p_c > p$ we a.s. have an infinite cluster.

The proof we will show here is due to Burton and Keave (1989).

Proof. Throughout the proof assume that $p \in (0,1)$. If $p = 0$ there $\mathbb{P}_p$-a.s. does not exist an infinite cluster and if $p = 1$, $\mathbb{P}_p$-a.s. every edge is open. We already know that there exists $k \in \mathbb{N}_0 \cup \{\infty\}$ such that $\mathbb{P}_p(N = k) = 1$.

Step 1 Prove $k \in \{0, 1, \infty\}$. Suppose that $1 < k < \infty$ meaning $\mathbb{P}_p(N = 1) = 0$. Define the event

$$F_n := \{\omega \in \Omega : \text{all } k \text{ infinite clusters in } \omega \text{ touch the box } \Lambda_n\}.$$

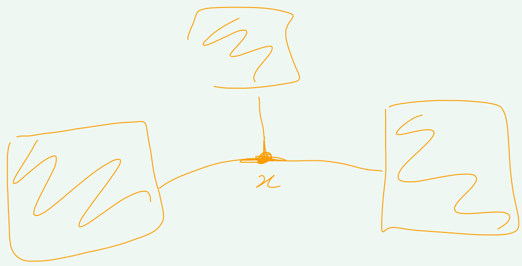
We already know that $\lim_{n\to+\infty} \mathbb{P}_p(F_n) = 1$ so there exists $n \in \mathbb{N}$ such that $\mathbb{P}_p(F_n) \geq \frac{1}{2}$. Now,

$$\begin{aligned}\mathbb{P}_p(N = 1) &\geq \mathbb{P}_p\left(F_n \cap \{\text{all edges in } \Lambda_n \text{ are open}\}\right)\\ &= \mathbb{P}_p(F_n)\mathbb{P}_p\left(\text{all edges in } \Lambda_n \text{ are open}\right)\\ &\geq \frac{1}{2} \cdot p^{|E(\Lambda_n)|} > 0\end{aligned}$$

if $p > 0$.

which is a contradiction.

Definition 2.2. $x \in \mathbb{Z}^d$ is a *trifurcation* in ω if



- x is incident to exactly 3 open edges in ω.
- if these 3 edges are removed from ω, $\mathcal{C}_x$ separates into 3 infinite clusters.

Denote
$$T_x := \{\omega \in \Omega : x \text{ is trifurcation in } \omega\}$$

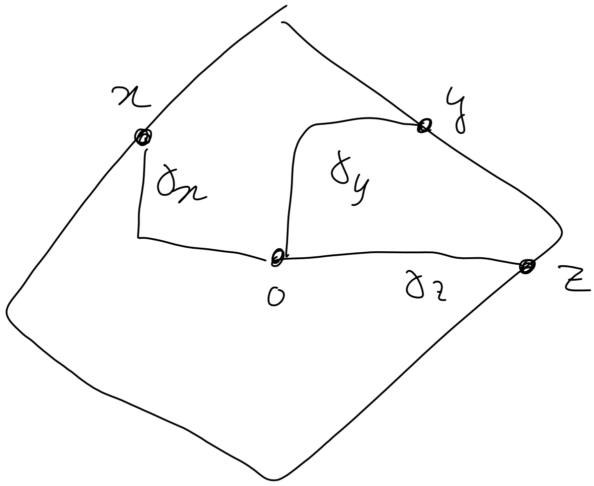
Step 2 If $\mathbb{P}_p(N \geq 3) > 0$ then $\mathbb{P}_p(T_0 > 0)$.. Define

$$\begin{aligned} B_n &:= \{x \in \mathbb{Z}^d : \|x\|_1 \leq n\} \\ E_n &:= \{\text{at least 3 infinite clusters touch } B_n\} \\ E_n(x,y,z) &:= \{\text{outside of } B_n,\ \mathcal{C}_x, \mathcal{C}_y, \mathcal{C}_z \text{ are infinite and disjoint}\}. \end{aligned}$$

For n large,

$$0 < \mathbb{P}_p(E_n) = \mathbb{P}_p\left(\bigcup_{\substack{x,y,z \in \partial B_n \\ x \neq y \neq z}} E_n(x,y,z)\right) \leq \sum_{x,y,z \in \partial B_m} \mathbb{P}_p(E_n(x,y,z))$$

so there exists $n \in \mathbb{N}$ and $x, y, z \in \partial B_n$ such that $\mathbb{P}_p(E_n(x,y,z)) > 0$. Take $\gamma_x, \gamma_y, \gamma_z$ as in the picture



such that the paths are pairwise disjoint. Then

$$\mathbb{P}_p(T_0) \geq \mathbb{P}_p\left(E_n(x,y,z) \cap \{\text{all edges in } \gamma_x, \gamma_y, \gamma_z \text{ are open and all other edges in } B_n \text{ are closed}\}\right)$$
$$= \mathbb{P}_p(E_n(x,y,z))\mathbb{P}_n(\ldots)$$

Step 3: For any $p \in (0,1), \mathbb{P}_p(N = \infty) = 0$. We assume by contradiction that $\mathbb{P}_p(N = \infty) = 1$. By the previous step $\mathbb{P}_p(T_0) = \text{const} > 0$. Define the random variable

$$\mathcal{T}_n := \#\text{trifurcation in } \Lambda_n.$$

Then

$$\mathbb{E}_p[\mathcal{T}_n] = \sum_{x \in \Lambda_n} \mathbb{P}_p\left(x \text{ is trifurcation}\right) = |\Lambda_n| \cdot \mathbb{P}_p(T_0) = \mathcal{O}(n^d). \tag{2.1}$$

We claim

$$\forall \omega \in \Omega \quad |\mathcal{T}_n(\omega)| \leq |\partial \Lambda_n| = \mathcal{O}(n^{d-1}). \tag{2.2}$$

If we prove this we get a contradiction to (2.1) and can conclude the proof.[todo: Rest of the proof peeling proceedure.]

Lemma 2.3. Let $T = (V, E)$ be a finite tree. Define

$$N_k := \{x \in V : \deg(x) = k\}$$

$$N_{\geq k} = \{x \in V : \deg(x) \geq k\}.$$

Then $|N_1| \geq 2 + |N_{\geq 3}| \geq |N_{\geq 3}|$.

After peeling $N_1 \subseteq \partial \Lambda_n$ and $N_{\geq 3} \supseteq \mathcal{T}_n$. So applying the above lemma to every component of the peeled forest yields

$$|\mathcal{T}_n| \leq |N_{\geq 3}| \leq |N_1| \leq |\partial \Lambda_n|.$$

This concludes the proof of the theorem. $\square$

Proof of Lemma 2.3. Observe that $|V| = |E| + 1$. Moreover,

$$2|E| = \sum_{x \in V} \deg(x) \geq |N_1| + 2|N_2| + 3|N_{\geq 3}|.$$

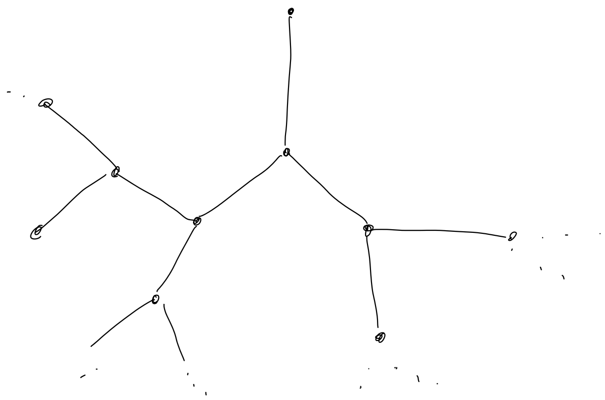
Also,

$$2|E| = 2(|V| - 1) = 2|N_1| + 2|N_2| + 2|N_{\geq 3}| - 2.$$

Substract the equations:

$$|N_1| \geq |N_{\geq 3}| + 2.$$ □

Percolation on $\mathbb{T}^d$. Let $\mathbb{T}^d$ denote an infinite tree where every vertex has degree 3.



Then one can prove that there exist $0 < p_1 < p_2 < 1$ such that

$$\mathbb{P}_p(N = 0) = 1, \quad \text{for } p < p_1$$

$$\mathbb{P}_p(N = \infty) = 1 \quad \text{for } p_1 < p < p_2$$

and

$$\mathbb{P}_p(N = 1) = 1, \quad \text{for } p > p_2.$$

The reason why the above proof does not work here is that the surface of the ball grows proportionally to the volume on $\mathbb{T}^d$.

2.1 Applications of the uniqueness of infinite clusters

We apply the uniqueness of infinite clusters to show several other interesting results. We start with a technical lemma. First for $k \in \{1, \ldots, n\}$ define

$$K_{k,n} = \# \text{ of disjoint clusters in } \Lambda_n \text{ connecting } \partial\Lambda_n \text{ and } \partial\Lambda_k \quad \text{and}$$
$$U_{k,n} := \{K_{k,n} \leq 1\}.$$

[todo: Image in the lecture notes]

Remark. The event $U_{k,n}$ is neither decreasing nor increasing as an event. However, $\mathbb{P}_p(U_{k,n})$ is increasing in n and decreasing in k (Exercise: check this).

Lemma 2.4. For any $\varepsilon > 0$ and $k \in \mathbb{N}$ there exists $n = n(k, \varepsilon) < +\infty$ such that

$$\mathbb{P}_p(U_{k,n}) > 1 - \varepsilon, \quad \text{for any } p \in [0, 1].$$

Proof. Fix $\varepsilon > 0$, $k \in \mathbb{N}$. Define

$$O_n := \{p \in [0, 1] \mid \mathbb{P}_p(U_{k,n}) > 1 - \varepsilon\}$$

which is open in $[0, 1]$ because the map $p \mapsto \mathbb{P}_p(U_{k,n})$ is continuous as $U_{k,n}$ depends only on finitely many edges. Then $O_n \subseteq O_{n+1}$ by the last remark. By the uniqueness of the infinite cluster (that is if it exists) we have

$$\text{for any } p \in [0, 1] \text{ there exist } n(p) \geq k \text{ such that } \mathbb{P}_p(U_{k,n(p)}) > 1 - \varepsilon.$$

(If the above statement is not true one must have two infinite clusters). By this we have that

$$[0, 1] = \bigcup_{m \geq k} O_m$$

since $[0, 1]$ is compact so there exists a finite sub-covering $(O_{m_i})_{i=1}^{i_0}$. The claim then follows by taking $n = \max_{i=1,\dots,i_0} m_i$ (because O_n are increasing sets we then have $O_n = [0, 1]$). □

Continuity of $\theta(p)$. We will show that $p \mapsto \theta(p)$ is continuous on $[0, 1] \setminus \{p_c\}$. Obviously θ is continuous on $[0, p_c)$ but the other part is shown by the following theorem.

Theorem 2.5. The map $p \mapsto \theta(p)$ is continuous on $(p_c, 1]$.

Proof. Define $\theta_n(p) := \mathbb{P}_p(0 \leftrightarrow \partial\Lambda_n)$. Since the event $0 \leftrightarrow \partial\Lambda_n$ depends only on finitely many edges θ_n is continuous. Also $\theta(p) = \lim \downarrow \theta_n(p)$ (which is a decreasing limit) so $\theta(p)$ is upper semicontinuous and since θ is decreasing, θ is left continuous (we won't need this in the proof).

Fix $p_0 > p_c$ arbitrary. We will show that $\theta_n \to \theta$ uniformly on $(p_0, 1]$. To this end, fix $\varepsilon > 0$ and pick $k \in \mathbb{N}$ large such that $\mathbb{P}_p(\partial\Lambda_k \leftrightarrow \infty) > 1 - \varepsilon$. Also pick $n > k$ large such that $\mathbb{P}_p(U_{k,n}) > 1 - \varepsilon$ for any $p \in [0, 1]$. Then

$$\begin{aligned}\theta(p) \geq &\mathbb{P}_p(\{0 \leftrightarrow \partial\Lambda_n\} \cap \{\partial\Lambda_k \leftrightarrow \infty\} \cap U_{k,n}) \\ \geq &\mathbb{P}_p(0 \leftrightarrow \partial\Lambda_n) - 2\varepsilon.\end{aligned}$$

From this we obtain that $\theta_n(p) \geq \theta(p) \geq \theta_n(p) - 2\varepsilon$ which is true for any p. Thus θ_n converges uniformly on $(p_0, 1]$. This concludes the proof. □

Box crossings.

Proposition 2.6. Let $p \in [0,1]$ be such that $\theta(p) > 0$. Then

$$\lim_{n\to+\infty} \mathbb{P}_p\left(\exists \text{path from left to right in } \Lambda_n\right) = 1.$$

[todo: Bild 15]

Proof. Let $\varepsilon > 0$ and pick k large such that $\mathbb{P}_p(\Lambda_k \leftrightarrow \infty) \geq 1 - \varepsilon$. (possible because we are in the supercritical phase). Then for any $k \geq n$,

$$\begin{aligned}\mathbb{P}_p(\Lambda_k \leftrightarrow \partial\Lambda_n) &= \mathbb{P}_p\left([\textit{todo}: \text{Bild } 16]\right) \\ &\geq 1 - \varepsilon.\end{aligned}$$

By the square root trick this implies that [todo: Bild 17]

[todo: Bild 18] □

3 Subcritical phase

Since $\theta(p) = 0$ for $p < p_c$ we know by the above proof that $\theta_n(p) \searrow 0$ for $p < p_c$. However, we know nothing about the speed of this convergence.

Theorem 3.1 (Menshikov, 1987, Aizenmann-Busky, 1987)**.** The following two statements hold:

1. For any $p < p_c$ there exists $c = c(p)$ such that

$$\mathbb{P}_p(0 \leftrightarrow \partial\Lambda_n) \leq \mathrm{e}^{-cn} \quad \text{for all } n \in \mathbb{N}$$

2. For all $p > p_c$ we have the *mean field lower bound*

$$\theta(p) \geq \frac{1}{2}(p - p_c).$$

From this one deduces the following corollary whose proof is left as an exercise to the reader.

Corollary 3.2. For any $p < p_c$ we have

$$\chi(p) := \mathbb{E}_p[\#\{x : x \leftrightarrow 0\}] < +\infty$$

Remark. Historically, people studied

$$p_c' = \inf\{p \in [0,1] : \chi(p) = +\infty\}$$

The previous theorem implies that $p_c' = p_c$ (show this as an exercise).

The proof we are presenting is due to H. Dummil-Copin and V. Tassion. We first need a few definitions and lemmas.

Definition 3.3. Let $S \subseteq \mathbb{Z}^d$ be a finite set with $0 \in S$. We define

$$\varphi_p(S) := p \sum_{\{x,y\} \in \Delta S} \mathbb{P}_p\left(0 \overset{S}{\leftrightarrow} x\right).$$

If $0 \notin S$ we set $\varphi_p(S) = 0$.

Remark. Note that the event "$\{x,y\}$ is open" is independent of $0 \overset{S}{\leftrightarrow} x$ for all $\{x,y\} \in \Delta S$ and has probability p. This implies that

$$\begin{aligned}\varphi_P(S) &= \sum_{\{x,y\}\in\Delta S} \mathbb{P}_p\left(\{x,y\} \text{ open}, 0 \overset{S}{\leftrightarrow} x\right) \\ &= \mathbb{E}_p\Big[\text{number of edges in } \Delta S \text{ connected to } 0\Big].\end{aligned}$$

Lemma 3.4. Suppose there is a finite set $S \subseteq \mathbb{Z}^d$ with $0 \in S$ and $\varphi_p(S) < 1$. Then there exists $c = c(p)$ such that

$$\theta_n(p) < \mathrm{e}^{-cn} \quad \text{for all } n \in \mathbb{N}.$$

Proof. Assume we have such S. Fix $k \in \mathbb{N}$ such that $S \subseteq \Lambda_k$. If $\{0 \leftrightarrow \partial\Lambda_{n\cdot k}\}$ occurs then there exists an edge $\{x,y\} \in \Delta S$ such that

$$0 \overset{S}{\leftrightarrow} x, \quad \{x,y\} \text{ is open and} \quad y \leftrightarrow \partial\Delta_{n\cdot k},$$

which are all events occuring disjointly. Now, using van-de Berg/Kesten inequality (Theorem 1.20),

$$\begin{aligned}\theta_{nk}(p) =& \mathbb{P}_p(0 \leftrightarrow \partial\Lambda_{nk}) \\ \leq& \sum_{xy\in\Delta S} \mathbb{P}_p\left(\left\{x \overset{S}{\leftrightarrow} 0\right\} \circ \{xy\text{open}\} \circ \{y \leftrightarrow \partial\Lambda_{nk}\}\right) \\ \leq& \sum_{xy} \mathbb{P}_p\left(0 \overset{S}{\leftrightarrow} x, xy \text{ open}\right) \qquad \underbrace{\mathbb{P}_p(y \leftrightarrow \partial\Lambda_{nk})}_{\leq \mathbb{P}_p(y\leftrightarrow\partial(y+\Lambda_{(n-1)k}))=\mathbb{P}_p(0\leftrightarrow\partial\Lambda_{(n-1)k})} \\ \leq& \underbrace{\varphi_p(S)}_{<1} \mathbb{P}_p(0 \leftrightarrow \partial\Lambda_{(n-1)k}).\end{aligned}$$

Iterate this to get the exponential decay. □

First we need another lemma:

Lemma 3.5. Let $\mathcal{S}_n := \{x \in \Lambda_n \mid x \not\leftrightarrow \partial\Lambda_n\}$. Then

$$\theta_n'(p) = \frac{1}{p(1-p)}\mathbb{E}_p\Big[\varphi_p(\mathcal{S}_n)\Big] \quad \text{for all } p \in [0,1],\ n \in \mathbb{N}.$$

Proof. Define $\mathcal{E}_n := \{\{x,y\} \mid x,y \in \Lambda_n\}$ and $A := \{0 \leftrightarrow \partial\Lambda_n\}$. Note that $\omega(e)$ and the event "e is pivotal for A" are independent. Using Russo's formula (Theorem **??**) we find

$$\begin{aligned}\theta_n'(p) &= \sum_{e\in\mathcal{E}_n} \mathbb{P}_p(e \text{ is pivotal for } A) \\ &= \frac{1}{1-p}\sum_{e\in\mathcal{E}_n} \mathbb{P}_p(e \text{ is closed}, e \text{ is pivotal for } A) \\ &= \frac{1}{1-p}\sum_{e\in\mathcal{E}_n} \mathbb{P}_p(A^{\complement} \cap \{e \text{ is pivotal for } A\}) = \dots\end{aligned}$$

Now, note that $A^\complement = \{0 \in \mathcal{S}_n\} = \bigcup_{S \subseteq \Delta_n, 0 \in S} \{\mathcal{S}_n = S\}$, thus

$$\cdots = \frac{1}{1-p} \sum_{\substack{S \subseteq \Delta_n \\ S \ni 0}} \sum_{e \in \mathcal{E}_n} \mathbb{P}_p(\mathcal{S}_n = S, e \text{ pivotal for } A).$$

If e is pivotal for A in ω and closed if

- one of the endpoints of e is connected to 0,
- he other one is connected to $\partial\Lambda_n$ and
- 0 is not connected to $\partial\Lambda_n$.

We deduce that

$$\begin{aligned}
\theta_n'(p) &= \frac{1}{1-p} \sum_{\substack{S \subseteq \Delta_n \\ S \ni 0}} \sum_{\{x,y\} \in \Delta S} \mathbb{P}_p\left(0 \overset{S}{\longleftrightarrow} x, \mathcal{S}_n = S\right) \\
&= \frac{1}{1-p} \sum_{\substack{S \subseteq \Delta_n \\ S \ni 0}} \sum_{\{x,y\} \in \Delta S} \mathbb{P}_p\left(0 \overset{S}{\longleftrightarrow} x\right) \mathbb{P}_p(\mathcal{S}_n = S) \\
&= \frac{1}{1-p} \mathbb{E}_p\left[\frac{1}{p} \varphi_p(\mathcal{S}_n)\right]
\end{aligned}$$

This concludes the proof of the lemma. $\square$

Proof of Theorem 3.1. First note that for any $p > p_c$ and $n \in \mathbb{N}$,

$$\theta_n'(p) = \underbrace{\frac{1}{p(1-p)}}_{\geq 1} \underbrace{\mathbb{E}_p\left[\varphi_p(\mathcal{S}_n)\right]}_{\geq 1 \text{ if } 0 \in \mathcal{S}_n} \geq \mathbb{P}_p(0 \in \mathcal{S}_n) = 1 - \theta_n(p).$$

Fix $p > \tilde{p}_c$. If $\theta_n(p) \geq \frac{1}{2}$ then $\theta_n(p) \geq \frac{1}{2}(p - \tilde{p}_c)$. If $\theta_n(p) < \frac{1}{2}$, then $\theta_n(q) < \frac{1}{2}$ for all $q \in [\tilde{p}_c, p]$ $\square$

(5.4) **Thm** (a) $\theta_n(p) = \mathbb{P}_p(0 \leftrightarrow \partial\Delta_n)$ decays exponentially in n $\forall p < p_c$

(b) $\forall p > p_c$: $\theta(p) \geq \frac{1}{2}(p - p_c)$

(5.6) **Lem** If $\underbrace{\exists S \subseteq \mathbb{Z}^d \text{ finite}, 0 \in S, \varphi_S(p) < 1}$ then $\forall n$ $\theta_n(p) < e^{-cn}$.

Define $\tilde{p}_c := \sup\{p \mid \;\;\}$. Goal: prove $\tilde{p}_c = p_c$ (actually we only need "$\geq$")

First we need another lemma:

(5.7) **Lem** Let $\mathcal{S}_n := \{x \in \Delta_n \mid x \not\leftrightarrow \partial\Delta_n\}$. Then

$$\theta_n'(p) \geq \frac{1}{p(1-p)} \mathbb{E}_p\left[\varphi_p(\mathcal{S}_n)\right] \quad \forall p \in [0,1], \forall n \in \mathbb{N}.$$

Pf: Define $E_n := \{\{x,y\} \mid x,y \in \Delta_n\}$, $A := \{0 \leftrightarrow \partial\Delta_n\}$. By Russo,

$$\theta_n'(p) = \sum_{e \in E_n} \mathbb{P}_p(e \text{ is pivotal for } A)$$ Note $\omega(e) \perp\!\!\!\perp e$ pivotal

$$= \sum_e \frac{1}{1-p} \mathbb{P}_p(\underbrace{e \text{ is closed}, e \text{ pivotal for } A}_{= \mathbb{P}_p(A^c \cap \{e \text{ pivotal for } A\})}) = \dots$$

independent

Note that $A^c = \{0 \in \mathcal{S}_n\} = \dot{\bigcup}_{\substack{S \subseteq \Delta_n \\ S \ni 0}} \{\mathcal{S}_n = S\}$ so

$$\dots = \frac{1}{1-p} \sum_{\substack{S \subseteq \Delta_n \\ S \ni 0}} \sum_{e \in E_n} \mathbb{P}_p(e \text{ pivotal for } A, \mathcal{S}_n = S)$$

If e is pivotal for A in ω then

- one of the vertices of e is connected to 0
- the other one is connected to Δ_n
- there is no connection of $0, \partial\Delta_n$ in ω_e.

On $\mathcal{S}_n = S$, e is pivotal iff

- $e \in \Delta \mathcal{S}^n$

•) one vertex of e must be connected to 0 in S.

$$\Rightarrow \theta_n'(p) = \frac{1}{1-p} \sum_{\substack{S\in\Delta_n \\ 0\in S}} \sum_{xy\in\Delta S} \mathbb{P}_p\big(\underbrace{0 \overset{S}{\longleftrightarrow} x}_{\in\sigma(\omega(e)\,:\, e=\{x,y\},\, x,y\in S\})}, \overbrace{\mathcal{J}^n = S}^{\sigma(\omega(e)\,:\, e=\{x,y\} \text{ at least one of } x,y \text{ is in } S^c)}\big)$$

independent!

$$= \frac{1}{1-p} \sum_{\substack{S\in\Delta_n \\ 0\in S}} \underbrace{\sum_{xy\in\Delta S} \mathbb{P}_p\left(0 \overset{S}{\longleftrightarrow} x\right)}_{=\frac{1}{p}\varphi_p(S)} \mathbb{P}_p\left(\mathcal{J}^n = S\right)$$

$$= \frac{1}{p(1-p)} \mathbb{E}_p\left[\varphi_p(\mathcal{J}_n)\right].$$

■

Pf of (5.4). $\forall p > p_c$, $\forall n\in\mathbb{N}$

$$\theta_n'(p) \geq \underbrace{\frac{1}{p(1-p)}}_{\geq 1} \mathbb{E}_p\Big[\underbrace{\varphi_p(\mathcal{J}^n)}_{\geq 1 \text{ if } 0\in\mathcal{J}^n}\Big] \geq \mathbb{P}_p(0\in\mathcal{J}^n) = 1-\theta_n(p)$$

Fix $p > \tilde{p}_c$. If $\theta_n(p) \geq \frac{1}{2}$ then $\theta_n(p) \geq \frac{1}{2}(p-\tilde{p}_c)$. If $\theta_n(p) < \frac{1}{2}$ then $\theta_n(q) < \frac{1}{2}$ $\forall q\in[\tilde{p}_c, p]$

$$\Rightarrow \forall q\in[\tilde{p}_c,p]: \quad \theta_n'(q) \geq \frac{1}{2}$$

$$\overset{\text{integrate}}{\Rightarrow} \quad \theta_n(p) \geq \frac{1}{2}(p-\tilde{p}_c)$$

$$\underset{n\to+\infty}{\Rightarrow} \quad \theta(p) \geq \frac{1}{2}(p-\tilde{p}_c).$$

so $\forall p > \tilde{p}_c$ $\exists$ a.s. an infinite cluster $\Rightarrow \tilde{p}_c \geq p_c$.

$\tilde{p}_c > p_c$ is not possible because if there's a p between them this p has exponential decay $\Rightarrow \theta(p) = 0$. ↯ ■

Corelation length: We know $p^n \leq \theta_n(p) \leq e^{-cn}$. Can we prove

$$\theta_n(p) \sim e^{-cn}$$

(6.1) Prop $e_1 = (1,0,0,\dots,0)$. Consider

$$-\left(\lim_{n\to+\infty} \frac{1}{n}\log \mathbb{P}_p(0 \longleftrightarrow ne_1)\right)^{1/n} =: \xi(p,d) \quad \text{: correlation length}$$

this quantity is well defined and $<+\infty$. This basically says
$$\mathbb{P}_p(0 \leftrightarrow ne_1) = e^{-\frac{n}{\xi}(1+o(1))}$$

<u>Fekete's lemma</u> $(u_n)_{n\in\mathbb{N}} \subseteq [-\infty, +\infty)$ $u_{n+m} \leq u_n + u_m$ (sub-additivity)
$$\Rightarrow \lim_{n\to+\infty} \frac{u_n}{n} = \inf \frac{u_n}{n}.$$

<u>Pf of (6.1)</u> $\mathbb{P}_p(0 \leftrightarrow (n+m)e_1) \geq \mathbb{P}_p(0 \leftrightarrow me_1, m \leftrightarrow (n+m)e_1)$

FKG + translation invariance $\geq \mathbb{P}_p(0 \leftrightarrow me_1)\,\mathbb{P}_p(0 \leftrightarrow ne_1)$

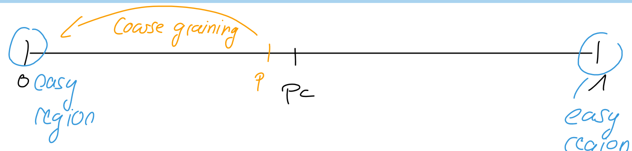
Define $u_m = -\log \mathbb{P}_p(0 \leftrightarrow me_1)$ and the result follows by Fekete. ■

ξ is kinda the largest expected local cluster "radius".

Exponential decay for volume
$$C_0 = \{x : x \leftrightarrow 0\}, \qquad \mathbb{P}_p(|C_0| \geq n) = ?$$

$$|C_0| \geq n \Rightarrow \operatorname{diam}_{L^\infty} C_0 \geq n^{1/d}$$
$$(5.4) \Rightarrow \mathbb{P}_p(|C_0| \geq n) \leq e^{-cn^{1/d}}$$

(6.3) <u>Thm</u> $p < p_c$. $\exists c = c(p,d) > 0$ s.t. $\mathbb{P}_p(|C_0| \geq n) = e^{-cn}$.



(6.4) <u>Lemma</u> Set $\mathcal{A}_n := \{C \subseteq \mathbb{Z}^d \mid 0 \in C,\ C \text{ connected},\ |C| = n\}$ (lattice animals). $|\mathcal{A}_n| \leq 16^{dn}$

Pf As on Exo 4(b). (by hardcore combinatorics) ■

Pf of (6.3) Fix k large s.t. $\mathbb{P}_p(\Lambda_k \leftrightarrow \partial\Lambda_{2k}) \leq \left(\frac{1}{16^d e}\right)^{3^d}$

(possible because of exponential decay)

Define $G_k := (V_k, E_k)$, $V_k = 2k\mathbb{Z}^d$, $E_k = \{\{2kx, 2ky\}, \|x-y\|_1 = 1\}$

box centered at u.

For $u \in V_k$, $\eta(u) := \begin{cases} 1 & \Delta_k(u) \longleftrightarrow \partial\Delta_{2k}(u) \\ 0 & \text{otherwise} \end{cases}$

By translation invariance, $\mathbb{P}_p(\eta(u) = 1) \le \left(\frac{1}{16^d e}\right)^{3^d}$. Problem: $\eta(u)$'s are not independent.

- $\|u-w\|_\infty > 4k \Rightarrow \text{int}\,\Delta_{2k}(u) \cap \text{int}\,\Delta_{2k}(w) = \emptyset \Rightarrow \eta(u) \perp\!\!\!\perp \eta(w)$.

$\overline{C_0}(w) := \{u \in V_k \mid C_0 \cap \Delta_k(w) \ne \emptyset\}$

•) $|C_0| \le |\Delta_k|\,|\overline{C_0}|$

•) $|\overline{C}|$ is connected in G_k

$\forall \overline{C} \subseteq V_k$ there are $v_1, \dots, v_n \in V_k \setminus \{0\}$ s.t.

•) $m \ge \frac{|\overline{C}|}{3^d}$

•) $i \ne j \Rightarrow \|v_i - v_j\|_\infty \ge 4k$.

For $|\overline{C}| \ge 2$

$$\mathbb{P}_p[\overline{C_0} = \overline{C}] \le \mathbb{P}_p[\forall u \in \overline{C} : \eta(u) = 1] \le \mathbb{P}_p[\underbrace{\forall i \le n : \eta(v_i) = 1}_{\text{independent}}]$$

$$= \prod_{i=1}^{n} \underbrace{\mathbb{P}_p(\eta(v_i) = 1)}_{\text{translation invariant}} = \mathbb{P}_p(\eta(v_1) = 1)^n \le \left(\left(\frac{1}{16^d e}\right)^{3^d}\right)^n$$

$$\le 16^{-d|\overline{C}|} e^{-|\overline{C}|}$$

For $n \ge 2|\Delta_k|$

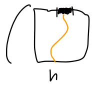
$$\mathbb{P}_p(|C_0| \ge n) \le \mathbb{P}_p\left(|\overline{C_0}| \ge \frac{n}{|\Delta_k|}\right)$$

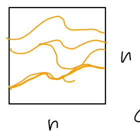
$$= \sum_{N \ge \frac{n}{|\Delta_k|}} \sum_{\substack{\overline{C} : \overline{C} \ni 0 \\ |\overline{C}| = N}} \mathbb{P}_p(\overline{C_0} = \overline{C}) = \sum_N \underbrace{|\mathcal{C}_N|}_{\le 16^{dN}} \cdot 16^{-dN} e^{-N}$$

∎

[todo: Lecture 07]

Continuation of the proof of thr (7.4):

Case 2: $\mathbb{P}_p\Big($  $\Big) \geq 1-(1-x)^{1/3}$

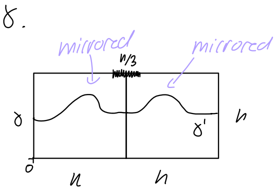
Consider  and let Γ' denote the lowest crossing of $[0,n]^2 \cap \mathbb{Z}^2$ and $\Gamma' = \emptyset$ if there is no crossing. (Note Γ' is well defined)

because we are in 2d think about it!

If γ is a (possibly closed!) L-R-path, the event $\{\Gamma' = \gamma\}$ is measurable (because it depends on finitely many edges)

$$R(\gamma) := \{ e \text{ edge in } [0,n]^2 \mid e \subseteq \gamma \text{ or below} \}$$

Fix such γ.



$$E_\gamma := \left\{ \left[\tfrac{5n}{6}, \tfrac{7n}{6}\right] \times \{n\} \longleftrightarrow \gamma \text{ in } \left[\tfrac{n}{2}, \tfrac{3n}{2}\right] \times [0,n] \text{ above } \gamma \cup \gamma' \right\}$$

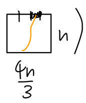
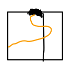
$E_{\gamma'} :=$ ________ γ' ________

Observe that $E_\gamma, E_{\gamma'} \perp\!\!\!\perp \{\Gamma' = \gamma\}$ and

$\mathbb{P}_p(E_\gamma \cup E_{\gamma'}) \geq \mathbb{P}_p\Big($  $\Big)$

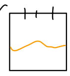
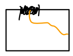
$\geq 1-(1-x)^{1/3}$

By square root trick: $\mathbb{P}_p(E_\gamma) = \mathbb{P}_p(E_{\gamma'}) \geq 1-(1-x)^{1/2}$

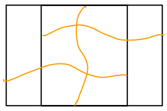
$\mathbb{P}_p\Big($  $\Big) \geq \sum_{\gamma \text{ LR crossings of } [0,n]^2} \mathbb{P}_p\Big(\Gamma' = \gamma,$  $\Big)$, where the box $\supseteq E_\gamma$

$$\geq \underbrace{\sum_\gamma \mathbb{P}_p(\gamma = \Gamma')}_{\mathbb{P}_p(\square_n) - x} \underbrace{\mathbb{P}_p(E_\gamma)}_{\geq 1-(1-x)^{1/2}}$$

$$\geq 1-(1-x)^{1/2}$$

$\mathbb{P}_p\Big($  $n\Big) \geq \mathbb{P}_p\Big($  $\cap$  $\Big)$

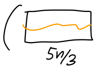
$$\overset{FKG}{\geq} \left(x\left(1-(1-x)^{1/2}\right)\right)^2 =: y$$



To force L-R connection in overlapping rectangle

We need: L-R connection in both and

T-B connection in intersection

FKG: $\mathbb{P}_p\Big($ 

$n\Big) \geq y^2 x.$

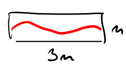
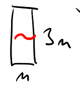
$\rightsquigarrow$ iterate this $\Rightarrow$ Result. ■

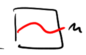
8/1

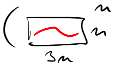
Critical behaviour in $d=2$:

Here always $p = p_c(\mathbb{Z}^2) = \frac{1}{2}$, we write $\mathbb{P}$ for $\mathbb{P}_{\frac{1}{2}}$.

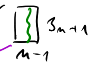
(8.1) **Theorem:** There is $c > 0$ s.t. $\forall n \geq 1$

$c \leq \mathbb{P}($  $) \leq \mathbb{P}($  $) \leq 1 - c$

Proof: • first inequality: (7.3) $\Rightarrow$ $x := \mathbb{P}($  $) \geq \frac{1}{2}$.

$\overset{(7.4)}{\underset{RSW}{\Longrightarrow}}$ $\mathbb{P}($  $) \geq c > 0$.

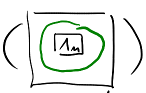
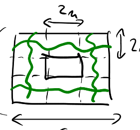
• 2^{nd} inequality: trivial

• 3^{rd} — " — : $\mathbb{P}($  $) \overset{\text{duality}}{=} 1 - \mathbb{P}($  $) \leq 1 - c' < 1$

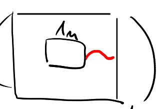
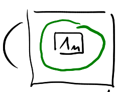
no connection LR $\Rightarrow$ T-B connection in dual   similarly to 1^{st} ineq. □

(8.1) **Corollary** [Upper bound on 1-arm probability]

$\forall n \geq 1$ $\quad \theta_n(\tfrac{1}{2}) = \mathbb{P}(0 \leftrightarrow \partial \Lambda_n) \leq \frac{1}{n^c}$

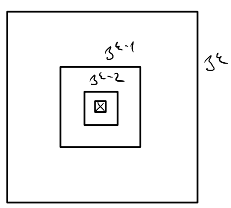
Proof: $\mathbb{P}($  $) \geq \mathbb{P}($  $) \underset{(8.1)}{\overset{\text{FKG}}{\geq}} c^4 =: c_0$

Hence

$\mathbb{P}($  $) = 1 - \mathbb{P}($  $) \leq 1 - c_0$.

Further: $\mathbb{P}(0 \leftrightarrow \partial\Lambda_{3^k}) \leq \mathbb{P}\left(\bigcap_{i=0}^{k-1} \{\partial\Lambda_{3^i} \leftrightarrow \partial\Lambda_{3^{i+1}}\}\right)$ independent over i

$= (1-c_0)^k$

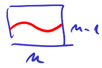


Choose $k = \lfloor \log_3 n \rfloor$ for $n \geq 3$. This finishes the proof □

8/2

(8.3) **Conjecture** $\exists c \in (0,\infty)$ s.t. $\theta_n(p) \sim \frac{c}{n^{5/48}}$

Rem: The exponent $n^{5/48}$ should not depend on 2-dim. lattice
It has been proved only for site percolation on triangular lattice.
here $p_c^{site}(\mathbb{T}) = 1/2$ due to Smirnov

Rem: It is easy to see that $\theta_n(p) \geq \frac{1}{2n}$
$\big(n P(0 \leftrightarrow \partial \Lambda_n) \geq P($  $) = \frac{1}{2} \big)$

Remark on critical exponents:

It is believed that many properties of percolation models at the criticality or near the criticality should be "universal", that is should not depend on the lattice/model details, but on the dimension d. (The value p_c of course depend on the model/lattice)

E.g. one expects:

$\theta(p) \asymp (p-p_c)^{\beta}$ — β: only depending on d — $p \downarrow p_c$

$\chi^f(p) := E_p\left(|C_0| \mathbb{1}_{|C_0|<\infty}\right) \asymp |p-p_c|^{-\gamma}$ — $p \to p_c$

$\kappa(p) := E_p\left(|C_0|^{-1}\right)$ $\quad \kappa'''(p) \asymp |p-p_c|^{-1-\alpha}$ — $p \to p_c$

$\xi(p) =$ corr. length $\asymp |p-p_c|^{-\nu}$ — $p \to p_c$

$P_{p_c}(|C| = n) \asymp n^{-1-\frac{1}{\delta}}$ — $n \to \infty$

$P_{p_c}(0 \leftrightarrow \partial \Lambda_n) \asymp n^{-1-\frac{1}{\rho}}$ — $n \to \infty$

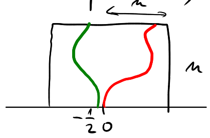
$P_{p_c}(0 \leftrightarrow x) \asymp |x|^{2-d-\eta}$ — $|x| \to \infty$.

- Their existence is mostly proved for $d \geq 11$ (or more) or in $d=2$ on triangular lattice.
- They are expected* to satisfy various relations:
  $\gamma = \nu(2-\eta)$; $d\rho = \delta+1$; $d\nu = 2-\alpha$, ...
  * or proved (assuming existence)
- For $d \geq 1$ they agree with exponents on trees, can be computed
  $\alpha = -1, \beta = 1, \gamma = 1, \delta = 2, \rho = \frac{1}{2}, \eta = 0, \nu = \frac{1}{2}$.

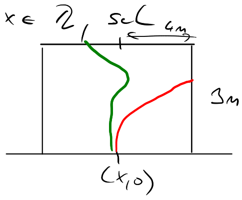
(see Grimmett, Chapters 9-10)

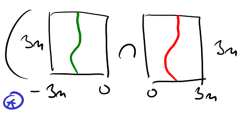
8/3

(8.4) **Corollary**: (A universal arm exponent) There is $c \in (0,1)$ s.t. $\forall m \geq 1$

$$c\,\frac{1}{m} \leq \mathbb{P}\Big($$  $$\Big) \leq \frac{1}{c}\,\frac{1}{m}$$

Proof sketch: For $x \in \mathbb{Z}$, set

$A_x = \Big\{$  $\Big\}$

Lower bound: $\frac{1}{4} \overset{\text{indep}}{\leq} \mathbb{P}\Big($  $\Big)$

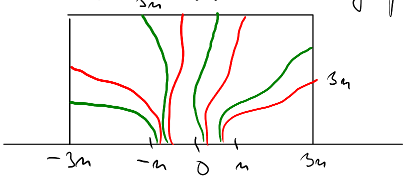
$\overset{\text{union bound}^{\circledast}}{\leq} \sum_{|x| \leq 3m} \mathbb{P}(A_x) \overset{\text{inv.}}{=} (6m+1)\,\mathbb{P}(A_0)$

translation invariance

⊛ consider the leftmost { and the rightmost {

$\Rightarrow \mathbb{P}\Big($  $\Big) \geq \mathbb{P}(A_0) \geq \frac{c}{m}$.

Upper bound: $N :=$ # disjoint open paths from $[-m,m]\times\{0\}$ to $\partial\Lambda_{3m}$ in the half plane $\{\mathbb{R}\times[0,\infty)\}\cap\mathbb{Z}^2$.



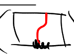
$N = 4$ here

Observe: $\sum_{|x| \leq m} \mathbb{1}_{A_x} \leq N$.

Hence: $(2m+1)\,\mathbb{P}(A_0) \leq \mathbb{E}[N] = \sum_{k \geq 0} \mathbb{P}(N \geq k)$

$$= \sum_{k\geq 0} \mathbb{P}\Big(\underbrace{(N\geq 1)\circ(N\geq 1)\circ\ldots\circ(N\geq 1)}_{k\text{-times}}\Big)$$

$$\overset{BK}{\leq} \sum_{k\geq 0} \mathbb{P}(N\geq 1)^k = \frac{1}{1-\mathbb{P}(N\geq 1)} = \frac{1}{1-\mathbb{P}\Big(\text{[diagram]}\Big)} \leq \frac{1}{c}$$



$\Rightarrow \mathbb{P}(A_0) \leq \frac{c}{m}$ ☐

We will skip the proo this in the lectures 8/4

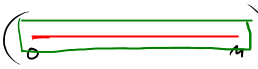
Supercritical behaviour in $d=2$

Key observation: $p>p_c(\mathbb{Z}^2) \Rightarrow 1-p<p_c(\mathbb{Z}^{2*})$

$\forall p>p_c$ $\forall x\in\mathbb{Z}^{2*}$ $\mathbb{P}\Big($  $\Big)\le e^{-cn}$.

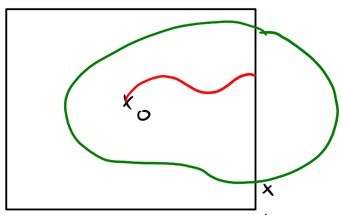
(3.5) **Theorem:** (Exp. decay of radius of a finite cluster)
Let $p\in(p_c,1)$. There are constants $0<c_1<c_0$ s.t.

$$e^{-c_0 n}\le \mathbb{P}_p(0\leftrightarrow\partial\Lambda_n,\ 0\not\leftrightarrow\infty)\le e^{-c_1 n}$$

Proof: LB: $\mathbb{P}_p(0\leftrightarrow\partial\Lambda_n,\ 0\not\leftrightarrow\infty)\ge \mathbb{P}_p\Big($  $\Big)$

$$\ge p^{(n-1)}(1-p)^{2(n+1)+2}\ge e^{-c_0 n}$$

UB:



$0\leftrightarrow\partial\Lambda_n$, $0\not\leftrightarrow\infty$ $\quad\Lambda_n$

$$\mathbb{P}_p(0\leftrightarrow\partial\Lambda_n,\ 0\not\leftrightarrow\infty)$$

$\le \mathbb{P}_p($∃ dual open circuit intersecting $\partial\Lambda_n$ of diameter $\ge n)$

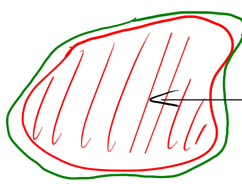
$\le \mathbb{P}_p(\exists x\in\partial\Lambda_n:\ x\leftrightarrow\partial\Lambda_n(x))$

$\le C n e^{-cn}\le e^{-c_1 n}$ for n large $\square$

(3.6) **Theorem:** [Stretched exp. decay of volume]
Let $p\in(p_c,1)$

$$\forall n\ge 1\qquad e^{-c_0\sqrt{n}}\le \mathbb{P}_p(n\le|C_0|<\infty)\le e^{-c_1\sqrt{n}}$$

Idea:



dual circuit of length $\sqrt{n}$.

cluster of volume $\sim n$

Proof: Upper bound: $|C_0|\ge n \Rightarrow 0\leftrightarrow\partial\Lambda_{\frac{\sqrt{n}}{3}}$

$$\mathbb{P}(n\le|C_0|<\infty)\le \mathbb{P}\big(0\leftrightarrow\partial\Lambda_{\frac{\sqrt{n}}{3}}\big)\le e^{-c_1\sqrt{n}}$$

8/5

<u>Lower bound</u> Set $k = \left\lceil \sqrt{\frac{n}{\theta(p)}} \right\rceil$, $N = \sum_{x \in \Lambda_k} 1_{x \leftrightarrow \partial\Lambda_k}$. Then

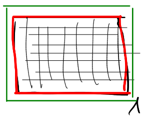
$$E_p(N) = \sum_{x \in \Lambda_k} P_p(x \leftrightarrow \partial\Lambda_k) \geq |\Lambda_k| \cdot \theta(p)$$

Hence

$$P\Big(\underbrace{|\Lambda_k| - N}_{\geq 0} \geq \big(1 - \tfrac{\theta(p)}{2}\big)|\Lambda_k|\Big) \overset{\text{Markov}}{\leq} \frac{1-\theta(p)}{1 - \frac{\theta(p)}{2}} \leq 1 - \frac{\theta(p)}{2}.$$

$E(\cdots) \leq |\Lambda_k|(1-\theta(p))$

and thus $$P_p\Big(N \geq \frac{\theta(p)}{2}|\Lambda_k|\Big) \geq \frac{\theta(p)}{2}$$

Now let $F = \Big\{$  $\Big\}$ Λ_k

$$P_p(F) = p^{|\partial\Lambda_k|}(1-p)^{\partial\Lambda_k} \geq e^{-ck}$$

Hence

$\geq \frac{4n}{\theta(p)}$ by def of k.

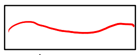
$$\begin{aligned}
P_p(|C_0| \geq n) &\geq P_p\Big(|C_0| \geq \frac{\theta(p)}{2}|\Lambda_k|\Big) \\
&\geq P_p\Big(0 \leftrightarrow \partial\Lambda_k,\ N \geq \frac{\theta(p)}{2}|\Lambda_k|,\ F\Big) \\
&\overset{\text{indep}}{=} P_p(F)\ P_p\Big(0 \leftrightarrow \partial\Lambda_k,\ N \geq \frac{\theta(p)}{2}|\Lambda_k|\Big) \\
&\overset{\text{FKG}}{\geq} e^{-ck} \cdot \theta(p) \cdot \theta(p)/2 \geq e^{-c_1\sqrt{k}}
\end{aligned}$$

$\square$

9/1

CONFORMAL INVARIANCE OF CRITICAL PERCOLATION (2D)

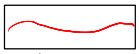
Motivation: We know $\forall k \geq 1 \ \exists \ 0 < c_1(k) < c_2(k) < 1$.



$$c_1(k) \leq P_{p_c}\Big(\cdot \, n \Big) \leq c_2(k)$$

(rectangle of width kn and height n)

Conjecture:



$$\lim_{n \to \infty} P_{p_c}\Big(\cdot \, n \Big) = c(k) \text{ exists.}$$

(rectangle of width kn and height n)

Actually, more is conjectured to be true $\longrightarrow$ **conformal invariance**

History:
- Conjectured in physics literature in 1992, confirmed by numerics.
- A "formula" (see below) proposed by Cardy (1992)
- Proved by Smirnov (2001) on triangular lattice

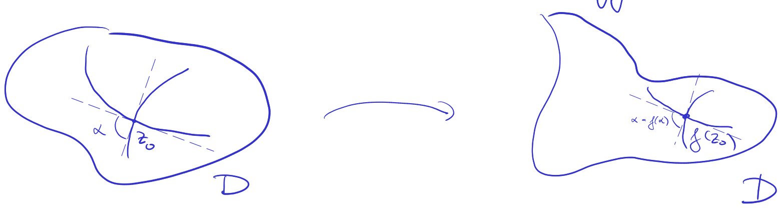
Short excursion to complex analysis

- **Domain** = non-empty connected open subset of $\mathbb{C}$
- For D, D' domains, **conformal map** from D to D' =
  = bijection $f: D \to D'$, analytic on D, $\forall z \in D \ f'(z) \neq 0$.
  (Consequence: the inversion $f^{-1}: D' \to D$ is analytic on D')

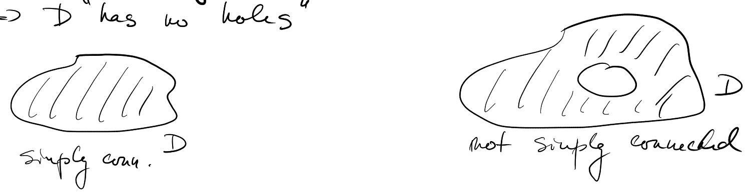
Rem: f is analytic $\rightarrow$ $\forall z_0 \in D$ $\quad f(z) = \underbrace{f(z_0)}_{\text{translation}} + \underbrace{f'(z_0)}_{\in \mathbb{C}}(z - z_0) + \dots$

rotation + scaling

$\longrightarrow$ f preserves angles $\Rightarrow$ terminology "conformal"



- D domain is **simply connected** if D and $\mathbb{C} \setminus D$ are connected
  $\Longleftrightarrow$ D "has no holes"

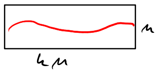


9/1

Conformal invariance of critical percolation (2D)

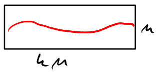
Motivation: We know $\forall k \geq 1 \ \exists \ 0 < c_1(k) < c_2(k) < 1$.

$$c_1(k) \leq \mathbb{P}_{p_c}\Big(\text{ } \Big) \leq c_2(k)$$



Conjecture:

$$\lim_{n\to\infty} \mathbb{P}_{p_c}\Big(\text{ } \Big) = c(k) \text{ exists.}$$



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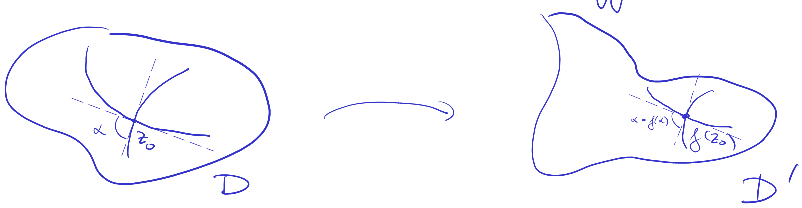
  (Consequence: the inversion $f^{-1}: D' \to D$ is analytic on D')

Rem: f is analytic $\to$ $\forall z_0 \in D$ $\quad f(z) = \underbrace{f(z_0)}_{\text{translation}} + \underbrace{f'(z_0)}_{\in \mathbb{C}\text{; rotation + scaling}}(z - z_0) + \dots$

$\longrightarrow$ f preserves angles $\Longrightarrow$ terminology "conformal"



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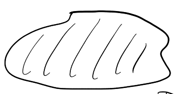
  $\Longleftrightarrow$ D "has no holes"



simply conn.

not simply connected

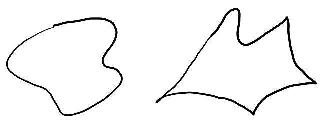
9/2

(9.1) **Theorem:** (Riemann mapping theorem)
Let $D \neq \mathbb{C}$ be a simply connected domain, $z \in D$. Then there is a unique conformal map $f : D \to \mathbb{D} = \{z : |z| \leq 1\}$ (disk) so that $f(z) = 0$, $f'(z) > 0$ (that is $f'(z) \in \mathbb{R}_+$)

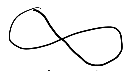
(We do not prove this theorem)

Consequences:
- there are ∞ many conformal maps $f : \mathbb{D} \to \mathbb{D}$
- $\forall D, D' \neq \mathbb{C}$ simply connected there are infinitely many conformal maps $f : D \to D'$.

- D domain: ∂D – **boundary** of D, $\bar{D}$ – **closure** of D $= D \cup \partial D$
- D is a **Jordan domain** if ∂D is a **Jordan curve** $= \partial D = \gamma(S^1)$ where $S^1 = \mathbb{R} \setminus \mathbb{Z}$ unit circle, γ: continuous injection.



Jordan

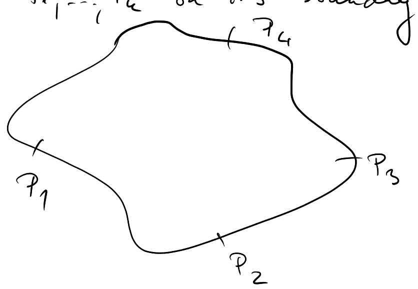


not-Jordan.

D Jordan $\Rightarrow$ D simply connected, $D \neq \mathbb{C}$

(9.2) **Theorem** (Caratheodory) D, D' Jordan domains, $f : D \to D'$ conformal. f can be uniquely extended to a cont. bijection of $\bar{D}$ and $\bar{D}'$.

- **k-marked domain** $(D, P_1, \dots, P_k)$ = Jordan domain D with k points $P_1, \dots, P_k$ on its boundary, ordered anti-clockwise

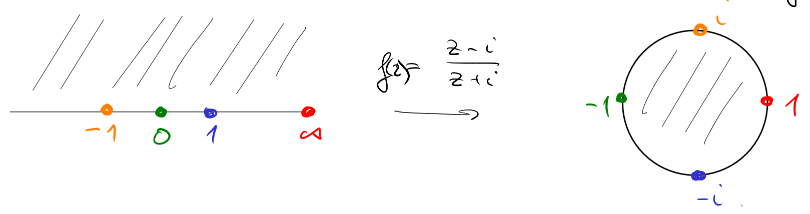


– 4-marked domain.

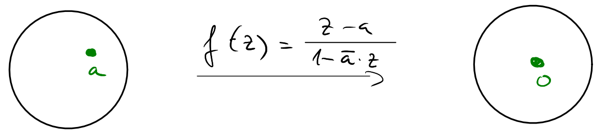
9/3

(9.3) **Theorem:** If $(D; P_1, P_2, P_3)$, $(D'; P_1', P_2', P_3')$ are 3-marked domains then there is a unique conformal map $f: D \to D'$ such that its extension satisfies $P_i' = f(P_i)$, $i = 1, 2, 3$.

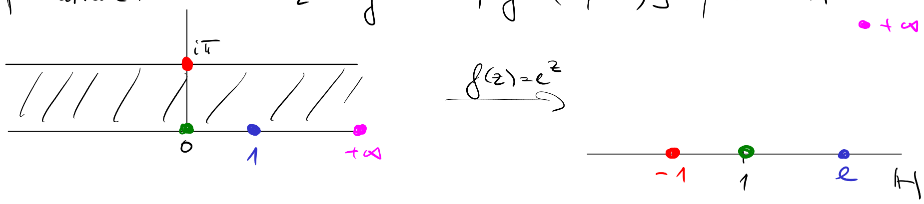
Examples: • $D = \mathbb{H} = \{z : \mathrm{Im}(z) > 0\}$, $D' = \mathbb{D}$ (Caley transform)



• Möbius automorphism $D = D' = \mathbb{D}$



• Exponential: $D = \{x+iy : x \in \mathbb{R}, y \in (0, \pi)\}$, $D' = \mathbb{H}$



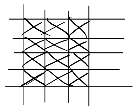
Conformal invariance conjecture:

topological rectangle

• $D_4 = (D, P_1, P_2, P_3, P_4)$, $D_4' = (D', P_1', P_2', P_3', P_4')$ are conformally equivalent if there is conformal map $f: D \to D'$ s.t. $P_i' = f(P_i)$ $\forall i = 1, \dots, 4$. (f is either unique, or it does not exist by (9.3))

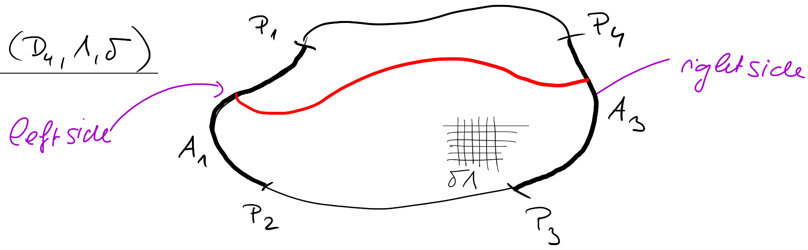
• $A_i :=$ arc of ∂D joining P_i with P_{i+1} $\quad (P_5 = P_0)$

9/4

Λ - any lattice (square, triangular, hexagonal,  , ...)
$\delta\Lambda$ - scaling of Λ by factor δ

- $\mathbb{P}^{\text{site}/\text{bond}}_{p,\Lambda}$ — site/bond percolation on Λ with parameter $p \in [0,1]$
- $p_c^{\text{bond}/\text{site}}(\Lambda) = \inf\{p : \mathbb{P}^{b,s}_{p(\Lambda)}(0 \longleftrightarrow \infty) > 0\}$.

<u>$LR(D_4, \Lambda, \delta)$</u>



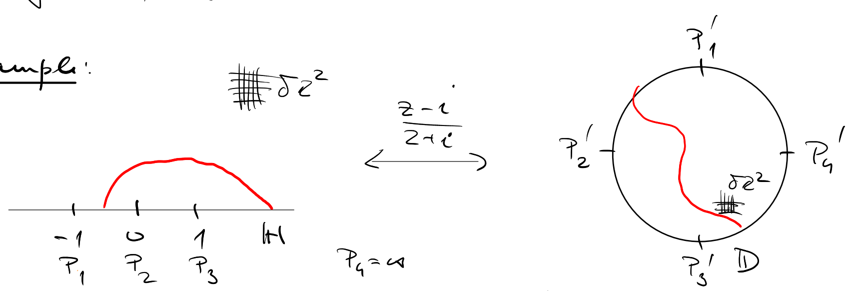
(9.4) <u>**Conjecture**</u>: The limit

$$\lim_{\delta \downarrow 0} \mathbb{P}_{p_c(\Lambda)}\big(LR(D_4, \Lambda, \delta)\big) = p(D_4) \in (0,1)$$

for every Jordan D. If D_4, D_4' are conf. equivalent, then $p(D_4) = p(D_4')$.

<u>Rem</u>: • If $p > p_c$ (or $p < p_c$) then the limit is 1 (resp. 0)
• By RSW, if the limit exists, it must be in $(0,1)$.

<u>Example</u>:

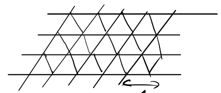


have same probas as $\delta \to 0$.

9/5

Smirnov's theorem:

Let $\mathbb{T}$ be the triangular lattice



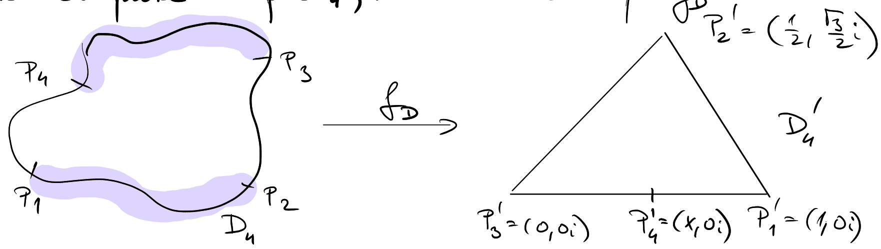
$P = P^{\text{site}}_{p_c^{\text{site}}(\mathbb{T})}(\mathbb{T})$ — site perc. on $\mathbb{T}$ at criticality.

(9.5) **Theorem** (Smirnov 2001) For every 4-marked domain D_4

$$\lim_{\delta \to 0} P\left(CR(D_4, \mathbb{T}, \delta)\right) = p(D_4)$$

exists and is conformally invariant.

How to "compute" $p(D_4)$: Consider map f_D:



s.t. f maps D to equilateral triangle, fixes

$$f_D(P_1) = 1,\ f_D(P_2) = \frac{1}{2} + \frac{\sqrt{3}}{2} i,\ f_D(P_3) = 0.$$

these two domains are conformally equivalent. (since we choose P_4 accordingly)

(9.6) **Theorem:** (Cardy's formula in Carleson formulation)

(i) On marked triangle D_4'

$$p(D_4') = x$$

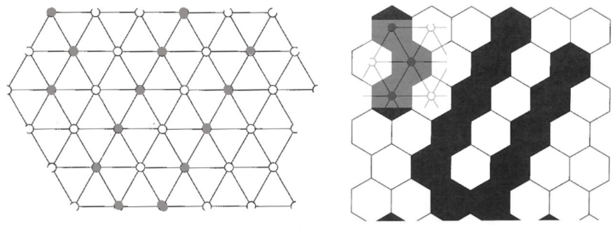
(ii) D_4 arbitrary, then $p_4(D_4) = f_D(P_4)$.

We now discuss the ingredients of the proofs of (9.5), (9.6).

9/6

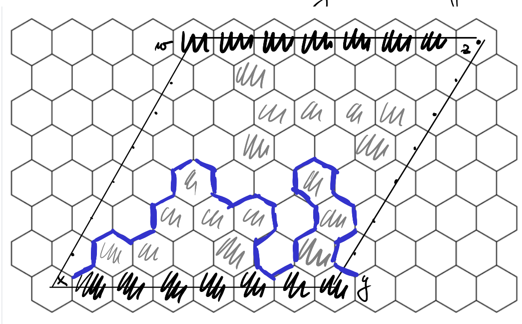
Representation by hexagonal tilings:

here: $p=\frac{1}{2}$



$\mathbb{T}$ (triangular) $\qquad$ $\mathbb{H}$ (hexagonal)

- Site percolation on $\mathbb{T}$ is better drawn as tilings of hexagons.
- There is even better duality than for $\mathbb{Z}^2$.

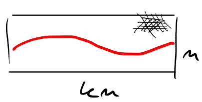


- there is either LR open crossing or TB closed crossing of any "rectangle." (consider the exploration as on the figure)
- in the rectangle on the figure $P_{\frac{1}{2}}(LR) = \frac{1}{2}$, for symmetry reasons.
- As in Kesten's theorem one can show that:

(9.7) **Proposition:** $\boxed{p_c^{site}(\mathbb{T}) = \frac{1}{2}}$

RSW theorem can be proved as well.

(9.8) **Proposition:** $\forall k \geq 1, n \in \mathbb{N}$ $\exists\, 0 < c_1(k) < c_2(k) < 1$ s.t.

$$c_1(k) \leq \mathbb{P}^{site}_{\frac{1}{2},\mathbb{T}}\Big($$  $$\Big) \leq c_2(k)$$

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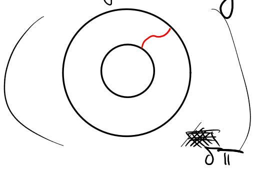
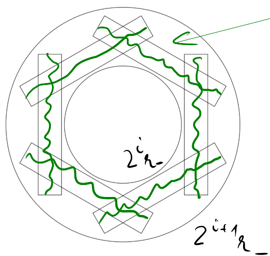
This can be used to show:

(9.9) **Lemma**: Let A be the annulus in $\mathbb{C}$, $A = \{z \in \mathbb{C} : r_- < |z| < r_+\}$.
There is $\alpha > 0$ s.t. for any $\delta < r_-/1000$ (say).

$P_{\frac{1}{2}}$  $\leq \left(\frac{r_-}{r_+}\right)^{\alpha}$

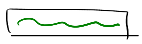
Proof sketch: • Let $(A_i)_{i=0}^{k-1}$ be a sequence of concentric annuli
A_i has inner radius $2^i r_-$, outer radius $2^{i+1} r_-$
$k = \left\lfloor \log_2 \frac{r_+}{r_-} \right\rfloor$. $\quad (\Rightarrow A_i \subset A \ \ \forall i = 0, \dots, k-1)$

• In every A_i, consider 6 rectangles as on the figure



closed crossings.

event E_i

• $P(E_i) \overset{FKG}{\geq} P($  $)^6 \geq c^6$

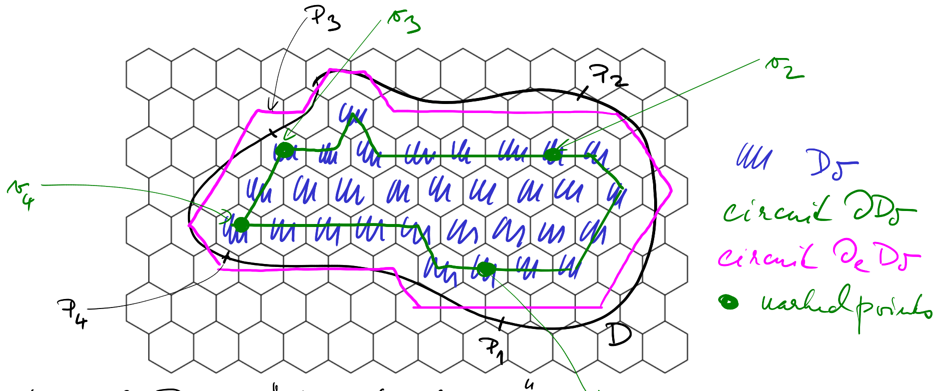
• $P($  $) \leq P_\phi\left(\bigcap_{i=0}^{k-1} E_i^c\right) \leq (1-c^6)^k \leq \left(\frac{r_-}{r_+}\right)^{\alpha}$. □

9/8

Color switching lemma:

For domain D_4 we consider its discrete approximation D_δ in $\mathbb{H}$ (or $\mathbb{T}$) s.t.

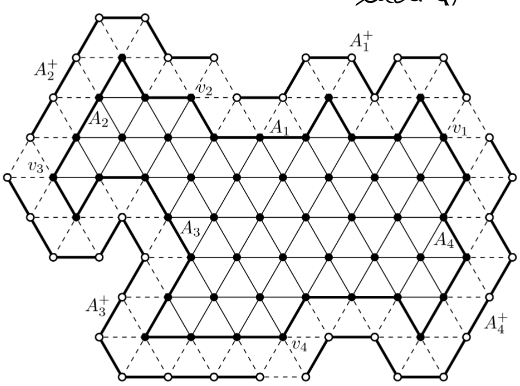
- union of the "hexagons in D_δ" is simply connected in $\mathbb{C}$
- ∂D_δ and $\partial_e D_\delta$ (inner and exterior boundary of D_δ in $\mathbb{T}_\delta$) correspond to circuits in $\mathbb{T}$

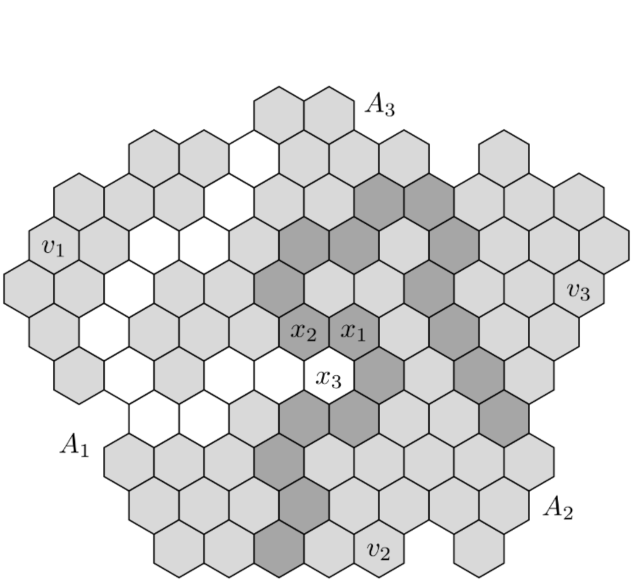


we call such D_δ a "discrete domain"

- v_i – points in ∂D_δ corresponding to P_i
  Assume that every v_i has two neighbours in $\partial_e D_\delta$

- A_i – part of ∂D_δ between v_i, v_{i+1} – both included
- We partition $\partial_e D_\delta$ into sets A_i^+ s.t.
  A_i^+ is a path in $\partial_e D_\delta$ starting next to v_i, ending next to v_{i+1}




A_3
v_1
v_3
x_2
x_1
x_3
A_1
A_2
v_2

This lecture will provide a very high level proof of Smirnov's theorem. 10/1

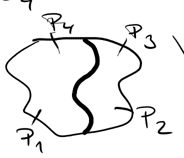
Smirnov's theorem:

We study site percolation on $\delta\mathbb{T}$ ($= \delta \times$ triangular lattice)

D – Jordan domain in $\mathbb{C}$

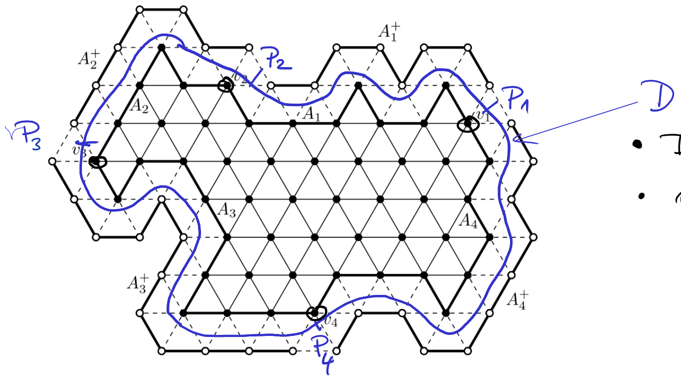
$D_4 = (D, P_1, \dots, P_4)$ – D marked with 4 points

Theorem: For every D_4

- $\lim\limits_{\delta \downarrow 0} \mathbb{P}_{\frac{1}{2}} \Bigg($  $\Bigg) = p(D_4)$
- $D_4 \mapsto p(D_4)$ is conformally invariant.

Color switching:

We approximate D_4 by a "discrete" domain.



- D_δ
- $v_j = P_i^\delta$

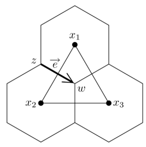
A_i – arc in ∂D_δ between $P_i^\delta, P_{i+1}^\delta$

A_i^+ – arc in $\partial_e D_\delta$ (are disjoint)

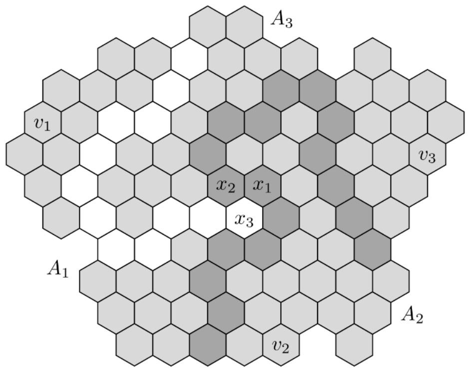
Ditto for 3-marked domain: have only $A_1, \dots, A_3, \dots$

10/2

Def: D_3 – 3-marked domain
x_1, x_2, x_3 – 3 points in $\operatorname{int} D_\delta$
in an anticlockwise triangle



$\underbrace{B_1B_2W_3}_{\text{This is one event}} = \begin{cases} \text{there are vertex disjoint paths } \gamma_1, \gamma_2, \gamma_3. \\ \gamma_i \text{ links } x_i \text{ with } A_i \\ \gamma_1, \gamma_2 \text{ are black, } \gamma_3 \text{ is white.} \end{cases}$



Event $B_1B_2W_3$ (gray hexagons are irrelevant)

Analogously, $B_1W_2B_2$, $W_1W_2B_3$, ...

Rem: Since $p = \frac{1}{2}$, $\mathbb{P}(W_1W_2B_3) = \mathbb{P}(B_1B_2W_3)$.

(10.1) **Lemma:** (colour switching)

$$\mathbb{P}(B_1B_2W_3) = \mathbb{P}(B_1W_2B_3) = \mathbb{P}(W_1B_2B_3)$$
$$\Big(= \mathbb{P}(W_1W_2B_3) = \mathbb{P}(W_1B_2W_3) = \mathbb{P}(B_1W_2W_3) \Big)$$

Vertical inequalities are trivial because we're working with $p = 1/2$ so swapping colours does not affect probability.

Rem: In general $\mathbb{P}(W_1W_2W_3) \neq \mathbb{P}(W_1W_2B_3)$.

10/3

Proof: We show $\mathbb{P}(B_1 W_2 B_3) = \mathbb{P}(B_1 W_2 W_3)$

(this is sufficient)

the other guys follow analoguosly.

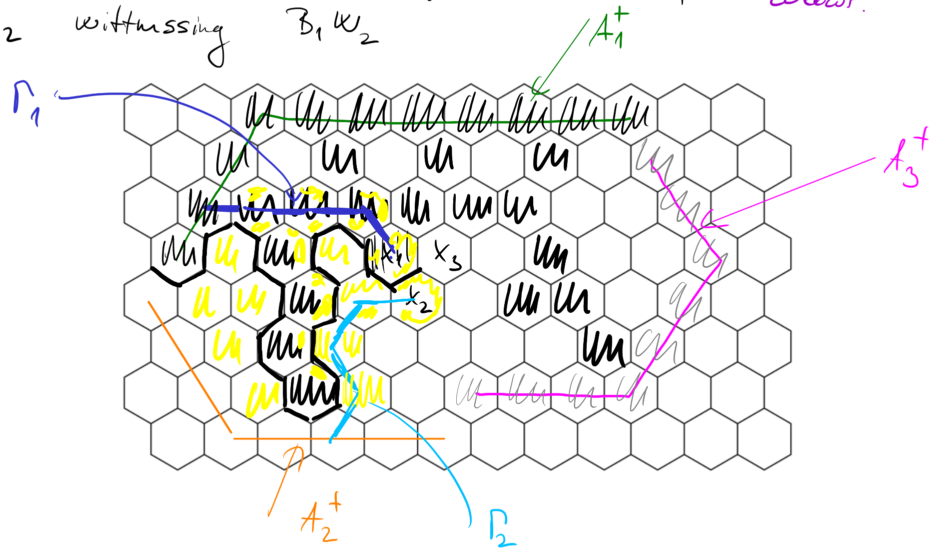
For this it suffices

(*) $$\mathbb{P}(B_1 W_2 B_3 \mid \underbrace{B_1 W_2}) = \mathbb{P}(B_1 W_2 W_3 \mid \underbrace{B_1 W_2})$$

first two do not change!

it is also important that the event we're conditioning on has two different colours.

If $B_1 W_2$ occurs one can find "innermost" paths Γ_1, Γ_2 witnessing $B_1 W_2$



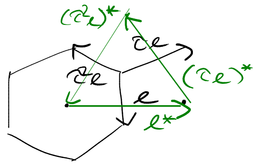
- For γ_1, γ_2 deterministic:
  The event $\{\Gamma_1 = \gamma_1, \Gamma_2 = \gamma_2\}$ depend only on hexagons on and inbetween of γ_1, γ_2 – yellow hexagons
- the black/white connection of $x_3\, A_3$ cannot cross Γ_1, Γ_2
  $\Rightarrow$ it depends on non-yellow hexagons
  $\Rightarrow \forall \gamma_1, \gamma_2$:
  $$\mathbb{P}(x_3 \overset{\text{white}}{\longleftrightarrow} A_3 \mid \Gamma_1 = \gamma_1, \Gamma_2 = \gamma_2)$$
  $$= \mathbb{P}(x_3 \overset{\text{black}}{\longleftrightarrow} A_3 \mid \Gamma_1 = \gamma_1, \Gamma_2 = \gamma_2)$$
- Sum over all γ_1, γ_2 to show (*). $\square$

10/4

Framework of the proof

$\tau = e^{\frac{2\pi i}{3}}$ = Rotation by 120°

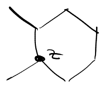
Notation:

- 

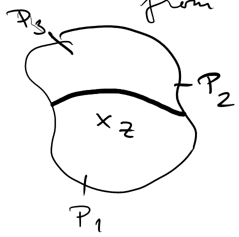
$\underbrace{\ }_{\text{rotated edge}}$

$$e^* + \tau(\tau e)^* + \tau^2(\tau^2 e)^* = 0$$

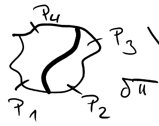
$$\left(\Leftarrow \ 1 + \tau^2 + \tau^4 = 0 \right)$$

- $z \in \mathbb{H} \cap D_\delta$ 

$E_{1,\delta}(z) =$ "$\exists$ simple black path separating z, P_1^δ from $P_2^\delta P_3^\delta$"



$E_{2,\delta}(z), E_{3,\delta}(z)$ – analogous (by just permutating indices)

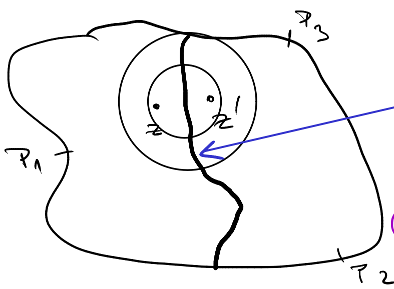
Rem: $\mathbb{P}\Big($  $\Big) = \mathbb{P}\left(E_{1,\delta}(P_4^\delta)\right)$

- $H_{j,\delta}(z) = \mathbb{P}\left(E_{j,\delta}(z)\right)$, extended continuously to D (by interpolation)

Claim: (Hölder continuity) $\exists\, k < \infty,\ \varepsilon > 0$
$\forall \delta < 1 \quad \forall z, z' \in D$

$$|H_{1,\delta}(z') - H_{1,\delta}(z)| \leq k\,|z - z'|^\varepsilon$$

Proof sketch



there is annulus crossing, proba

$\geq \left(\frac{r_-}{r_+}\right)^\alpha \approx |z - z'|^\alpha$

$\mathbb{P}$(crossing of annulus)

$\square$

10/5

By Arzela–Ascolli: $\{H_{1,\delta} : \delta > 0\}$ is relatively compact
$\Rightarrow$ Every H_{1,δ_n}, $\delta_n \downarrow 0$, has a converging subsequence.

Goal: All limits of converging subsequences ~~are the same~~ have the same limit.

Let $\delta_n \downarrow 0$ be such that
$$H_{i,\delta_n} \xrightarrow{\|\cdot\|_\infty} h_i \quad , \quad h_i : \bar{D} \to [0,1].$$

Set:

real valued functions

$$H_{\delta_n} = H_{1,\delta_n} + \tau H_{2,\delta_n} + \tau^2 H_{3,\delta_n} \xrightarrow{n\to\infty} h = h_1 + \tau h_2 + \tau^2 h_3$$
$$S_{\delta_n} = H_{1,\delta_n} + H_{2,\delta_n} + H_{3,\delta_n} \longrightarrow s = h_1 + h_2 + h_3 .$$

Rem: h and s determine h_1, h_2, h_3.

Discrete integration:

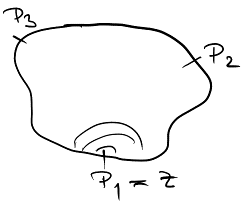
Claim: h & s are holomorphic on D
(That is $\Delta \operatorname{Re} h = \Delta \operatorname{Im} h = 0$, Cauchy-Riemann, ditto for s)

$z = x + iy$

$$\frac{\partial \operatorname{Re} h}{\partial x} = \frac{\partial \operatorname{Im} h}{\partial y}$$
$$\frac{\partial \operatorname{Re} h}{\partial x} = -\frac{\partial \operatorname{Im} h}{\partial y}$$

Cor: $s \equiv 1$

Proof: s is holomorphic, real valued
$\xRightarrow{CR}$ s is constant.



Almost obvious:
$$h_2(P_1) = 0 = h_3(P_1)$$
$$h_1(P_1) = 1$$
$\parallel$
probability that P_1 and P_1 are separated from the other points

Proof of the claim:

Theorem: (Morera) $f : D \to \mathbb{C}$ is holomorphic $\Leftrightarrow$
$\forall \gamma$ smooth loop in D $\oint_\gamma f(z)\,dz = 0$.

10/6

- Fix γ a smooth loop.
- Discretise γ along $\delta\mathbb{H} \longrightarrow \gamma_\delta$
- $N(\delta) = \#$ vertices of $\gamma_\delta \lesssim \delta^{-1}$, $\gamma_\delta = (x_1, \dots, x_{N(\delta)} \overset{=x_0}{})$
- Discrete contour integral:

$$I_\delta(\gamma) = \sum_{i=0}^{N(\delta)-1} \frac{1}{2}\left(H_\delta(x_i) + H_\delta(x_{i+1}) \right)(x_{i+1} - x_i)$$

It holds $\quad I_{\delta_n}(\gamma) \longrightarrow \oint_\gamma h(z)\,dz.$

Goal: $\quad I_{\delta_n}(\gamma) \longrightarrow 0.$

Notation: We omit δ sometimes.

$e = \{x, y\} \qquad H(e) = \frac{1}{2}(H(x) + H(y))$

$\qquad\qquad\qquad \partial_e H = H(y) - H(x).$

- $$I(\gamma) = \sum_{e \in \gamma_\delta} H(e)\, e = \sum_{f \in \mathrm{int}\,\gamma_\delta} \sum_{e \in \partial f} H(e) \cdot e$$

oriented edges $e = \vec{e}$

all hexagons inside of γ

- Easy calculation yields:

center of f.

$$\sum_{e \in \partial f} H(e) e = \sum_{e = \{x,y\} \in \partial f} \partial_e H \left(\frac{x+y}{2} - c(f) \right)$$

$$\Longrightarrow \quad I(\gamma) = \sum_{f \in \mathrm{int}\,\gamma} \sum_{e = \{x,y\} \in \partial f} \partial_e H \left(\frac{x+y}{2} - c(f) \right)$$

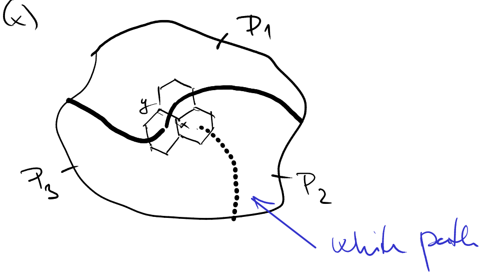
$O(\delta^{-1} \cdot \delta^{1+\varepsilon}) \to 0$

$$= \sum_{e \subset \mathrm{int}\,\gamma} \partial_e H \cdot e^* + \text{boundary term}$$

- $\partial_e H = \partial_e H_1 + \tau\, \partial_e H_2 + \tau^2 \partial_e H_3$
- $$\partial_e H_{1,\delta} = \partial_e P(E_{1,\delta}) = P(E_{1,\delta}(y) \setminus E_{1,\delta}(x)) - P(E_{1,\delta}(x) \setminus E_{1,\delta}(y)) =: f_{1,\delta}(e) + f_{1,\delta}(-e) \qquad e = (x,y)$$

10/7

• $E_{1,\delta}(y) \setminus E_{1,\delta}(x)$ =



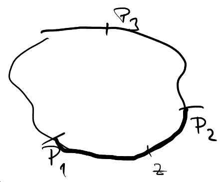
color switch

$\Rightarrow \quad f_{1,\delta}(e) = f_{2,\delta}(\tau e) = f_{3,\delta}(\tau^2 e)$

• Putting all together $\Rightarrow \quad I_{\delta_n}(y) \longrightarrow 0$ as required □.

Determining h: To prove that the limit h is unique we need to find boundary conditions for h.

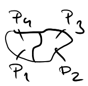
• If z is on P_1P_2, then
$h_3 = 0 \quad (h_3 = \lim P(E_{3,\delta}(z)))$
$h_1 + h_2 = 1 \quad$ (exploration started from P_3)
$\Rightarrow \quad h(z) = h_1(z) + \tau h_2(z) + \tau^2 h_3(z)$
$\in [1, \tau]$ ← complex segment.



• Similarly: z on $P_2P_3 \quad \Rightarrow \quad h(z) \in [\tau, \tau^2]$
on $P_3P_1 \quad \Rightarrow \quad h(z) \in [\tau^2, 1]$

Further complex analysis implies that
h is the conformal map from D to triangle $(1, \tau, \tau^2)$
s.t. $P_1 \overset{h}{\longmapsto} 1, \; P_2 \overset{h}{\longmapsto} \tau, \; P_3 \longmapsto \tau^2$
$\Rightarrow h$ is unique.

. . . □

Finally, $h_1(z) = \dfrac{2\,\mathrm{Re}\, h + 1}{3}$ determines $P($  $)$.

4 Random Walks in Random Environment

4.1 Reminders about Markov-Chains

In this section we will see several reminders on Markov-chains. We will consider Markov-chain on at most countable state spaces $(E, \mathcal{E})$, simplifying a lot of the theory.

Definition 4.1. Let $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\in\mathbb{N}_0}, \mathrm{P})$ be a probability space and E an at most countable set with $\mathcal{E} := \mathcal{P}(E)$ its power set. A *Markov-Chain* in discrete time is a process $X = (X_n)_{n\in\mathbb{N}_0}$ which takes values in E such that

1. X is $\mathcal{F}_n$ adapted and
2. The Markov-property holds, that is for any $n \in \mathbb{N}_0$ and $A \subseteq E$,

$$\mathrm{P}(X_{n+1} \in A \mid \mathcal{F}_n) = \mathrm{P}(X_{n+1} \in A \mid \sigma(X_n)).$$

Remark. Typically $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$, the natural filtration of X. The above statement is equivalent to the following statement: for any $n \in \mathbb{N}_0$ and $x_0, \dots, x_{n+1} \in E$ we have

$$\mathrm{P}(X_{n+1} = x_{n+1} \mid X_0 = x_0, \dots, X_n = x_n) = \mathrm{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

Definition 4.2. A Markov chain X is time homogenous if

$$\mathrm{P}(X_{n+1} = y \mid X_n = x) =: p_{x,y}$$

does not depend on n.

Remark. The map p from above can be identified with a kernel from E to E by

$$p(x, A) = \sum_{y\in A} \mathrm{P}(x, y).$$

Definition 4.3. A Markov-Chain X is *time-homogenous* if there is $p : E^2 \to [0,1]$ such that for any $n \in \mathbb{N}_0$

$$\mathrm{P}(X_{n+1} = y \mid X_n = x) = p(x,y) \overset{\text{Notation}}{=} p_{x,y}.$$

Remark. Note that p can be identified with a kernel from E to E by

$$p(x,A) = \sum_{y \in A} \mathrm{P}(x,y).$$

Example 4.4. Let ξ_i be i.i.d. uniformly distributed on $[0,1]$. Let $f : E \times [0,1] \to E$ be a measurable function and X_0 be a given random variable. The process defined by $X_{n+1} = f(X_n, \xi_n)$ is a Markov-chain.

We introduce the following notation: Given $x \in E$ and $p : E \times E \to [0,1]$ we write P_x for the law of X given X_0, that is $\mathrm{P}_x(\cdot) = \mathrm{P}(\cdot \mid X_0 = x)$.

In the following we list some elementary facts about Markov chains, whose proof we leave as an exercise to the reader. Let X be a time homogenous Markov-chain. Then:

- For any $x, y \in E$ and $m, n \in \mathbb{N}_0$ we have that the *Chapman–Kolmogorov equation* holds:
$$p^{m+n}_{x,y} = \sum_z p^n_{x,z} p^m_{z,y},$$
where $p^n_{x,y} = \mathrm{P}_x(X_n = y)$.
- We have the *(weak) Markov-property*: For any $f : E^{\mathbb{N}} \to \mathbb{R}$ bounded and measurable and $n \in \mathbb{N}_0$ we have that
$$\begin{aligned}\mathrm{E}\Big[f(X_n, X_{n+1}, \dots) \mid \mathcal{F}_n\Big] &= \mathrm{E}\Big[f(X_n, X_{n+1}, \dots) \mid \sigma(X_n)\Big] \\ &= \mathrm{E}_{X_n}\Big[f(X_0, X_1, \dots)\Big]\end{aligned}$$
- We have the *strong Markov-property*: For any stopping time $T : \Omega \to \mathbb{N}_0 \cup \{+\infty\}$ we have
$$\begin{aligned}\mathrm{E}\Big[f(X_T, X_{T+1}, \dots) \mid \mathcal{F}_T\Big] &= \mathrm{E}\Big[f(X_T, X_{T+1}, \dots) \mid \sigma(X_T)\Big] \\ &= \mathrm{E}_{X_T}\Big[f(X_0, X_1, \dots)\Big]\end{aligned}$$

Definition 4.5. A Markov chain is *irreducible* if for any $x, y \in E$ there exists $n = n(x,y)$ such that $\mathrm{P}_x(X_n = y) > 0$.

With Markov-chains one is often interested if the chain returns to a point and how often. To this end, we define the *hitting time* of X as

$$H_x = \inf\{n \geq 0 \mid X_n = x\}$$

and the *entrance time* of x as

$$\tilde{H}_x := \inf\{n \geq 1 \mid X_n = x\}.$$

The number of visits in x is denoted by N_x and given by

$$N_x = \sum_{n \geq 0} \mathbf{1}_{X_n = x}.$$

Definition 4.6. Let $x \in E$. The point x is *recurrent* if

$$\mathrm{P}_x(\tilde{H}_x < +\infty) = 1 \iff N_x \overset{\text{a.s.}}{=} +\infty$$

If x is not recurrent we call x *transient*.

The following proposition is sometimes useful. Its proof can be found in the lecture notes for the course Probability Theory.

Proposition 4.7. (a) X is transient iff $N_x \sim \mathsf{Geom}(\mathrm{P}_x(\tilde{H}_x = +\infty))$ iff

$$\sum_{n \geq 0} p^n(x, x) < +\infty.$$

(b) If X is irreducible and there exists a $x \in E$ that is recurrent/transient then all $y \in E$ are recurrent/transient.

As we saw, the standard random walk on $\mathbb{Z}^d$ is recurrent if $d = 1, 2$ and transient if $d \geq 3$.

Generators and harmonic functions. We introduce the operator $L : B(E) \to B(E)$[1], $L = \mathrm{Id} - p$, mapping bounded functions on E to bounded functions on E that is,

$$(Lf)(x) = f(x) - \sum_{y \in E} p(x, y) f(y)$$

and call L *Generator* of X.

Definition 4.8. Let $f : E \to \mathbb{R}$. We call f *harmonic* on $A \subseteq E$ if for any $x \in A$ we have $(Lf)(x) = 0$. This is equivalent to saying for any $x \in A$ we have $\mathrm{E}_x[f(X_1)] = f(x)$.

If f is harmonic on A, then $M_n = f(X_{n \wedge H_{A^\complement}})$ is a discrete-time martingale, where $H_{A^\complement}$ denotes the hitting time of the complement of A.

[1]With $B(E)$ we denote the set of all measurable and bounded functions $f : E \to E$. Since our σ-algebra is $\mathcal{E} = \mathcal{P}(E)$, every function $E \to E$ is measurable.

Definition 4.9. The Markov-chain X is *reversible* if there exists $\pi : E \to \mathbb{R}_{\geq 0}$ such that

$$\pi(x)p(x,y) = \pi(y)p(y,x). \tag{4.1}$$

Example 4.10.
- The standard random walk on $\mathbb{Z}^d$ reversible (take e.g., $\pi \equiv 1$).
- If π satisfies (4.1), then it is "invariant" that is

$$\pi(x) = \sum_{y \in E} p(y,x)\pi(y).$$

(Exercise: Show this).

4.2 Introduction to Random Walks in Random Environment

The practical goal of this section is to understand stochastic processes on $E = \mathbb{Z}^d$ that are not "spatially homogenous" as standard random walk i.e., the rule how X moves may depend on the position $x \in \mathbb{Z}^d$.

Environment. Let ω denote the collection of those rules:

- In full generality, $\omega = (\omega_x)_{x \in \mathbb{Z}^d}$ where for every x, ω_x is a probability measure on $\mathbb{Z}^d$, $\omega_x(y) = \mathrm{P}_x(X_1 = x + y)$.
- In the following we will first consider the *nearest neighbour case*: ω_x is a probability measure on neighbours of $x \in \mathbb{Z}^d$; or better: $\omega(x)$ is a measure on $\{\pm e_i\}_{i=1}^d := \mathcal{E}_d$, where e_i denotes the i-th standard basis vector. Then

$$\omega_x \in S_{2d-1} = \left\{ (p_e)_{e \in \mathcal{E}_d} \in [0,1]^{\mathcal{E}_d} \;\middle|\; \sum_{e \in \mathcal{E}_d} p_e = 1 \right\}.$$

so $\omega \in \Omega := S_{2d-1}^{\mathbb{Z}^d}$.

A *random walk in environment* ω is a $\mathbb{Z}^d$ valued Markov-chain such that

$$\mathrm{P}(X_{n+1} = x + e \mid X_n = x) = \omega_x(e), \quad x \in \mathbb{Z}^d, e \in \mathcal{E}_d.$$

We will introduce the following notation: Let P_x^ω denote the distribution of the above random walk started at x in ω. If $x = 0$ we denote $\mathrm{P}_0^\omega = \mathrm{P}^\omega$.

In general selecting "interesting" non-constant environments is hard, so we will consider a *random environment*. To this end, let $\mathcal{A}$ denote the cylinder σ-algebra on Ω (where Ω is given by the above) and let $\mathbb{P}$ be a probability measure on $(\Omega, \mathcal{A})$.

There are many possible choices for such $\mathbb{P}$, however a quite standard requirement is that $\mathbb{P}$ is *stationary*, that is

$$\text{for any } x \in \mathbb{Z}^d, \quad (\omega_{y+x} : y \in \mathbb{Z}^d) \overset{\text{law}}{=} (\omega_y : y \in \mathbb{Z}^d).$$

Example 4.11. (a) *Product measure*: Fix μ a probability measure on S_{2d-1} and take $\mathbb{P} = \mu^{\otimes \mathbb{Z}^d}$, or equivalently, $(\omega_x : x \in \mathbb{Z}^d)$ is an i.i.d. sequence with $\omega_x \sim \mu$ for all x.

If we take $d = 1$, then $\omega(x) = (\omega(x - e), \omega(x + e))$ and taking e.g.,

$$\mu = p\delta_{(\frac{1}{4},\frac{3}{4})} + (1-p)\delta_{(\frac{2}{3},\frac{1}{3})}$$

we have the following picture [todo: Bild]

(b) *Random conectedness:* Let E_d denote the set of edges of $\mathbb{Z}^d$. To each edge we associate a number c_e which we call the *conductance of the edge e*. [todo: Finish this example see comment for link to page]

(c) *Random walk on the percolation cluster (the ant in a labyrinth)*: Pick

$$e_x = \begin{cases} 0 & \text{with probability } (1-p) \\ 1 & \text{with probability } p \end{cases}$$

[todo: finish example]

We now consider a random walk X in a ranom environment $\omega \in \Omega$.

Definition 4.12.
- For a fixed realisation ω of the random environment, the law P^ω of X is called *quenched law.*
- Taking ω randomly with distribution $\mathbb{P}$ we define the *averaged/annealed* law of X by the "semidirect procut" P, that is for any $A \in \mathcal{A}$, $B \in \mathcal{F} = \sigma(X_0, X_1, \dots)$ and $x \in \mathbb{Z}^d$ we define
$$\mathrm{P}_x(A \times B) = \int_A \mathrm{P}_x^\omega(B)\mathbb{P}(\mathrm{d}\omega),$$
again with the convention that $\mathrm{P} := \mathrm{P}_0$.

Note that this above procedure corresponds to defining the process in two steps:

(1) pick ω randomly according to $\mathbb{P}$ and

(2) then run X in the evironment ω according to P_x^ω.

Remark.
- Under the qunеched law, X is a Markov-chain (quite a simple process) but its transition rules are x dependent, making it complicated.
- Under the averaged measure, X is *not* a Markov-chain but its transition rules are stationary.

Example 4.13. [todo: Example in Lecture 11 Page 6]

Definition 4.14. The random environment is called *elliptic* resp. satisfies the *ellipticity assumption*, if

$$\mathbb{P}\text{-a.s. for all } x \in \mathbb{Z}^d \text{ and } e \in \mathcal{E}_d \; \omega_x(e) > 0 \tag{E}$$

We say that it is *uniformly elliptic* if

$$\text{there exists } \varepsilon_0 > 0 \text{ sucht that } \mathbb{P}\text{-a.s. for any } x \in \mathbb{Z}^d \text{ and all } e \in \mathcal{E}_d \; \omega_e > 0 \tag{UE}$$

In the following we alway assume:

Assumption 1. Considering a nearest neighbour walk on $\mathbb{Z}$ ($d = 1$) we will assume that

- $(\omega_x : x \in \mathbb{Z})$ are i.i.d. under $\mathbb{P}$ and
- the evironment $\omega \in \Omega$ is uniformly elliptic (satisfies (**UE**)).

Exercise 4.15. Show that (**UE**) implies that X is irreducible under P^ω for $\mathbb{P}$-a.e. $\omega \in \Omega$.

Since $\omega_x(-e) = 1 - \omega_x(e)$ we can "define" $\omega_x := \omega_x(e)$ the probability of going to the right and view it as an element of $[0,1]^{\mathbb{Z}}$ (or $[\varepsilon, 1-\varepsilon]^{\mathbb{Z}}$, using (**UE**)).

Harmonic functions. It is useful to know harmonic functions for X (under P^ω). Here, f is harmonic iff

$$\text{for all } x \in \mathbb{Z}, \quad f(x) = \omega_x f(x+1) + (1-\omega_x) f(x-1). \tag{4.2}$$

Introducing

$$\rho_x = \frac{1-\omega_x}{\omega_x} \quad \text{and} \quad \Delta f(x) = f(x) - f(x-1),$$

(4.2) is equivalent to

$$\Delta f(x+1) = \rho_x \Delta f(x). \tag{4.3}$$

Equation (4.3) determines harmonic function up to two constants as proved in the next lemma:

Lemma 4.16. A function $f : \mathbb{Z} \to \mathbb{R}$ is harmonic if and only if for some $A, B \in \mathbb{R}$ we have

$$f(x) = A + \begin{cases} B \sum_{i=1}^{x} \prod_{j=0}^{i-1} \rho_i, & x > 0 \\ (-B) \sum_{i=x+1} \prod_{j=-i}^{-1} \rho_i, & x \leq 0; \end{cases} \tag{4.4}$$

that is, the space of all harmonic function is isomorphic to $\mathbb{R}^2$.

Exercise 4.17. Prove Lemma 4.16.
Hint: Prove a similar formula for Δf and then use that (**UE**) implies that $\rho_x, \rho_x^{-}1 \neq 0$ for all $x \in \mathbb{Z}$, $\mathbb{P}$-a.s.

Recurrence/Transience. Consider the mean drift of X at 0, given by

$$d := \mathbb{E}\Big[\omega_0 \cdot 1 + (1-\omega_0)\cdot(-1)\Big] = 2\mathbb{E}[\omega_0] - 1.$$

One would believe that X is recurrent under P^ω for $\mathbb{P}$-a.e. $\omega \in \Omega$ iff $d = 0$. However, this guess is *false*!

Theorem 4.18 (Solomon, 1975)**.** Under the assumption 1 we have the following:

(a) If $\mathbb{E}[\log \rho_0] = 0$ then

$$\limsup_{n\to+\infty} X_n = -\liminf_{n\to-\infty} X_n = +\infty,$$

that is, X is $\mathbb{P}$-a.s. recurrent.

(b) If $\mathbb{E}[\log \rho_0] < 0$ then $\lim X_n = +\infty$ and if $\mathbb{E}[\log \rho_0] > 0$ we have $\lim X_n = -\infty$. In both cases X is $\mathbb{P}$-a.s. transient.

Remark.
- The assumption (**UE**) is in fact not necessary. One only needs that $\mathbb{E}[\log \rho_0] \in [-\infty, \infty]$ is well-defined.
- The assumption of $(\omega_x)_{x\in\mathbb{Z}}$ being i.i.d. can be weakened to $(\omega_x)_{x\in\mathbb{Z}}$ being *ergodic*.

Proof. Consider f as in (4.4) with $A = 0$, $B = 1$. Then define $V(x)$ by

$$V_x = \log(\Delta f(x)).$$

which leads to

$$f(x) = \begin{cases} 0, & x = 0; \\ \sum_{i=1}^{x} \mathrm{e}^{V_i}, & x > 0; \\ \sum_{i=x+1}^{0} \mathrm{e}^{-V_i} & x < 0. \end{cases} \tag{4.5}$$

Then note that f is increasing and for any $x \in \mathbb{Z}$, $\Delta f(x) > 0$. Since f is harmonic for P^ω, $M(n) = f(X_n)$ is a non-negative martingale (in discrete time) under P^ω. ¨

(b): $\mathbb{E}[\log \rho_i] > 0$. By the strong law of large numbers,

$$V_x = x\Big(\mathbb{E}[\log \rho_i] + o(1)\Big), \quad \text{as } x \to \pm\infty,$$

hence

$$\lim_{x\to+\infty} f(x) = +\infty, \quad \text{and} \quad \lim_{x\to-\infty} f(x) = c(\omega) > -\infty, \quad \mathbb{P}\text{-a.s.}$$

By monotone convergence the limit $\lim_{n\to+\infty} M(n)$ exists P^ω-a.s. and is finite. That is only possible if $\lim_{n\to+\infty} M(n) = -c(\omega)$, that is $\lim_{n\to+\infty} X_n = -\infty$.

(b): $\mathbb{E}[\log \rho_i] < 0$. This case follows by the prevous one by reversing left and right.

(a): $\mathbb{E}[\log \rho_i] = 0$. In this case we have $\lim_{x\to+\infty} f(x) = +\infty$ and $\lim_{x\to-\infty} f(x)$ so $M(n)$ does not necessarily converge. We thus fix $A \in \mathbb{N}$ and consider

$$M^+(n) := M(n \wedge H_A), \quad \text{and} \quad M^-(n) = M(n \wedge H_{-A}).$$

Then M^+ (resp. M^-) is a martingale bounded from above (resp. from below). Hence $M^\pm$ converge as $n \to +\infty$, P^ω-a.s. The only possible value for the limit is A (resp $-A$), so

$$\mathrm{P}^\omega\left[H_A < +\infty, H_{-A} < +\infty\right] = 1$$

implying the claim. □

Law of large numbers. We now wish to establish a law of large numbers, that is to understand

$$v = \lim_{n\to+\infty} \frac{X_n}{n},$$

the "asymptotic speed of X".

The questions are:

- Does this limit exist?
- In which sense, under which measure ($\mathbb{P}$, P^ω) does it exist (a.s.)?
- Does it (and if yes, how) depend on the environment ω?

Theorem 4.19 (Solomon, 1975)**.** Under Assumption 1 the limit $v = \lim_{n\to+\infty} \frac{X_n}{n}$ exists P-a.s. with

$$v = \begin{cases} \frac{1-\mathbb{E}(\rho_0)}{1+\mathbb{E}(\rho_0)} > 0 & \text{if } \mathbb{E}(\rho_0) < 1; \\ 0 & \text{if } \frac{1}{\mathbb{E}(1/\rho_0)} \leq 1 \leq \mathbb{E}(\rho_0) \\ \frac{\mathbb{E}(1/\rho_0)-1}{1+E(1/\rho_0)} < 0 & \text{if } 1 < \frac{1}{\mathbb{E}(1/\rho_0)} \end{cases}$$

Remark. (1) This above law of large number uses the averaged measure P. However, as v is P-a.s. constant, it also implies

$$\text{for } \mathbb{P}\text{-a.s. } \omega, \quad \lim_{n\to+\infty} \frac{X_n}{n} = v \quad \mathrm{P}^\omega\text{-a.s.}.$$

(2) By Jensen's inequality,

$$\frac{1}{\mathbb{E}[1/\rho_0]} \leq \mathbb{E}[\rho_0],$$

so the above cases cover all the possibilities.

(3) Jensen's inequality also implies

$$\mathbb{E}[\log \rho_0] \leq \log \mathbb{E}[\rho_0]$$

with a strict inequality whenever $\mathbb{P}$ is non-degenerate. Hence there are examples when

$$\mathbb{E}[\log \rho_0] < 0 \text{ that is } \lim_{n\to+\infty} X_n = +\infty,$$
$$\mathbb{E}[\rho_0] \geq 1 \text{ that is } \lim_{n\to+\infty} \frac{X_n}{n} = 0.$$

(4) Another application of Jensen's inequality implies that

$$\mathbb{E}[\rho_0] = \mathbb{E}\left[\frac{1}{\omega_0} - 1\right] = \mathbb{E}[\omega_0^{-1}] - 1 \geq \frac{1}{\mathbb{E}[\omega_0]} - 1.$$

Since $0 < u \mapsto \frac{1-u}{1+u}$ is decreasing, in the first case,

$$0 < v = \frac{1-\mathbb{E}[\rho_0]}{1+\mathbb{E}[\rho_0]} \leq \frac{2-\frac{1}{\mathbb{E}[\rho_0]}}{\frac{1}{\mathbb{E}[\rho_0]}} = 2\mathbb{E}[\rho_0] - 1 = d$$

with inequality being strict if $\mathbb{P}$ is non-degenerate.

We now start to prepare for the proof of Theorem 4.19. We concentrate on the first two cases, the third follows by left-right flip.

Lemma 4.20. Suppose that $\limsup_{n\to+\infty} X_n = +\infty$ (that is $\mathbb{E}[\rho_0] \leq 0$). Then

$$\lim_{n\to+\infty} \frac{H_n}{n} = c \in [1,+\infty] \implies \lim_{n\to+\infty} \frac{X_n}{n} = \frac{1}{c} \in [0,1].$$

Proof. Set $X_n^\star = \max_{k=0,\dots,n} X_n$. Suppose that $c < +\infty$ (the other case is left as an exercise to the reader). By definition $H_{X_n^\star} \leq n \leq H_{X_n^\star+1}$ so

$$c \xleftarrow[n\to+\infty]{} \frac{H_{X_n^\star}}{X_n^\star} \leq \frac{n}{X_n^\star} \leq \frac{H_{X_n^\star+1}}{X_1^\star+1} \cdot \frac{X_n^\star+1}{X_n^\star} \xrightarrow[n\to+\infty]{} c \cdot 1$$

as $X_n^\star \to +\infty$ by assumption. Since we consider the nearest neighbour random walk,

$$X_n^\star - X_n \leq n - H_{X_n^\star},$$

thus,

$$0 \leq \limsup_{n\to+\infty} \frac{X_n^\star - X_n}{n} \leq 1 - \liminf_{n\to+\infty} \frac{H_{X_n^\star}}{X_n^\star} \cdot \frac{X_n^\star}{n} = 0. \qquad \square$$

We now set

$$\tau_k = H_k - H_{k-1}, \qquad \text{for } k \geq 1 \tag{4.6}$$

and write $H_n = \sum_{k=1}^{n} \tau_K$. We would like to use some sort of law of large numbers but the τ_k's are

- identically distributed but not independent under P and
- independent but not identically distributed under P^ω.

So what to do?

4.3 More Ergodic Theory

Definition 4.21. A sequence of random variables $(X_n)_{n\in\mathbb{N}}$ of S-valued random variables, where S is some arbitrary state space is called *stationary* if for all $k \in \mathbb{N}$,

$$(X_{k+n})_{n\in\mathbb{N}_0} \overset{(d)}{=} (X_n)_{n\in\mathbb{N}_0}$$

The goal is to show that for $f : S \to \mathbb{R}$ such that $\mathrm{E}|f(X_0)| < +\infty$ the value

$$\lim_{n\to+\infty} \frac{1}{n} \sum_{i=0}^{n-1} f(X_i) \quad \text{exists a.s...}$$

Example 4.22. The following sequences are stationary:

(a) $(X_n)_{n\geq 0}$ is an i.i.d. sequence.

(b) $X = (X_n)_{n\in\mathbb{N}_0}$ a Markov chain wiht transition probability $p(x, A)$ and stationary distribution π, that is

$$\mathrm{P}(X_{n+1} \in A \mid X_n = x) = p(x, A), \qquad \int p(x, A)\pi(\mathrm{d}x) = \pi(A)$$

and X_0 is π-distributed (one can show that $X_i \sim \pi$ for all $i \in \mathbb{N}$, using Markov-property).

(c) *Rotations of circle*: Take $\Omega = [0, 1)$ equiped with the Borel σ-algebra and P the Lebesgue measure. Fix $\theta \in S$ and $X_0(\omega) = \omega \sim \mathrm{P}$ and

$$X_{n+1}(\omega) = X_n(\omega) + \theta \mod 1$$

This is a Markov chain. It should be intuitively clear why this is is stationary.

(d) $\mathbf{1}_{\{k\cdot e_1 \text{ is a trifurcation}\}}$.

(e) The random variables τ_i given by (4.6) form a stationary sequence.

All the above instances are described by the following: Take $(\Omega, \mathcal{A}, \mathrm{P})$ a probability space and $\varphi : \Omega \to \Omega$ measurable and measure preserving, that is $P(\varphi^{-1}A) = P(A)$ for all $A \in \mathcal{A}$. Let $\varphi^n = \varphi \circ \varphi^{n-1}$ with $\varphi^0 = \mathrm{Id}$. Let X be a random variable. Set

$$X_n(\omega) = X(\varphi^n(\omega)). \tag{4.7}$$

Exercise 4.23. Show that X_n defined by (4.7) is stationary.

This encompasses all examples from Example 4.22 by taking $\Omega = S^{\mathbb{N}_0}$, $\mathcal{A} = \mathcal{S}^{\otimes \mathbb{N}_0}$, P the law of $(X_0, X_1, \dots)$ where $X_i(\omega) = \omega_i$ and let φ the shift to the left, that is

$$\varphi(\omega_0, \omega_1, \dots) = (\omega_1, \omega_2, \dots)$$

Definition 4.24. A set $A \in \mathcal{A}$ is invariant if $\mathrm{P}(A \triangle \varphi^{-1}(A)) = 0$ or in slightly sloppy notation: $\varphi^{-1}(A) \overset{\text{a.s.}}{=} A$. Define

$$\mathcal{I} := \{A \in \mathcal{A} \mid A \text{ is invariant}\},$$

which is a σ-algebra. A function $f : \Omega \to \mathbb{R}$ is invariant if $f \circ \varphi = f$ ($\Leftrightarrow$ f is $\mathcal{I}$ measurable). The system $(\Omega, \mathcal{A}, \mathrm{P}, \varphi)$ is called *ergodic* if $\mathcal{I}$ is P-trivial in the sense that $\mathrm{P}(A) \in \{0, 1\}$ for all $A \in \mathcal{I}$.

The concepts of ergodicity and irreducibility are related in the sense that if φ is not ergodic there exists A such that $\mathrm{P}(A) \in (0, 1)$ such that $\varphi(A) \subseteq A$ and $\varphi(A^{\complement}) \subseteq A^{\complement}$.

Example 4.25.

- Let X be an i.i.d. sequence, where $\Omega = S^{\mathbb{N}_0}$ and φ is the shift map. If A is an invariant set, then $A \in \bigcap_{n \geq 0} \sigma(X_n, X_{n+1}, \dots)$ - the tail σ-algebra so $(\Omega, \mathcal{A}, \mathrm{P}, \varphi)$ is ergodic by Kolmogorov's 0-1 law.
- Consider Example 4.22 (3). If $\theta = \frac{p}{q} \in \mathbb{Q}$, then φ is not ergodic because any set $\bigcup_{i=0}^{q-1} \varphi^i A$ is invariant. If θ is irrational, then φ is ergodic.

Exercise 4.26. Prove that the Markov-chain from Example 4.22 is ergodic if and only if its irreducible.

Theorem 4.27 (Birkhoff's pointwise Ergodic Theorem, 1931)**.** If $(\Omega, \mathcal{A}, \mathrm{P}, \varphi)$ is a measure preserving system (that is, φ is measure preserving) and $X : \Omega \to \mathbb{R}$ is in $L^1(\mathrm{P})$. Then

$$\frac{1}{n} \sum_{i=0}^{n-1} f((X \circ \varphi^n)(\omega)) \xrightarrow[n \to +\infty]{\text{a.s.}, L^1} \mathrm{E}[X \mid \mathcal{I}](\omega).$$

Remark.

- Note that if the system $(\Omega, \mathcal{A}, \mathrm{P}, \varphi)$ is ergodic, then $\mathrm{E}[X \mid \mathcal{I}] = \mathrm{E}[X]$ a.s.. This means that this result implies the strong law of large numbers.

- If $(X_n)_{n\in\mathbb{N}_0}$ is an irreducible Markov-chain started from π and $f \in L^1(\pi)$, then this theorem implies that

$$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \xrightarrow[n\to+\infty]{} \int_S f(x)\,\mathrm{d}\pi(x).$$

- If θ is irrational and $f = \mathbf{1}_A$ then

$$\frac{1}{n}\sum_{k=1}^{n} \mathbf{1}_A((\omega + k\theta) \bmod 1) \xrightarrow[n\to+\infty]{a.s.,L^1} \mathrm{P}(A)$$

  this is similar to Weyl's equidistribution theorem.

Proof (Not done in the lecture!) see Peterson 2000 (very simple proof of ergodic theorem, see Adam [todo:] □

Definition 4.28. The measure preserving map φ is *strongly mixing* if for any $A, B \in \mathcal{A}$

$$\lim_{n\to+\infty} \mathrm{P}(A \cap \varphi^{-n}(B)) = \mathrm{P}(A)\mathrm{P}(B).$$

Analoguous to the proof of Theorem 1.29 we get the following:

Theorem 4.29. If φ is strongly mixing, then $(\Omega, \mathcal{A}, \mathrm{P}, \varphi)$ is ergodic.

Proof. The proof is left as an exercise to the reader

Hint: Adapt the proof of Theorem 1.29 to show this. □

Lemma 4.30. If $\limsup_{n\to+\infty} X_n = +\infty$ P-a.s., that is if $\mathbb{E}[\log\rho_0] \leq 0$ then $(\tau_i)_{i\geq 1}$ is a stationary strong mixing (and thus ergodic system) under $\mathrm{P} = \mathrm{P}_0$ (the anihiled measure).

Proof. The *stationarity* follows directly from the stationarity of the random environment (ω is i.i.d.) and the Markov-property under P^ω.

We now show *strong mixing*: To this end, we start with "cylinder events",

$$A = \bigcap_{i=1}^{k}\{\tau_i \in A_i\}, \quad B = \bigcap_{i=1}^{\ell}\{\tau_i \in B_i\}, \quad A_i, B_i \subseteq \mathbb{N}.$$

We know that $\limsup X_n = +\infty$ so $\mathrm{P}(\tau_i > M) \searrow 0$ as $M \nearrow +\infty$. So for given $\varepsilon > 0$ we can fix $M = M(\varepsilon)$ such that

$$B^{m,M} = \bigcap_{i=1}^{\ell}\{\tau_{i+m} \in B_i^M\}, \quad \text{where } B_i^M := B_i \cap [0, M]$$

satisfies $\mathrm{P}\left(B^n \backslash B^{m,M}\right) \leq \varepsilon$. One has that $B^{m,M}$ depends on $\omega_{m-M}, \dots, \omega_{m+\ell-1}$ whereas A depends only on $\omega_{k-1}, \omega_{k-2}, \dots$ i.e., for $m \geq M+k$ the events $B^{m,M}$ and A depend on different ω_i's. Since we choose ω_i to be i.i.d. we have that $B^{m,M}$ and A are then idependent i.e.,

$$\mathrm{P}(A \cap B^{m,M}) = \mathrm{P}(A)\mathrm{P}(B^{m,M}) = \mathrm{P}(A)\mathrm{P}(B^{0,M})$$

this implies the lemma for cylindrical events. By Lemma 1.28 any event can be approximated by cylinders which implies the strong mixing. □

Lemma 4.31. If $\mathbb{E}[\log \rho_0] \leq 0$, then

$$\mathrm{E}^\omega(\tau_1) = \frac{1}{\omega_0} + \sum_{k=1}^{+\infty} \frac{1}{\omega_{-k}} \rho_{-k+1} \dots \rho_0 = 1 + 2\sum_{k=0}^{+\infty} \rho_{-k}\rho_{-k+1} \dots \rho_0. \tag{4.8}$$

Proof. Under the assumption that the following manipulations are ok, that is, the following expectations are finite, we have

$$\begin{aligned}\mathrm{E}_0^\omega(\tau_1) =& \mathrm{E}_0^\omega(H_1) = \omega_0 \cdot 1 + (1-\omega_0)(1 + \mathrm{E}_{-1}^\omega(H_1)) \\ =& 1 + (1-\omega_0)(\mathrm{E}_{-1}^\omega(H_0) + \mathrm{E}_0^\omega(H_1)),\end{aligned}$$

so

$$\mathrm{E}_0^\omega[\tau_1] = \frac{1}{\omega_0} + \rho_0 \mathrm{E}_{-1}^\omega[\tau_0].$$

Thus iterating the above procedure one obtains that

$$\mathrm{E}_0^\omega[\tau_1] = \frac{1}{\omega_0} + \sum_{k=1}^{n} \frac{1}{\omega_{-k}} \rho_{-k+1} \dots \rho_0 + \rho_{-n} \dots \rho_0 \mathrm{E}_{-n-1}^\omega[\tau_{-n}].$$

For both we truncate. Fix $M < +\infty$. Then,

$$\begin{aligned}\mathrm{E}_0^\omega[\tau_1 \wedge M] =& 1 + (1-\omega_0)\mathrm{E}_{-1}^\omega[(1+H_1) \wedge M] \\ \leq& 1 + (1-\omega_0)(\mathrm{E}_{-1}^\omega[\tau_0 \wedge M] + \mathrm{E}_0^\omega(\tau_1 \wedge M)) \\ &\vdots \\ \leq& \frac{1}{\omega_0} + \sum_{k=1}^{n} \frac{1}{\omega_{-k}} \rho_{-k+1} \dots \rho_0 + \rho_{-n} \dots \rho_0 \underbrace{\mathrm{E}_{-n-1}^\omega[\tau_{-n} \wedge M]}_{\leq M}.\end{aligned}$$

Case 1: $\lim_{n \to +\infty} \rho_{-n} \dots \rho_0 = 0$. This case is equivalent to saying that $\mathbb{E}[\log \rho_0] < 0$. In this case there is nothing more to do; we can just send $n \to +\infty$ and then $M \to +\infty$. For a lower bound we can just forget the last term $\rho_{-n} \dots \rho_0 \mathrm{E}_{-n-1}^\omega[\tau_{-n} \wedge M]$.

Case 2: The RHS of (4.8) is infinite. If the RHS of (4.8) is infinite, one can easily show that $\mathrm{E}_0^\omega(\tau_1) = +\infty$ by contradiction, using the above. This concludes the proof. □

Remark.
- For the proof of Theorem 4.19 we only consider the case $\limsup_{n\to+\infty} X_n = +\infty \iff \mathbb{E}[\log\rho_0] \le 0$.
- By Lemma 4.20, $\frac{X_n}{n} \to v \in [0,1]$ is equivalent to $\frac{H_n}{n} \to \frac{1}{v} \in [1,+\infty]$, where

$$H_n = \inf\{k \ge 0 \mid H_k = n\}.$$

- We wrote

$$H_n = \sum_{i=1}^n \tau_i, \quad \text{where } \tau_i = H_i - H_{i-1},\ H_0 \equiv 0.$$

The random variables τ_i are stationary and ergodic under P by Lemma 4.30. By Birkhoff's ergodic theorem 4.27,

$$\lim_{n\to+\infty} \frac{1}{n} H_n = \lim \frac{1}{n}\sum_{i=1}^n \tau_i = \mathrm{E}(\tau_1) \in [1,+\infty].$$

Proof of Theorem 4.19. We only consider the case $\limsup_{n\to+\infty} X_n = +\infty$ being equivalent to $\mathbb{E}[\log\rho_0] \le 0$ and we only need to check the formula for v. By Jensen we have

$$\log\mathbb{E}\left[\frac{1}{\rho_0}\right] \ge \mathbb{E}\left[\log\frac{1}{\rho_0}\right] = -\mathbb{E}[\log\rho_0] \ge 0$$

which implies that $\mathbb{E}\left[\frac{1}{\rho_0}\right] \ge 1$, so we only need to very the first two lines in Theorem (4.19). By Assumption 1 and previous remarks

$$\begin{aligned} v =& \frac{1}{\mathrm{E}_0(\tau_1)} = \frac{1}{\mathbb{E}[\mathrm{E}_0^\omega(\tau_1)]} \\ =& \left(1 + \mathbb{E}\left[2\sum_{k\ge0}\rho_{-k}\dots\rho_0\right]\right)^{-1} \\ =& \left(1 + \sum_{k\ge0} 2\mathbb{E}[\rho_0]^{k+1}\right) \\ =& \begin{cases} 0, & \text{if } \mathbb{E}[\rho_0] \ge 1 \text{ and} \\ \left(1 + 2\cdot\frac{\mathbb{E}[\rho_0]}{1-\mathbb{E}[\rho_0]}\right)^{-1} = \frac{1-\mathbb{E}[\rho_0]}{1+\mathbb{E}[\rho_0]}, & \text{if } \mathbb{E}[\rho_0] < 1, \end{cases} \end{aligned}$$

where we used Lemma 4.31 and the fact that ρ_i's are i.i.d. □

Remark.
- This law of large numbers also holds without the i.i.d. assumption on the environment. Ergodicity is sufficient but the formula for v is less explicit.
- One can prove more results:
  - averaged CLT: If $\mathbb{E}[\rho_0^2] < 1$, then $\frac{X_n - vn}{\sigma\sqrt{n}} \xrightarrow{\mathrm{P}} \mathcal{N}(0,1)$, where σ depends on $\mathbb{P}$.
  - quenched CLT: If $\mathbb{E}[\rho_0^2] < 1$, then $\frac{X_n - vn + Z_n(\omega)}{n\sqrt{n}} \xrightarrow{\mathrm{P}^\omega} \mathcal{N}(0,1)$.

4.4 Sinai's model: non standard limit laws

We now want to understand the situation when X is recurrent. Note that this section is about very advanced maths thus it will be higher level than usual in this lecture.

Assumption 2. We assume that $\mathbb{E}[\log \rho_0] = 0$, (**UE**), i.e., $\log \rho_0$ is bounded and that $(\omega_x)_{x\in\mathbb{Z}}$ is an i.i.d. sequence of random variables.

Recall that the harmonic function f given by (4.5). Here $V(x)$ is a sum of centered i.i.d. random variable, so one can use Donsker's theorem to understand its behaviour.

Claim 4.32. Let

$$W^n(t) = \frac{1}{\log n} V(\lfloor t \log^2 n \rfloor), \quad t \in \mathbb{R}.$$

Then

$$(W^n(t))_{t\in\mathbb{R}} \xrightarrow{d} (\sigma B_t)_{t\in\mathbb{R}}$$

where $\sigma^2 = \mathbb{E}[\log(\rho_0)^2]$ and $(B_t)_{t\in\mathbb{R}}$ is a two sided Brownian motion, that is $(B_t)_{t\geq 0}$ and $(B_{-t})_{t\geq 0}$ are two independent Brownian motion.

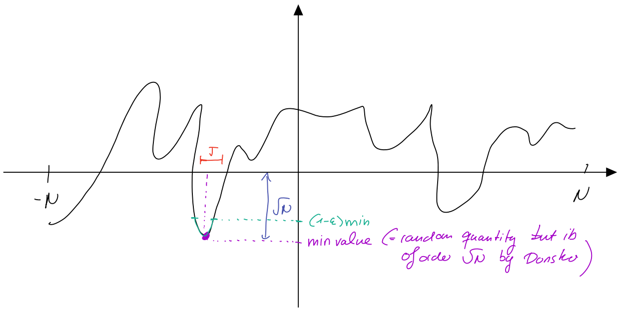
Proof. Use Donsker's theorem. $\square$

Remark. The role of the potential V is more visible when one studies the invariant measures of X under P^ω.

From the exercises we know that if the measure is invariant, then X is reversible. That is, there exists some function $\pi : \mathbb{Z} \to [0, +\infty)$ such that $\pi(x)\omega_x = \pi(x+1)(1-\omega_{x+1})$. Setting $\pi(0) = 1$ we get by iterating

$$\begin{aligned}\pi(x) &= \begin{cases} \prod_{i=0}^{x-1} \frac{\omega_i}{1-\omega_{i+1}}, & x>0 \\ 1, & x=0 \\ \prod_{i=x}^{-1} \frac{1-\omega_{i+1}}{\omega_i}, & x<0 \end{cases} \\ &= \begin{cases} \frac{1-\omega_0}{1-\omega_x} \cdot \prod_{i=0}^{x-1} \frac{1}{\rho_i} = \frac{1-\omega_0}{1-\omega_x} \mathrm{e}^{-V(x)}, & x>0 \\ 1, & x=0 \\ \frac{1-\omega_0}{1-\omega_x} \prod_{i=x}^{-1} \rho_i = \underbrace{\frac{1-\omega_0}{1-\omega_x}}_{\text{bounded}} \mathrm{e}^{-V(x)}, & x<0 \end{cases}.\end{aligned}$$

Imagine now that we restrict our random walk X to the interval $[-N, N]$ by setting $\omega_N = 0$, $\omega_{-N} = 1$.



If $x \notin J$ then $\frac{\pi(x)}{\pi(\min)} \cong \frac{\mathrm{e}^{-V(x)}}{\mathrm{e}^{-V(\min)}} \asymp \mathrm{e}^{-\varepsilon\sqrt{N}}$, thus

$$\frac{\pi(J^{\complement})}{\pi(J)} \leq 2N\mathrm{e}^{-\varepsilon\sqrt{N}} \xrightarrow{N\to+\infty} 0,$$

hence π is very far from being a unform distribution, that is X spends most of its time in J.

Definition 4.33. A triple (a, b, c) with $a < b < c$ is a *valley* of W^n if $W^n(b) = \min_{a\leq t\leq c} W^n(t)$ and

$$W^n(a) = \max_{a\leq t\leq b} W^n(t) \quad \text{and} \quad W^n(c) = \max_{b\leq t\leq c} W^n(t).$$

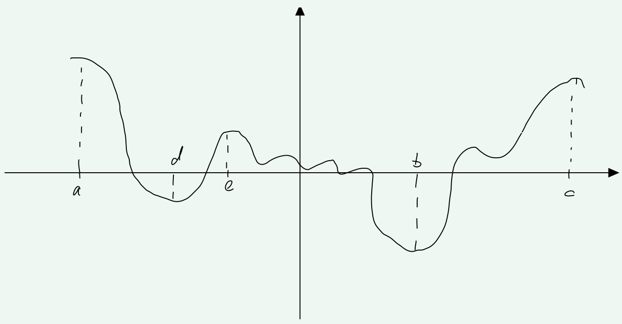
We denote the *depth* of the valley by

$$d(a, b, c) = \min\{W^n(a) - W^n(b), W^n(c) - W^n(b)\}.$$

If $a < d < e < b$ are such that

$$W^n(e) - W^n(d) = \max_{a\leq x<y\leq b} W^n(y) - W^n(x)$$

(the maximal increase), then (a, d, e) and (e, b, c) are again valleys. This procedure is called *left refinement* of (a, b, c).



Right refinement is defined analoguously.

By left/right refinement one splits the valey in two new valleys. We now construct a very "specific" valley. Let

$$\begin{aligned} c_0^n &= \min\{t \geq 0 : W^n(t) \geq 1\} \\ a_0^n &= \max\{t < 0 : W^n(t) \geq 1\} \\ b_0^n &= \text{the position of the global minimum of } W^n \text{ in } [a_0^n, c_0^n]. \end{aligned}$$

Note that b_0^n is not well defined as there might be multiple minima. However with probability tending to 1 as $n \to +\infty$ there is only one minimum, so we will ignore this case. By refining this valley one can construct the "smallest" valley $(a_n^\delta, b_n^\delta, c_n^\delta)$ such that $d(a_n^\delta, b_n^\delta, c_n^\delta) \geq 1 + \delta$ and the origin is contained in the valley. For $\delta = 0$ we just write $(a_n^\delta, b_n^\delta, c_n^\delta) = (a_n, b_n, c_n)$.

Theorem 4.34 (Sinai, 1982)**.** It holds that

$$\frac{X_n}{\log^2 n} - b_n \xrightarrow[n\to+\infty]{\mathrm{P}_0} 0.$$

Remark.
- As b_n is of order 1, typically, this implies that $|X_n| \approx \log^2 n$. Compare it with the usual symmetric random walk with $|X_n| \approx \sqrt{n}$.
- 4.33[todo: check this] implies that given ω, X_n is "essentially deterministic" and located (with high probability) at $b_n \cdot \log^2 n + o(\log^2 n)$.

Proof. We will only sketch the proof as it is quite long. Let $(a_n, b_n, c_n) = (a_n^1, b_n^1, c_n^1)$ denote the smallest valley such that $d(a_n, b_n, c_n) \geq 1$ such that $a_n < 0 < c_n$.

Step 1 (a_n, b_n, c_n) is with high probability (w.h.p.) non-degenerate. We first need to define the collection of ω's such that (a, b, c) is non-degenerate. To this end, let $A_n^{J,\delta}$ be the collection of all $\omega \in \Omega$ such that

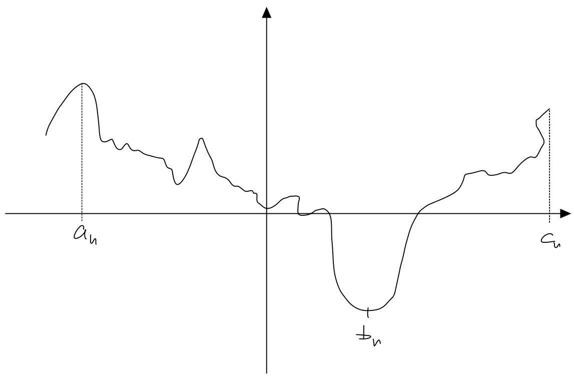
- $b_n = b_n^\delta$
- any refinement of $(a_n^\delta, b_n^\delta, c_n^\delta)$ has depth $< 1 - \delta$.
- $|a_n^\delta| + |c_n^\delta| < J$ and
- we have
$$\min_{t\in[a_n,c_n]\setminus[b_n-\delta,b_n+\delta]} W^n(t) - W^n(b_n) \geq \delta^3.$$

Using the properties of Brownian motion (actually one needs to work quite a bit to achieve that, using reflection principle etc.) one checks that

$$\lim_{\delta\searrow 0} \lim_{J\to+\infty} \lim_{n\to+\infty} \mathrm{P}(A_n^{J,\delta}) = 1.$$

This shows the first step.

For simplicity from now on we assume that $A_n^{J,\delta}$ holds and $b_n \geq 0$.



We set $A_n^\delta = \lfloor (\log^2 n) a_n^\delta \rfloor$, $B_n^\delta = \lfloor (\log^2 n) b_n^\delta \rfloor$, $C_n^\delta = \lfloor (\log^2 n) c_n^\delta \rfloor$.

Step 2 $\mathrm{P}_0^\omega[H_{B_n} \le n] \xrightarrow{n \to +\infty} 1$. Since f is harmonic, we have

$$\mathrm{P}_0^\omega(H_{A_n^\delta} \le H_{B_n}) = \frac{f(B_n) - f(0)}{f(B_n) - f(A_n^\delta)} = \frac{\sum_{x=1}^{B_n} \mathrm{e}^{V(x)}}{\sum_{x=A+1}^{B_n} \mathrm{e}^{V(x)}}.$$

Due to the second property of $A_n^{J,\delta}$ the sum in the denominator contains a term $\mathrm{e}^{V(x_0)}$ such that $V(x_0) \ge \delta \log(n) + V(y)$ for all $y \in [1, B_n]$. Hence

$$\mathrm{P}_0^\omega(H_{A_n^\delta} \le H_{B_n}) \le \frac{\overbrace{|B_n|}^{\le J \log^2 n} \ \max_{y \in [1, B_n]} \mathrm{e}^{V(y)}}{\mathrm{e}^{V(x_0)}} \le J \log^2(n) \mathrm{e}^{-\delta \log n} \le (J \log^2 n) n^{-\delta} \xrightarrow{n \to +\infty} 0. \tag{4.9}$$

Consider now RWRE "reflected at A_n^δ". Then as in Lemma 4.31,

$$\mathrm{E}_0^\omega[\tau_1] = \frac{1}{\omega_0} + \cdots + \frac{1}{\omega_{A_n^\delta + 1}} \rho_{A_n^\delta + 2} \cdots \rho_0 + \rho_{A_n^\delta + 1} \cdots \rho_0.$$

Since the reflection does not change the law of $H_{\{A_n^\delta, B-n\}} := \overline{H_n}$,

$$\begin{aligned}
\mathrm{E}_0^\omega[\overline{H_n}] = & \overbrace{\widetilde{\mathrm{E}_0^\omega}}^{\text{reflection}} [\overline{H_n}] \le \widetilde{\mathrm{E}_0^\omega}[H_{B_n}] = \widetilde{\mathrm{E}_0^\omega}\left[\sum_{i=1}^{B_n} \tau_i\right] \\
= & \sum_{i=1}^{B_n} \sum_{x=A_n^\delta}^{i-1} \underbrace{\frac{1}{\omega_x}}_{\le c} \underbrace{\rho_{x+1} \cdots \rho_{i-1}}_{\mathrm{e}^{V(i) - V(x)}} \\
\le & c \sum_{i=1}^{B_n} \sum_{x=A_n^\delta}^{i-1} \mathrm{e}^{\log n \left(W^n\left(\frac{i}{\log^2 n}\right) - W^n\left(\frac{x}{\log^2 n}\right)\right)} \\
\le & C (J \log^2 n)^2 n^{(1-\delta)} \le n^{1 - \frac{\delta}{2}},
\end{aligned}$$

where we used that $W^n\left(\frac{i}{\log^2 n}\right) - W^n\left(\frac{x}{\log^2 n}\right) \le 1 - \delta$ due to the second property of $A_n^{J,\delta}$. By Markov inequality,

$$\mathrm{P}(\overline{H_n} > n) \xrightarrow[n \to +\infty]{} 0, \tag{4.10}$$

hence combining (4.9) and (4.10) one deduces

$$\mathrm{P}(\overline{H_n} < n, X_{H_n} = B_n) \xrightarrow[n\to+\infty]{} 1,$$

proving the claim.

Step 3 . Similarly to the proof of (4.9) one proves

$$\begin{aligned}
\mathrm{P}^{\omega}_{B_n-1}[H_{B_n} < H_{A_n^\delta}] \geq& 1 - n^{-\left(1+\frac{\delta}{2}\right)},\\
\mathrm{P}^{\omega}_{B_n+1}[H_{B_n} < H_{C_n^\delta}] \geq& 1 - n^{-\left(1+\frac{\delta}{2}\right)},
\end{aligned}$$

that is the probability of how likely it is to reach the bottom of the valley before leaving the valley. We ommit this proof here as it is quite similar to the proof of (4.9).

Step 4 Conclusion. Steps 2 and 3 imply that on $A_n^{J,\delta}$ with high probability,

- X reaches B_n before time n
- After reaching B_n, X will need at least $n^{1+\frac{\delta}{2}}$ steps to exit $[A_n^\delta, C_n^\delta]$.

So instead of looking at X we may look at its reversion that is refelcted at A_n^δ and C_n^δ (i.e., $\omega_{A_n^\delta} = 1$ and $\omega_{C_n^\delta} = 0$). We use $\tilde{\mathrm{E}}^\omega, \tilde{\mathrm{P}}^\omega$ for this reversion. Its advantage is that it is a "finite state space" Markov chain, which is reversible. □

5 Random Walks on Weighted Graphs

In the following few section we will be following the book of Barlow [todo: Literature]

This field has a lot of similarities with the analysis of parabolic PDEs. It is easier than RWRE so we will also cover $d \geq 2$. This will be the last big part of the lecture.

Example 5.1. Consider the random walk among random conductivity, that is let $G = (\mathbb{Z}^d, E_d)$ and let $\mu = (\mu_e)_{e \in E_d}$ the conductances which we assume to be independent under $\mathbb{P}$. We assume ellipticity, that is there exists some $\varepsilon > 0$ such that

$$\mathbb{P}\left(\varepsilon \leq \mu_e \leq \frac{1}{\varepsilon}\right) = 1 \quad \text{for all } e \in E_d.$$

Consider a random walk X such that, given μ,

$$P^\mu(x, y) = \mathrm{P}^\mu(X_1 = y \mid X_0 = x) = \frac{\mu_{x,y}}{\sum_{z \sim x} \mu_{x,z}}.$$

This is always reversible.

Fact. The random walk in random environment in the recurrent situation grows really slowly,

$$\frac{X_n}{\log^2 n} \sim b_n$$

RWRC is always recurrent and

$$\frac{X_{nt}}{\sqrt{n}} \to B_t,$$

where B is a Brownian motion. Compared to RWRE where we had $\pi(x) \asymp \mathrm{e}^{V(x)}$ and $V(x)$ is a random walk again, the invariant measure of RWRC is $\pi(x) = \mu_{x,x+1} + \mu_{x,x-1}$.

Notation 1. The notation will at some times be slightly different than in the previous lectures. Here $G = (V, E)$ will denote a graph (which will often be infinite) and we will write $x \sim y$ iff $\{x, y\} \in E$. Here we will use the word path for a finite squence $x_0, \ldots, x_n \in V$ with $x_i \sim x_{i-1}$ (note that a path does not need to be self avoiding here!). Moreover we define the degree of a vertex $v \in V$ by

$$\deg(v) := \#\{z \in V \mid z \sim x\}.$$

The ball of radius $r > 0$ with center $x \in V$ is given by

$$B_r(x) = \{y \in V \mid d(x,y) \leq n\}$$

where

$$d(x,y) = \inf\{n = \text{length}(\pi) \mid \pi = x_0, \dots x_n \text{ is a path from } x \text{ to } y\}$$

We say that G has *bounded geometry* iff y

$$\sup\{\deg(x) \mid x \in V\} < +\infty.$$

Assumption. We assume that our graph $G = (V,E)$ is *locally finite*.

Definition 5.2. A weight (conductance) is a function $\mu : V \times V \to \mathbb{R}_{\geq 0}$, $(x,y) \mapsto \mu_{x,y}$ with the following properties:

- μ is symmetric, that is $\mu_{x,y} = \mu_{y,x}$.
- If $x \neq y$, $\mu_{x,y} > 0$ if and only if $x \sim y$, that is $\{x,y\} \in E$.

Moreoever we define

$$\pi(x) := \sum_{z \in V} \mu_{x,z} = \sum_{z \sim x} \mu_{x,z} + \mu_{x,x}.$$

We call the pair (G,μ) *weighted graph*.

Remark. A natural choice for a weight would be $\mu_{x,y} = \mathbf{1}_{x \sim y}$. If we do not specify a weight function we implicitely use this.

Definition 5.3. A weighted graph (G,μ) has bounded weights if there exists $C \in \mathbb{R}_{\geq 0}$ such that

$$\text{for any } e \in E, \quad \mu_e \in [C^{-1}, C].$$

We say that (G,μ) has controlled weights if there exists $C > 0$ such that

$$\text{for any } x \sim y \quad \frac{\mu_{x,y}}{\pi(x)} \geq C.$$

At last we define $P(x,y) = \frac{\mu_{x,y}}{\pi(x)}$, where $P(x,y) > 0$ implies that either $x \sim y$ or $x = y$.

Exercise 5.4. If we have controlled weights then G has bounded geometry.

Let $(X_n)_{n \in \mathbb{N}_0}$ be a random walk on (G,μ) with transition matrix P (it is only a matrix if G is finite). Moreover we denote by P_x the law of X given $X_0 = x$.

Lemma 5.5.
- X is reversible and π is its reversible measure.
- For any $x_0, \dots, x_n \in V$ we have

$$\pi(x_0)\mathrm{P}_{x_0}(X_1 = x_1, \dots, X_n = x_n) = \pi(x_n)\mathrm{P}_{x_n}(X_1 = x_n, \dots, X_n = x_0).$$

Proof. The easy proof is left as an exercise to the reader. □

Remark. Every reversible chain is RWRC.

[todo: 16.7]

We know that the Random Walk on $\mathbb{Z}^d$ is recurrent if and only if $d \leq 2$.

Question 5.6.
- Is RW on a triangular/hexagonal lattice in $d = 2$ recurrent?
- Is RW on $(\mathbb{Z}^d, \mu)$ where μ are bounded weights recurrent if $d \leq 2$?
- Do the above Random Walks have a Brownian motion limit?

We will look at the so-called *heat kernel bound*:

$$p_n(x, y) \asymp \frac{c_1}{n^{d/2}} \mathrm{e}^{-c_2 \frac{|x-y|^2}{n}}.$$

We define the *transition densities* by

$$p_n(x, y) = \frac{1}{\pi(y)} P^n(x, y) = \frac{1}{\pi(y)} \mathrm{P}_x(X_n = y).$$

Lemma 5.7.
- The Chapman Kolmogorov-equation holds:

$$p_{n+m}(x, y) = \sum_{z \in V} p_n(x, z) p_n(z, y) \pi(z) = \int_V p_n(x, z) p_n(z, y) \pi(\mathrm{d}z).$$

- It holds that $p_n(x, y) = p_n(y, x)$ for any $n \in \mathbb{N}_0$ and $x, y \in V$.
- The p_n's are normalized in the sense that

$$\int_V p_n(x, z) \pi(\mathrm{d}z) = 1.$$

Definition 5.8. Denote $C(V) = \{f : V \to \mathbb{R}\}$ (C because all functions $V \to \mathbb{R}$ are continuous if we equip V with the discrete topology). Denote $C_0(V)$ the set of all functions with finite support. and denote $L^p = L^p(V, \pi)$. As usual L^2 is a Hilbert space and we denote

$$\langle f, g \rangle := \langle f, g \rangle_\pi := \langle f, g \rangle_{L^2(\pi)} = \sum_{x \in V} f(x) g(x) \pi(x).$$

The transition densities can be viewed as an operator given by

$$P_n f(x) = \sum_{y \in V} P^n(x,y) f(y) = \mathrm{E}_x[f(X_n)].$$

Definition 5.9. We define the *probabilistic Laplace operator* by $\Delta = P - \mathrm{Id}$ i.e.,

$$\begin{aligned}\Delta f(x) =& (Pf)(x) - f(x) = \mathrm{E}_x[f(X_1)] - f(x) \\ =& \sum_{y \in V} p(x,y)(f(y) - f(x))\pi(y) \\ =& \frac{1}{\pi(x)} \sum_{y \in V} \mu_{x,y}(f(x) - f(y)).\end{aligned}$$

This is the "generator" of the Markov-chain X.

We have the following elementary properties of P and Δ whose proof are left as an exercise to ther reader.

Lemma 5.10. (a) The function $f \equiv 1$ is an eigenfunction of P i.e., $\Delta 1 = 0$.

(b) P and Δ are self-adjoint in $L^2(\pi)$ that is for any $f, g \in L^2$,

$$\langle f, Pg \rangle = \langle Pf, g \rangle, \quad \langle f, \Delta g \rangle = \langle \Delta f, g \rangle,$$

(c) The operator-norms of P and Δ satisfy

$$\|P\|_{L^p \to L^p} \leq 1, \quad \|\Delta\|_{L^p \to L^p} \leq 2.$$

5.1 Dirichlet (Energy)-form

Definition 5.11. Let $f, g \in C(V)$. The quadratic form

$$\mathcal{E}(f,g) = \frac{1}{2} \sum_{x,y \in V} \mu_{x,y}(f(x) - f(y))(g(x) - g(y)) \tag{5.1}$$

is defined if the above series is absolutely convergent and in this case we call $\mathcal{E}$ the *Dirichlet (Energy)-form.*

Remark. To see why it is called Dirichlet-Energy, compare it with the expression

$$\int_\Omega \nabla f(x) \cdot \nabla g(x) \, \mathrm{d}x.$$

where the discrete gradients are given by $f(x) - f(x-1)$. We will see later why we named it that way. At this point in time the name is random.

Definition 5.12. We define the space

$$H^2 = H^2(G, \mu) = \{f \in C(V) \mid f \text{ is such that (5.1) converges absolutely}\}.$$

The norm on H^2 is given by

$$\|f\|_{H^2}^2 = \mathcal{E}(f, f) + f(o)^2,$$

where o is a base point of the graph (like an arbitrarily fixed origin).

Lemma 5.13. (a) If $x \sim y$ we have

$$|f(x) - f(y)| \leq \sqrt{\frac{\mathcal{E}(f, f)}{\mu_{x,y}}}.$$

(b) $\mathcal{E}(f, f) = 0$ if and only if $f \equiv \text{const}$.

(c) For $f \in L^2$ we have that $\mathcal{E}(f, f) \leq 2\|f\|_{L^2}^2$, so $L^2 \subseteq H^2$.

(d) The space H^2 is a Hilbert space with the scalar product

$$\langle f, g\rangle_{H^2} = \mathcal{E}(f, g) + f(o)g(o).$$

(e) Convergence in H^2 implies convergence pointwise convergene *everywhere*.

Proof. We have that

$$\mathcal{E}(f, f) = \frac{1}{2}\sum_{x,y} \mu_{x,y}(f(x) - f(y))^2 \leq 2\sum_x f(x)^2 \sum_y \mu_{x,y} = 2\|f\|_{L^2}^2,$$

which shows (c). The rest of the points are left as an exercise to the reader. $\square$

Remark. Point (b) in Lemma 5.13 explains why we needed to add $f^2(o)$ in the definition of $\|\cdot\|_{H^2}^2$.

In the following we will establish an analogon of the famous Green-Gauss theorem from analysis.

Theorem 5.14 (Green, Gauss)**.** If $f, g \in C(V)$ are such that

$$\sum_{x,y \in V} |g(x)| \cdot |f(x) - f(y)|\mu_{x,y} < +\infty, \tag{5.2}$$

then

$$\mathcal{E}(f,g) = -\langle \Delta f, g\rangle_{L^2(\pi)}.$$

Proof. Note that (5.2) ensures that all the series used in the following proof converge absolutely. Using that $\mu_{x,y} = \mu_{y,x}$

$$\begin{aligned}
\mathcal{E}(f,g) =& \frac{1}{2}\sum_{x,y}(f(y)-f(x))g(y)\mu_{x,y} - \frac{1}{2}\sum_{x,y}(f(y)-f(x))g(x)\mu_{x,y} \\
=& \frac{1}{2}\sum_{x,y}(f(x)-f(y))g(x)\mu_{x,y} - \frac{1}{2}\sum_{x,y}(f(y)-f(x))g(x)\mu_{x,y} \\
=& -\sum_{x} g(x) \underbrace{\sum_{y}(f(y)-f(x))\frac{\mu_{x,y}}{\pi(x)}}_{=\Delta f}\pi(x) \\
=& -\langle g, \Delta f\rangle_{\pi}.
\end{aligned}$$

□

Proposition 5.15. Suppose that $\pi(V) < +\infty$. Then the random walk X on the weighted graph (G,μ) is recurrent.

Proof. Fix $z \in G$ and define $\varphi(x) := \mathrm{P}_x(H_z = +\infty)$. Note that φ is bounded and since $\pi(V) < +\infty$ $\varphi \in L^2 \subseteq H^2$. For $x \neq z$,

$$\varphi(x) = \sum_{y\in V}\mathrm{P}_x(X_n = y, H_z = +\infty) = \sum_{y\in V} p(x,y)\varphi(y),$$

that is, φ is harmonic on $V\setminus\{z\}$ with $\varphi(z) = 0$. Since φ is bounded we can apply discrete Green Gauss Theorem ([todo: todo numbering]):

$$\mathcal{E}(\varphi,\varphi) = -\langle\varphi,\Delta\varphi\rangle_\pi = \underbrace{\varphi(z)}_{=0}\Delta\varphi(z)\pi(z) = 0$$

which implies by (**??**) that φ must be constant, i.e., $\varphi \equiv 0$. □

Killed process. Given a graph $G = (V,E)$ with weights μ let $A \subseteq V$. We set

$$T_A = \inf\{n \in \mathbb{N}_0 \mid X_n \notin A\}$$

the exit time of A and

$$p_n^A(x,y) = \frac{1}{\pi(y)}\mathrm{P}_x(X_n = y, T_A > n)$$

the transition density of X killed on exiting A. We define the following operators:

$$\begin{aligned}
\mathrm{Id}_A f(x) =& f(x)\mathbf{1}_A(x) \\
P_n^A f(x) =& \sum_{y\in A} p_n^A(x,y)\pi(y)f(y) \overset{\text{if } x\in A}{=} \mathrm{E}_x[f(X_1)\mathbf{1}_{T_A>1}], \quad n\geq 1 \\
\Delta_A =& P^A - \mathrm{Id}_A\,.
\end{aligned}$$

Remark. In general P^A is a "sub-Markovian" operator, i.e., Mass can disappear for example when X hits A.

P^A has many properties similar to P.

Lemma 5.16. The following statements hold:

1. $p_n^A(x,y) = 0$ if $x \notin A$ or $y \notin A$ and $p_n^A(x,y) = p_1(y,x)$ for all $x, y \in A$.
2. We have that

$$\begin{aligned} p_{n+m}^A(x,y) &= \sum_{z \in V \text{ or } z \in A} p_n^A(x,z) p_n^A(z,y) \pi(z) \\ p_{n+1}^A(x,y) - p_n^A(x,y) &= \Delta_y p_n^A(x,y), \end{aligned}$$

the discrete heat equation (second line).

3. $P_n^A = (P^A)^{\circ n}$ and $P_A f = \mathrm{Id}_A P \,\mathrm{Id}_A f$, $\Delta_A f = \mathrm{Id}_A \Delta \,\mathrm{Id}_A f$.

Proof. Similar to to the case where $A = V$. $\square$

Green's function. Analoguous to Stochastic Analysis we define Green's function.

Definition 5.17. The function $g_A : V \times V \to \mathbb{R}_{\geq 0} \cup \{+\infty\}$ given by

$$g_A(x,y) = \sum_{n=0}^{+\infty} p_n^A(x,y)$$

is called *Green's function* or *Green's density*. As a convention: $g := g_V$.

We record some elementary properties of Green's function whose proofs are left as an exercise to the reader.

- g_A is symmetric, that is $g_A(x,y) = g_A(y,x)$ for all $x, y \in V$.
- We have
$$g_A(x,y) = \frac{1}{\pi(y)} \mathrm{E}_x \left[\sum_{n=0}^{+\infty} \mathbf{1}_{X_n = y, n < T_A} \right]$$
where the sum in the expectation is the number of visits to y before exiting A.
- If (G,μ) is recurrent, $g(x,y) = +\infty$ for all $x, y \in V$. If (G,μ) is transient, $g(x,y) < +\infty$ for all $x, y \in V$.
- If $A \neq V$, then $g_A(x,y) < +\infty$.

Next is a useful idendity for g_A

Theorem 5.18. Suppose that either $A \neq V$ or X is transient, that is, $g(x,y) < +\infty$ for all $x, y \in V$. Then

1. $g_A(x,y) = \mathrm{P}_x(H_y < T_A) g_A(y,y)$ for all $x, y \in V$ and
2. $\pi(y) g_A(y,y) = \frac{1}{\mathrm{P}_y(T_A < \tilde{H}_y)} = \frac{1}{\mathrm{P}_y(T_A \leq \tilde{H}_y)}$ for $y \in A$.

Proof. [todo: See notes] □

Exercise 5.19. The formulas of the above theorem are very intuitive, explain why.

Theorem 5.20. If $x, y \in A$ then

$$\Delta_y g_A(x,y) = -\frac{\delta_{x,y}}{\pi(x)},$$

where δ denotes the Kronecker-Delta and Δ_y denotes the Laplacian applied in the y-argument of the function.

Proof. Note that

$$\begin{aligned} P g_A(x,y) &= \sum_{z \in V} \overbrace{p(y,z)\pi(z)}^{\mathrm{P}_y(X_1=z)} g_A(x,z) \\ &= \sum_{z \in V} \sum_{n \in \mathbb{N}_0} p(y,z)\pi(z) p_n^A(x,z) \\ &\overset{!}{=} \sum_{n \in \mathbb{N}_0} p_{n+1}^A(x,y) \\ &= g_A(x,y) - p_0^A(x,y) = g_A(x,y) - \frac{\delta_{x,y}}{\pi(y)}, \end{aligned}$$

where we used at ! that every term in the above sums is positive, that is we can use monotone convergence to swap sums and also the Chapmann-Kolmogorov-equality. The claim follows by using $\Delta = P - \mathrm{Id}$. □

Definition 5.21. Let $A \subseteq V$ and $f : \bar{A} = A \cup \partial A \to \mathbb{R}$. f is *harmonic* on A if $\Delta f(x) = 0$ for all $x \in A$. We call f *superharmonic* if $-\Delta f(x) \geq 0$ for all $x \in V$ and subharmonic if $-\Delta f(x) \leq 0$ for all $x \in V$.

Theorem 5.22 (Maximum principle)**.** Let $A \subseteq V$ be connected and $h : \bar{A} \to \mathbb{R}$ subharmonic on A.

(1) If there exists some $x \in A$ such that $h(x) = \max_{\bar{A}} h$ then h is constant.

(2) If A is finite then h attains its maximum on $\bar{A}$ and the maximum is attained on the boundary ∂A.

Proof. *(1):* Let x be the value where the maximum is attained. Let $B = \{y \in V \mid h(y) = h(x)\}$. Then for $y \in B \cap A$ we have for all $z \sim y$ that $h(z) \leq h(y)$. Also,

$$\begin{aligned} 0 \leq & \pi(y)\Delta(y) \\ = & \sum_{z \sim y} \mu_{y,z} \underbrace{(h(z) - h(y))}_{\leq 0} \leq 0 \end{aligned}$$

that is $h(z) = h(y)$ for all $z \sim y$. This means we can repeat the above argument with all neighbouring points. Since A is connected, h must be constant on A, that is $B = A$.

(2): [todo:] □

Theorem 5.23. Assume that (G, μ) is recurrent. If f is a nonnegative function that is superharmonic on V, then f is constant.

Proof. Since f is superharmonic and nonnegative, $M_n = f(X_n)$ is a non-negative supermartingale, hence the limit

$$\lim_{n \to +\infty} f(X_n) \tag{5.3}$$

exists and is finite P_x-a.s. On the other hand X visits every point infinitely often P_x-a.s. so f must be constant (otherwise the limit (5.3) does not exist). □

Theorem 5.24 (Forster's criterion)**.** (G, μ) is recurrent if and only if there exists some $A \subseteq V$ finite and $h : V \to [0, +\infty)$ such that h is superharmonic on $V \setminus A$ and for any $M < +\infty$ the sublevel set

$$\{x \in V \mid h(x) \leq M\}$$

is finite.

Proof. $\Longleftarrow$: Suppose we have found such A and h as in the theorem. Then $M_n = h(X_{n \wedge H_A})$ is a non-negative supermartingale, hence the limit $\lim_{n \to +\infty} M_n \in [0, +\infty)$ exists P_x-a.s. for all $x \in V$. Suppose by contradiction that X is transient. Then there exists $x \in V$ such that $\mathrm{P}_x(H_A < +\infty) < 1$. But then $\mathrm{P}_x(\lim M_n = +\infty) > 0$ because the sublevel set of h is finite (since you hit every point a.s. finitely many times the limit has to be infinite).

$\Longrightarrow$: Let $B_n = B(A, n)$ the ball of size n around A. Let

$$h_n(x) = \mathrm{P}_x(T_{B_n} < H_A).$$

Then h is superharmonic on $V \setminus A$ and $h \equiv 1$ on $B_n^{\complement}$ and $h_n(x) \searrow 0$ as $n \to +\infty$ by recurrence (exercise: fill in the details). Let h_{n_k} be a subsequence such that $h_{n_k}(x) \leq 2^{-k}$ for all $x \in B_k$. Let

$$h = \sum_{k \geq 0} h_{n_k}.$$

Then h is superharmonic on $V \setminus A$ and $h \geq 0$ and $h(x) < +\infty$ for all $x \in V$, $h(x) \geq M$ for all $x \in B_{n_k}^{\complement}$, that is h has the required properties. $\square$

6 Random walks and electrical resistance

Note: This section is based on the lecture notes of Barlow; see ADAM. [todo: Bild 29]

Electrical Current. We have Ohm's law, that is there exists some $U : V \to \mathbb{R}$ such that for any x, y, the resistance of $\{x, y\}$ is given by

$$U(y) - U(x) = \frac{I_{x,y}}{\mu_{x,y}}$$

(by reversing the electrical flow, replacing $I_{x,y}$ with $-I_{x,y}$ is possible, this is just a convention). We further assume Kirchhoff's law (marked with '!'),

$$\sum_{y \in V} I_{x,y} = \sum_{x \sim y} I_{x,y} \overset{!}{=} 0 \quad \text{for all } x \notin \{x_1, x_0\}.$$

Note that Ohm's law gives

$$0 = \sum_y I_{x,y} \overset{\text{Ohm}}{=} \sum_y \mu_{x,y}(U(y) - U(x)) = \pi(x)\Delta U(x),$$

i.e., U is harmonic. We want to solve the following discrete *Dirichlet problem*:

$$\begin{cases} \Delta U(x) &= 0 \quad \text{on } \{x_0, x_1\}^{\complement} \\ U(x_0) &= 0, \quad U(x_1) = 1 \end{cases} . \tag{6.1}$$

Lemma 6.1. If the graph G is finite, (6.1) has a unique solution given by

$$U(x) = \mathrm{P}_x(H_{x_1} < H_{x_0}).$$

Remark. By finiteness of G one could use linear algebra to prove this. We will, however, use a different proof.

Proof. Set $\varphi(x) = \mathrm{P}_x(H_{x_1} < H_{x_0})$. Then $\varphi(x_i) = i$ for $i = 0, 1$. Also by (**??**) we have that φ is harmonic on $\{x_1, x_0\}^{\complement}$. Suppose that U_1, U_2 are solutions to (6.1). Then $u = U_1 - U_2$ satisfies (6.1) with homogenous boundary conditions. Using discrete Green Gauss (Theorem 5.14, which we can use because we are on a *finite* graph) we have

$$\mathcal{E}(u, u) = -\langle u, \Delta u \rangle = \sum_{x \in V} \pi(x) u(x) \Delta u(x)$$

$$= \sum_{x \in V \setminus \{x_0, x_1\}} \pi(x) u(x) \Delta u(x) + \sum_{x \in \{x_0, x_1\}} \pi(x) u(x) \Delta u(x)$$
$$= 0,$$

which implies that u is constant. Using the initial condition $u(x_0) = 0$ we obtain $u \equiv 0$. □

Remark. If (G, μ) is transient, the function

$$h^B(x) = \mathrm{P}_x(H_B = +\infty) \not\equiv 0, \quad \text{where } B = \{x_0, x_1\}.$$

Also h^B is harmonic on $B^\complement$ and $h^B(x_0) = h^B(x_1) = 0$, that is $\varphi + \lambda h^B$ solves (6.1) for all $\lambda \in \mathbb{R}$, so we cannot guarantee the uniqueness of a solution of (6.1).

Definition 6.2. Define

$$F(I, x_0) := \sum_{y \sim x_0} I_{x_0, y} = \sum_{y \in V} I_{x_0, y}$$

the *total current from* x_0. This is also the total current in the circuit. Also define the *effective resistance between* x_0 *and* x_1 by

$$R_{\text{eff}}(x_0, x_1) = R(x_0, x_1) = \frac{U}{I} = \frac{1}{F(I, x_0)}$$

and

$$C_{\text{eff}}(x_0, x_1) = C(x_0, x_1) = \frac{1}{R_{\text{eff}}(x_0, x_1)} = F(I, x_0).$$

Exercise 6.3.
- Prove that if G is finite $F(I, x_0) = -F(I, x_1)$.
- Show that $R_{\text{eff}}(x_0, x_1) = R_{\text{eff}}(x_1, x_0)$ that is resistance is independent of direction (obivous from a physical perspective).

Theorem 6.4. Suppose that G is finite. For any $x_0, x_1 \in V$, $x_1 \neq x_0$ it holds that

$$\pi(x_0) \underbrace{\mathrm{P}_{x_0}(H_{x_1} < \tilde{H}_{x_0})}_{\text{“escape probability”}} = C_{\text{eff}}(x_0, x_1).$$

Proof. Can be checked by next calculation.

$$C_{\text{eff}}(x_0, x_1) = \sum_{y \in V} I_{x_0, y} \overset{\text{Ohm}}{=} \sum_y \mu_{x_0, y}(U(y) - U(x_0))$$
$$\overset{\text{Lemma}}{=} \pi(x_0) \sum_y \underbrace{\frac{\mu_{x_0, y}}{\pi(x_0)}}_{= P(x_0, y)} \mathrm{P}_y(H_{x_1} < H_{x_0}) \overset{\text{Markov}-\text{property}}{=} \pi(x_0) \mathrm{P}_{x_0}(H_{x_1} < \tilde{H}_{x_0}). \quad □$$

Remark. This will be helpful because C_{eff} is monotone but it is a priori very hard to make any claims about monotonicity of the probability.

Theorem 6.5 (Commute time idendity)**.** Consider G finite and $x_0 \neq x_1$. Then the commute time (LHS) satisfies

$$\mathrm{E}_{x_0}[H_{x_1}] + \mathrm{E}_{x_1}[H_{x_0}] = R_{\text{eff}}(x_0, x_1)\pi(V).$$

Corollary 6.6. Under the assumptions of Theorem 6.5 the map $(x, y) \mapsto R_{\text{eff}}(x, y)$ is a metric on V, if one defines $R_{\text{eff}}(x, x) = 0$.

Proof. Clearly we have $R_{\text{eff}}(x, x) = 0$ by definition (exercise: prove that $R_{\text{eff}}(x, y) = 0$ implies $x = y$). Symmetry follows from the above exercise. Lastly, for x, y, z we have

$$\begin{aligned} \mathrm{E}_x[H_z] &\leq \mathrm{E}_x[H_y] + \mathrm{E}_y[H_z] \\ \mathrm{E}_z[H_x] &\leq \mathrm{E}_y[H_x] + \mathrm{E}_z[H_y]. \end{aligned}$$

Summing these two equations and using Theorem 6.5 yields

$$R_{\text{eff}}(x, z) \leq R_{\text{eff}}(x, y) + R_{\text{eff}}(y, z).$$

□

Proof of Theorem 6.5. Set $R = R_{\text{eff}}(x_0, x_1)$ and

$$\varphi_1(x) = \mathrm{P}_x(H_{x_1} < H_{x_0}) = 1 - \varphi_0(x)$$

and $f(x) = \mathrm{E}_x[H_{x_0}]$. Then applying Green-Gauss twice gives us $\langle \varphi_1, -\Delta f\rangle_\pi = \langle -\Delta\varphi_1, f\rangle_\pi$. Also $f(x_0) = 0$, $\Delta f(x) = -1$ for any $x \neq x_0$, φ_1 satisfies (6.1) and

$$R^{-1} = \pi(x_0)\Delta\varphi_1(x_0) = -\pi(x_1)\Delta\varphi_1(x_1).$$

Thus we have

$$\langle \varphi_1, -\Delta f\rangle_\pi = \sum_{x \neq x_0} \pi(x)\varphi_1(x) = \sum_x \pi(x)\varphi_1(x)$$

and

$$\langle -\Delta\varphi_1, f\rangle_\pi = -\Delta\varphi(x_1)\pi(x_1)f(x_1) = \frac{1}{R} \cdot \mathrm{E}_{x_1}[H_0]$$

By interchanging 0 and 1 the resistance does not change and we can use the above to arrive at the conclusion (by using $\varphi_1 + \varphi_0 \equiv 1$). □

Given a set $A \subseteq V$ we can obtain a new set by collapsing A: (G', μ') is defined by

$$V' = (V \setminus A) \cup \{a\},$$

$$E' = \{\{x,y\} \in E \mid x,y \notin A\} \cup \{\{a,x\} \mid \exists y \in A : \{x,y\} \in E, x \neq a\}$$

and the weights are given by

$$\mu'_{x,y} = \begin{cases} \mu_{x,y}. & x,y \in A \\ \sum_{z\in A} \mu_{z,y} & x = a, y \notin A \end{cases}.$$

Definition 6.7.
- $R_{\text{eff}}(B_0, B_1) = R'_{\text{eff}}(x_0, x_1)$ is defined to be the resistance between x_0, x_1 in G', obtained by collapsing B_0 to x_0, B_1 to x_1.
- If G is infinite and B_0 finite we take $A_n \nearrow V$, A_n finite and set

$$R_{\text{eff}}(B_0, +\infty) := \lim_{n\to+\infty} R_{\text{eff}}(B_0, A_n^{\complement}).$$

The limit exists because the sequence on the RHS is increasing.

Theorem 6.8. Let (G, μ) be inifinite and $x_0 \in V$. Then

$$\pi(x_0)\mathrm{P}_{x_0}(\tilde{H}_x = +\infty) = R_{\text{eff}}^{-1}(x_0, +\infty) = C(x_0, +\infty)$$

The theorem has a useful corollary:

Corollary 6.9. x_0 is recurrent (that is, X is recurrent) if and only if $C(x_0, +\infty) = 0$.

Proof of Theorem 6.8. Let $A_n = B_n(x_0)$. Collapse $A_n^{\complement}$ to x_1. Then

$$\pi(x_0)\mathrm{P}_{x_0}(\tilde{H}_{x_0} = +\infty) \xleftarrow{n\to+\infty} \pi(x_0)\mathrm{P}_{x_0}(H_{x_1} < \tilde{H}_{x_0}) = \frac{1}{R_{\text{eff}}(x_0, x_1)} \xrightarrow{n\to+\infty} \frac{1}{R(x_0, +\infty)}.$$

This proves the formula. $\square$

Exercise 6.10. Prove Corollary 6.9.

Variational methods to estimate $C(x_0, +\infty)$. Calculating the resistances on a huge graph can be quite cumbersome. [todo: Write this better] Let (G, μ) be a finite weighted graph and let $B_0, B_1 \subseteq V$ non-empty and disjoint. Define

$$\mathcal{F}(B_0, B_1) := \{f \in H^2 \mid f|_{B_i} \equiv i\}.$$

Moreover, set

$$\bar{C}_{\text{eff}}(B_0, B_1) = \min\{\mathcal{E}(f, f) : f \in \mathcal{F}(B_0, B_1)\}.$$

Theorem 6.11. Suppose G is finite and take B_0 and B_1 as above

$$\bar{C}_{\text{eff}}(B_0, B_1) = C_{\text{eff}}(B_0, B_1) \quad \left(= \frac{1}{R_{\text{eff}}(B_0, B_1)}\right)$$

Proof. Set $\varphi(x) = \mathrm{P}_x(H_{B_1} < H_{B_0})$ and $I_{x,y} = \mu_{x,y}(\varphi(y) - \varphi(x))$. Then

$$\begin{aligned} R_{\text{eff}}(B_0, B_1) =& F(I, B_0) := \sum_{x \in B_0} \sum_{y \in V} I_{x,y} \quad \text{(justify this by collapsing)} \\ &= \sum_{x \in B_0} \sum_{y} \mu_{x,y}(\varphi(y) - \varphi(x)) = \sum_{x \in B_0} \pi(x) \Delta\varphi(x). \end{aligned}$$

Also, using discrete Green-Gauss (Theorem 5.14),

$$\mathcal{E}(\varphi, \varphi) = \mathcal{E}(1 - \varphi, 1 - \varphi) = \langle -\Delta\varphi, f - \varphi \rangle_\pi = \sum_{x \in B_0} \pi(x) \Delta\varphi(x),$$

where the last equality comes from the fact that φ is harmonic outside of $B_0 \cup B_1$ and $\varphi \equiv 0$ on B_1. Hence $\mathcal{E}(\varphi, \varphi) = R_{\text{eff}}(B_0, B_1)^{-1}$. Now, if $f \in \mathcal{F}(B_0, B_1)$, Then

$$\mathcal{E}(\varphi, f - \varphi) = \langle -\Delta\varphi, \varphi - \varphi \rangle = 0,$$

again using that $\Delta\varphi \equiv 0$ outside of $B_0 \cup B_1$ and $f - \varphi \equiv 0$ on $B_0 \cup B_1$. We have thus shown that $\mathcal{E}(\varphi, \varphi) = -\mathcal{E}(\varphi, f)$ and so

$$0 \leq \mathcal{E}(f - \varphi, f - \varphi) = \mathcal{E}(f, f) - \mathcal{E}(\varphi, \varphi) \quad \text{i.e.,} \quad \mathcal{E}(f, f) \geq \mathcal{E}(\varphi, \varphi).$$

Using that $\varphi \in \mathcal{F}(B_0, B_1)$ we can take infiumum over all such f and obtain the desired equality. $\square$

Note that $E = UI \overset{\text{Ohm}}{=} \frac{U^2}{R} = U^2 \cdot C \overset{\text{Ohm}}{=} I^2 R$ (from physics). This translates to

$$\mathcal{E}(f, f) = \frac{1}{2} \sum_{x,y} \underbrace{\mu_{x,y}}_{C} \underbrace{(f(y) - f(x))^2}_{U^2}.$$

In the following we want to minimize the energy and this intuition helps understanding things.

Definition 6.12. $J = (J_{x,y})_{x,y \in V}$ is called *flow from B_0 to B_1* if

1. $\sum_{y \in V} J_{x,y} = 0$ if $x \notin B_1 \cup B_0$
2. $J_{x,y} = -J_{x,y}$
3. $J_{x,y} = 0$ if $x \nsim y$.

The *total flow out of* B_0 is given by

$$F(J,B_0)=\sum_{x\in B_0}\sum_{y\sim x}J_{x,y},$$

sometimes also called the flux of J.

Exercise 6.13. Show that if G is finite then $F(J,B_0)=-F(J,B_1)$. Show that it does not hold in general when G is infinite (then one can "move things to infinity").

What is the difference between flow and current?[todo: Write something here]

Definition 6.14. We define

$$E(I,J)=\frac{1}{2}\sum_{x}\sum_{y\sim x}I_{x,y}J_{x,y}\frac{1}{\mu_{x,y}}$$

We'd like to study the problem:

$$R_{\text{eff}}(B_0,B_1)=\inf\left\{E(J,J):J\text{ is flow from }B_0\text{ to }B_1\text{ with }F(J,B_0)=1\right\}. \tag{6.2}$$

Theorem 6.15 (Thompson's principle). If (G,μ) is finite, then $R_{\text{eff}}(B_0,B_1)$ is indeed the resistance between B_0 and B_1.

Proof. Let G be finite and let J be as in (6.2) and $f\in\mathcal{F}(B_0,B_1)$. Then

$$\begin{aligned}\frac{1}{2}\sum_{x}\sum_{y}J_{x,y}(f(y)-f(x))=&\frac{1}{2}\sum_{x,y}-J_{x,y}f(x)-\frac{1}{2}\sum_{x,y}J_{x,y}f(x)\\ =&-\sum_{x}\underbrace{f(x)}_{\in\mathcal{F}(B_0,B_1)}\sum_{y\sim x}J_{x,y}\\ =&-\sum_{x\in B_1}\underbrace{f(x)}_{\equiv 1}\sum_{y\sim x}J_{x,y}\\ =&-F(J,B_1)\overset{\text{Exercise}}{=}F(J,B_0)=1.\end{aligned}$$

Hence if $\varphi(x)=\mathrm{P}_x(H_{B_1}<H_{B_0})=U(x)$ solves (6.1) and $I_{x,y}=\mu_{x,y}(\varphi(y)-\varphi(x))$, then

$$E(I,J)=\frac{1}{2}\sum_{x}\sum_{y}\frac{1}{\mu_{x,y}}I_{x,y}J_{x,y}=\frac{1}{2}\sum_{x,y}J_{x,y}\overbrace{(\varphi(y)-\varphi(x))}^{\in\mathcal{F}(B_0,B_1)}=1.$$

I is not feasable for (6.2) since $F(B_0,I)=C_{\text{eff}}(B_0,B_1)$ so we set $I'=R_{\text{eff}}(B_0,B_1)\mathrm{I}$. Then

$$E(I',J)=R(B_0,B_1)\quad\text{and thus also}\quad E(I',I')=R_{\text{eff}}(B_0,B_1),$$

hence for any J as in (6.2),

$$0 \leq E(J-I', J-I') = \cdots = E(J,J) - E(I',I'),$$

proving the statement. $\square$

Example 6.16. [todo: See notes] in the picture every edge between x_0 and x_1 has resistance 1 hence $R(x_0, x_1) = 3$ and $C(x_0, x_1) = \frac{1}{3}$. On binary tree we then have resistance $\left(\frac{1}{2}\right)^{d(x_0,\cdot)}$ or something like this. [todo: make this understandable].

Corollary 6.17. Let (G, μ) be finite and B_0, B_1 as usual.

1. $\mu \mapsto C(B_0, B_1)$ is coordinate wise increasing in μ.
2. "Cutting". Deleting an edge (i.e., setting its conductance to zero) decreases the conductance.
3. "Shorting". Identifiying verticies x, y (that is, formally, setting $\mu_{x,y} = +\infty$) increases the conductance.

Proof. [todo:] $\square$

Example 6.18. SRW on $\mathbb{Z}^2$ is recurrent if and only if $C_{\text{eff}}(0, +\infty) = 0$. Short $\partial\Lambda_n$ to one vertex (for all $n \in \mathbb{N}$) but then [todo: $R = \frac{1}{4}, R = \frac{1}{12}, R = \frac{1}{20}$ on $\partial\Lambda_1, \partial\Lambda_2, \partial\Lambda_3$ we obtain $R_{\text{eff}} = +\infty$ hence the conductance is 0.]

[todo: Rest of lecture 19]

6.0.1 Transience, Recurrence and rough isometries

How does the " long time behaviour" of RW relate to the "large scale geometry" of the graph? If two graphs G_i, $i = 1, 2$ are similar, do the random walks on G_1, G_2 have similar properties?

To explain what "similar" means we need the concept of rough isometries which is one of the simplest kinds of similarity. First we introduce a similar notion:

Definition 6.19. Let (G, μ) be a weighted graph. Then (G', μ') is obtained from (G, μ) by bounded pertubation of weights if $G = G'$ and there exists $C \in (0, +\infty)$ such that

$$\text{for any } x, y \in V, \quad \frac{1}{C}\mu_{x,y} \leq \mu'_{x,y} \leq C\mu_{x,y}.$$

Often we also want to modify G itself. Recall, G can be viewed as a metric space (G, d), where G is the natural distance on G.

Definition 6.20. Let (X_1, d_1), (X_2, d_2) be metric spaces. A map $\varphi : X_1 \to X_2$ is called *rough isometry* if there exist some $C_1, C_2 \in (0, 1\infty)$ such that

(a) for all $x, y \in X_1$ we have

$$\frac{1}{C_1}\Big(d_1(x,y) - C_2\Big) \leq d_2\Big(\varphi(x), \varphi(y)\Big) \leq C_1\Big(d_1(x,y) + C_2\Big)$$

and

(b)

$$\bigcup_{x \in X_1} B_{d_2}(\varphi(x), C_2) = X_2.$$

We say that X_1, X_2 are *roughly isometric* if such a map φ exists.

Exercise 6.21. Using the axiom of choice, show that rough isometry is an equivalence relation.

For weighted graph it is suitable to "combine" these two definitions. [todo: Continue typing this lecture]